

Chapter 5

Fields and Galois Theory

5.1 Field Extensions

5.1.1 The Basics

Definition 5.1.1. A *field* is a commutative division ring. By convention, the zero ring is not a field.

Recall that the *characteristic* of a ring is the least positive integer n such that $\underbrace{1 + 1 + \cdots + 1}_n = 0$ (where we say the ring has characteristic 0 if no such n exists).

Proposition 5.1.2. If k is a field, then $\text{char } k = 0$ or a prime.

Proof. Since the 0 ring isn't a field, suppose otherwise that $\text{char } k$ is composite, say $\text{char } k = mn$. Then

$$0 = \underbrace{1 + 1 + \cdots + 1}_{mn} = \left(\underbrace{1 + 1 + \cdots + 1}_m \right) \left(\underbrace{1 + 1 + \cdots + 1}_n \right),$$

but the right hand side is the product of two nonzero elements of k . ■

If k and F are fields, and we have a homomorphism $\phi : k \rightarrow F$, then $\ker \phi$ does not contain 1 (as $\phi(1) = 1$, and the zero ring is not a field). Thus, $\ker \phi \subsetneq k$ is a proper ideal, and so must be 0. Thus, *homomorphisms between fields are injective*. For this reason, we often think of these as equivalent to inclusions $k \subseteq F$. We say F is a *field extension* of k , and also write F/k .

If we have extensions F/k and E/k , we say a homomorphism $\phi : F \rightarrow E$ is a homomorphism of extensions if ϕ restricts to the identity on k . Now, if we have a field extension F/k , then we may view F as a vector space over k . We take advantage of this fact quite often.

Definition 5.1.3. The *degree* of a field extension F/k is $[F : k] = \dim_k F$. A field extension is said to be *finite* if $[F : k]$ is finite, and *infinite* otherwise.

Proposition 5.1.4. Let $F/E/k$ be a tower of field extensions. Then F/k is finite if and only if F/E and E/k are, and in this case $[F : k] = [F : E] \cdot [E : k]$.

Proof. If F/k is finite, then clearly F/E and E/k are finite. Conversely, suppose that F/E and E/k are finite. Let $b_1, \dots, b_n \in F$ be an E -basis, and $c_1, \dots, c_m \in E$ be a k -basis. Fix an element $a \in F$, and write $a = \sum_{i=1}^n a_i b_i$ for $a_i \in E$. Then we can write $a_i = \sum_{j=1}^m \alpha_{i,j} c_j$ for $\alpha_{i,j} \in k$. This gives $a = \sum_{i=1}^n \sum_{j=1}^m \alpha_{i,j} (c_i b_j)$, so that $\{b_i c_j : 1 \leq i \leq n, 1 \leq j \leq m\}$ spans F as a k -vector space. In particular, F/k is finite. To see that we have a basis, suppose we have a linear dependence relation

$$0 = \sum_{i=1}^n \sum_{j=1}^m \beta_{i,j} b_i c_j = \sum_{i=1}^n \left(\sum_{j=1}^m \beta_{i,j} c_j \right) b_i,$$

where the $\beta_{i,j}$ lie in k . Since $\sum_{j=1}^m \beta_{i,j} c_j \in E$ for all i , the E -linear independence of the b_i mean that $\sum_{j=1}^m \beta_{i,j} c_j = 0$ for all i . But the c_j are k -linearly independent, so $\beta_{i,j} = 0$ for all i, j , as desired. In particular, $[F : k] = nm = [F : E] \cdot [E : k]$. ■

A special kind of extension is a so-called *simple* extension.

Definition 5.1.5. Let F/k be a field extension and $\alpha \in F$. We write $k(\alpha)$ for the smallest subfield of F containing both k and α . The extension is said to be *simple* if $F = k(\alpha)$.

We also write $k[\alpha]$ for the smallest *subring* of F containing both k and α (note that $k[\alpha] \subseteq k(\alpha)$, but this inclusion may be strict).

More generally, given intermediate fields $k \subseteq E, K \subseteq F$, we write EK for the smallest subfield of F containing both E and K (this exists by considering the intersection of all subfields containing both E and K). It is not hard to see that $E \cap K$ is the largest subfield of F contained in both E and K , so the subfields of the extension F/k form a lattice.

Remark 5.1.6. Every field has a *prime subfield* that is the unique minimal subfield. If $\text{char } F = p > 0$, this field is isomorphic to $\mathbb{Z}/p\mathbb{Z}$, and if $\text{char } F = 0$, this field is isomorphic to $\mathbb{Q}$. There is exactly one way of embedding the prime subfield of F into F (since we must send 1 to 1, and everything else is uniquely determined). In particular, it doesn't hurt to *only* consider maps of field extensions $F/k \rightarrow L/k$, since we can take k to be the prime subfield of F and essentially consider all maps $F \rightarrow L$.

5.1.2 Algebraic Extensions

Definition 5.1.7. Let F/k be a field extension. We say $\alpha \in F$ is *algebraic over k* if there is a nonzero polynomial $f \in k[x]$ such that $f(\alpha) = 0$. Otherwise, we say α is *transcendental over k* . We say the extension F/k is *algebraic* if each $\alpha \in F$ is algebraic over k .

Since the polynomial ring $k[x]$ has a lot of structure, algebraic extensions are easier to understand.

Proposition 5.1.8. *Finite extensions are algebraic.*

Proof. Let F/k be a finite extension of degree n , and let $\alpha \in F$. Then $1, \alpha, \alpha^2, \dots, \alpha^n$ are k -linearly dependent, and so there is a polynomial in $k[x]$ with α as a root. ■

Now, if $\alpha \in F$ is algebraic over k , then $\{f \in k[x] : f(\alpha) = 0\}$ is a nonzero ideal of $k[x]$. Since $k[x]$ is a PID, this ideal is generated by a single element, unique up to units.

Definition 5.1.9. Let F/k be a field extension, and let $\alpha \in F$ be algebraic over k . The *minimal polynomial* of α is the unique monic generator of $\{f \in k[x] : f(\alpha) = 0\}$, denoted m_α .

Now, the evaluation homomorphism $k[x] \rightarrow k[\alpha]$ has kernel $\langle m_\alpha \rangle$, and so $k[x]/\langle m_\alpha \rangle$ is a domain. Thus, $\langle m_\alpha \rangle$ is a nonzero prime ideal of $k[x]$, and so maximal. This means m_α is irreducible and $k[\alpha]$ is a field. In particular, we see that $k[\alpha] = k(\alpha)$.

Proposition 5.1.10. *Let F/k be a field extension, and $\alpha \in F$. The following are equivalent.*

- (1) α is algebraic over k
- (2) $k[\alpha] = k(\alpha)$
- (3) $k(\alpha)/k$ is a finite extension

Furthermore, $[k(\alpha) : k] = \deg m_\alpha$.

Proof.

(1) $\Rightarrow$ (2) This follows from our above discussion.

(2) $\Rightarrow$ (1) Since $k[\alpha]$ is a field, and elements of $k[\alpha]$ are of the form $f(\alpha)$ for $f \in k[x]$, we can find some $g \in k[x]$ with $\alpha^{-1} = g(\alpha)$. Then α is a root of $xg(x) - 1$.

(1) $\Rightarrow$ (3) We have that $k(\alpha) = k[\alpha] \cong k[x]/\langle m_\alpha \rangle$, so that $1, \alpha, \dots, \alpha^{\deg m_\alpha - 1}$ is a basis.

(3) $\Rightarrow$ (1) The extension $k(\alpha)/k$ is finite, and so algebraic, meaning α is algebraic over k . ■

On the other hand, if α is transcendental, then the evaluation map $k[x] \rightarrow k[\alpha]$ has trivial kernel, so that $k[\alpha] \cong k[x]$.

Corollary 5.1.11. *Let $F/E/k$ be a tower of field extensions with E/k algebraic. If $\alpha \in F$ is algebraic over E , then α is algebraic over k . In particular, if F/E is algebraic, then so is F/k .*

Proof. Let $f(x) \in E[x]$ such that $f(\alpha) = 0$, and write $f(x) = \sum_{i=0}^n a_i x^i$. Since each a_i is algebraic over k , it is also algebraic over $k(a_0, \dots, a_{i-1})$. Thus, since we have a chain of fields

$$k(a_0, \dots, a_n) \supseteq k(a_0, \dots, a_{n-1}) \supseteq \dots \supseteq k(a_0) \supseteq k$$

with each step finite, we see that $k(a_0, \dots, a_n)/k$ is finite. Since α is algebraic over $k(a_0, \dots, a_n)$, we also have that $k(\alpha, a_0, \dots, a_n)/k$ is finite, and so $k(\alpha)/k$ is finite. Thus, α is algebraic over k . ■

Corollary 5.1.12. *Let F/k be a field extension. Then*

$$A = \{\alpha \in F : \alpha \text{ algebraic over } k\}$$

is a subfield of F .

Proof. Let $\alpha, \beta \in F$ be algebraic over k . Then $k(\alpha, \beta)/k$ is finite, since we have a finite tower $k(\alpha, \beta) \supseteq k(\alpha) \supseteq k$. Thus, $\alpha - \beta$ and $\alpha\beta^{-1}$ are algebraic over k . ■

5.1.3 Simple Extensions

The following result is key in our development of field theory.

Theorem 5.1.13. *Let $\iota : k_1 \rightarrow k_2$ be an isomorphism of fields, and let $k_1(\alpha_1)/k_1$ and $k_2(\alpha_2)/k_2$ be finite simple extensions. If $\iota(m_{\alpha_1}) = m_{\alpha_2}$, then ι extends uniquely to an isomorphism $k(\alpha_1) \rightarrow k(\alpha_2)$ sending α_1 to α_2 .*

Proof. We have a chain of maps

$$k_1[x] \longrightarrow k_2[x] \longrightarrow \frac{k_2[x]}{\langle m_{\alpha_2} \rangle} \xrightarrow{\sim} k_2(\alpha_2)$$

with kernel $\langle m_{\alpha_1} \rangle$ (since $\iota(m_{\alpha_1}) = m_{\alpha_2}$). Thus, we get

$$k(\alpha_1) \xrightarrow{\sim} \frac{k_1[x]}{\langle m_{\alpha_1} \rangle} \xrightarrow{\sim} \frac{k_2[x]}{\langle m_{\alpha_2} \rangle} \xrightarrow{\sim} k_2(\alpha_2),$$

and this restricts to ι . Since a homomorphism on $k(\alpha_1)$ is determined by how it acts on k and α_1 , this extension is unique. ■

More generally, if F/k is any field extension such that F contains a root β of m_α , then we get a unique homomorphism of field extensions $k(\alpha)/k \hookrightarrow F/k$ with $\alpha \mapsto \beta$. Note that there may be more than one map $k(\alpha)/k \hookrightarrow F/k$ (in fact, there are as many maps as there are roots of m_α in F). For instance, we have two distinct maps $\mathbb{Q}(\sqrt{2}) \hookrightarrow \mathbb{R}$, given by $\sqrt{2} \mapsto \pm\sqrt{2}$. These have the same image, but in general this need not be true: consider the three distinct maps $\mathbb{Q}(\sqrt[3]{2}) \hookrightarrow \mathbb{C}$, each of which has a different image. We will see that such images will always coincide when F/k is what is known as a normal extension.

We'll now provide a classification of simple extensions, though we first prove a result of independent interest. Recall that Euler's totient function is given by $\varphi(n) = \#\{1 \leq d \leq n : \gcd(d, n) = 1\}$, and that we have $\sum_{d|n} \varphi(d) = n$.

Proposition 5.1.14. *Let G be a finite subgroup of $F^\times$. Then G is cyclic.*

Proof. For each n dividing $|G|$, $x^n - 1$ has at most n roots in F , and so at most n roots in G . Conversely, the group homomorphism $G \rightarrow G$ $g \mapsto g^{|G|/n}$ has a kernel of size $\leq |G|/n$, and so an image of size $\geq n$. Since each element of the image is a root of $x^n - 1$, we see that $x^n - 1$ has exactly n roots in G .

Now, for each n , we claim that G contains $\varphi(n)$ elements of order n . We show this by induction on n , with $n = 1$ being clear: G contains $\varphi(1) = 1$ element of order 1. Since there are n elements of order dividing n , induction shows that there are

$$n - \sum_{\substack{d|n \\ d < n}} \varphi(d) = \varphi(n)$$

elements of order exactly n . In particular, G contains an element of order exactly $|G|$, as desired. ■

This fact has consequences in number theory, since it means that the field $\mathbb{Z}/p\mathbb{Z}$ has a cyclic unit group. We will however take advantage of this to prove a very different result.

Theorem 5.1.15 (Artin). *Let F/k be a finite field extension. Then $F = k(\alpha)$ if and only if there are finitely many intermediate fields $k \subseteq E \subseteq F$.*

Proof. Suppose first that $F = k(\alpha)$. For each intermediate field $F/E/k$, we have a polynomial g_E that is the minimal polynomial of α over E . We claim that this polynomial is unique (i.e. if $g_E = g_{E'}$ then $E = E'$). If the coefficients of g_E are $a_0, \dots, a_n$, it suffices to show that $E = k(a_0, \dots, a_n)$, as this allows us to uniquely recover E from g_E . Write $L = k(a_0, \dots, a_n)$. Since g_E is irreducible in $E[x] \supseteq L[x]$, it is irreducible in $L[x]$, and so is the minimal polynomial of α over L . Thus,

$$[F : E] = \deg g_E = [F : L],$$

and since $E \supseteq L$, we have equality. We have therefore built an injection from the set of intermediate fields $F/E/k$ to the set of monic polynomials in $k[x]$ dividing $m_\alpha = g_k$ (since g_k lies in $E[x]$ and has α as a root for all $F/E/k$), and so there are finitely many intermediate fields.

Conversely, suppose that we have finitely many intermediate fields $F/E/k$. We proceed by induction on $[F : k]$, with $[F : k] = 1$ being clear. If $[F : k] > 1$, take $\alpha \in F \setminus k$, so that $[F : k(\alpha)] < [F : k]$. Since there are finitely many intermediate fields $F/E/k(\alpha)$ (as these are in particular intermediate fields $F/E/k$), induction gives $F = k(\alpha, \beta)$ for some β .

First, suppose that k is infinite. Let $E_t = k(\alpha + \beta t)$ for $t \in k$. Since we only have finitely many intermediate fields but infinitely many elements of k , $E_t = E_s$ for some $t \neq s$. But then $\alpha + s\beta, \alpha + t\beta \in E_t$, and so $\beta(s - t) \in E_t$. Since $s \neq t$, this means $\beta \in E_t$ and $\alpha \in E_t$ as well, so that

$$F = k(\alpha, \beta) \subseteq E_t \subseteq F$$

and $F = k(\alpha + t\beta)$ is a simple extension.

Suppose otherwise that k is finite. Since $[F : k]$ is finite, F is finite as well, and so $F^\times$ is cyclic, say generated by α . But then $F = k(\alpha)$, and so we are done. ■

5.1.4 Normal Extensions

Definition 5.1.16. A polynomial $f \in k[x]$ is said to *split* if we can write f as a product of linear polynomials.

Definition 5.1.17. Let $f \in k[x]$. A *splitting field extension* for f is an extension F/k such that $f \in F[x]$ splits, and $F = k(\alpha_1, \dots, \alpha_n)$, where the α_i are the roots of f .

If we have an irreducible $f \in k[x]$, then $\langle f \rangle \subseteq k[x]$ is maximal, and so $k[x]/\langle f \rangle$ is a field. Further, the coset $\bar{x}$ is easily seen to be a root of f , so we can construct field extensions that contain a root of a given irreducible polynomial. Iterating this gives the following.

Theorem 5.1.18. *Let $f \in k[x]$ be nonzero of degree n . Then there exists a field extension L/k such that L is a splitting field for f , and $[L : k] \leq n!$.*

Proof. We induct on n , with $n = 0$ being trivial. If $n > 1$, let $h \mid f$ be an irreducible factor of f , and construct a field extension k'/k containing a root α of h . Now, $f/(x - \alpha) \in k'[x]$ has degree $n - 1$, so induction gives a field extension L/k' with $[L : k'] \leq (n - 1)!$. Now,

$$[L : k] = [L : k'] \cdot [k' : k] = [L : k'] \cdot \deg h \leq (n - 1)! \cdot n = n!. \quad \blacksquare$$

Theorem 5.1.19 (Uniqueness of Splitting Fields). *Let $\iota : k_1 \rightarrow k_2$ be an isomorphism of fields. Let $f_i \in k_i[x]$ with $\iota(f_1) = f_2$, and let L_i/k_i be a splitting field of f_i . Then there is an isomorphism $L_1 \cong L_2$ restricting to ι on k_1 .*

Proof. We induct on $\deg f_1$. If $\deg f_1 = 1$, then $L_i = k_i$, and so ι is the necessary extension of itself. Take $\deg f_1 = n > 1$. If $L_1 = k_1$, then ι is once again its own extension, so suppose $L_1 \supsetneq k_1$. Let $\alpha \in L_1 \setminus k_1$ be a root of f_1 . Then $m_\alpha \mid f_1$, so $\iota(m_\alpha) \mid \iota(f_1) = f_2$, meaning that $\iota(m_\alpha)$ has a root $\beta \in L_2$, since f splits. We may choose $\beta \notin k_2$ (since otherwise m_α would split over k_1). Theorem 5.1.13 gives an isomorphism $k_1(\alpha) \rightarrow k_2(\beta)$ extending ι . Now, L_1 is the splitting field of $f_1/(x - \alpha)$ over $k_1(\alpha)$ and L_2 is the splitting field of $f_2/(x - \beta)$ over $k_2(\beta)$, so induction gives the result. ■

Splitting fields turn out to be precisely equivalent to so-called *normal* extensions.

Definition 5.1.20. A field extension F/k is said to be *normal* if every irreducible $f(x) \in k[x]$ with a root in F splits over F .

Requiring that a field extension be normal seems a quite restrictive condition. For instance, $\mathbb{Q}(\sqrt[3]{2})$ is not normal, since $t^3 - 2$ has only one root in it. However, it is not quite as bad as one would expect.

Theorem 5.1.21. Let F/k be a finite field extension. The following are equivalent.

- (1) F/k is normal
- (2) F is the splitting field of some $f \in k[x]$
- (3) For any extension L/F , every embedding $F \hookrightarrow L$ has image F .

Proof.

- (1) $\Rightarrow$ (2) Since F/k is finite, we can write $F = k(\alpha_1, \dots, \alpha_n)$ for some $\alpha_1, \dots, \alpha_n \in F$. Let m_i be the minimal polynomial of α_i , and write $f = m_1 \cdots m_n$. Since F/k is normal and each m_i has a root in F , f splits over F . Further, by construction, F is generated by the roots of f , so F is a splitting field of f over k .
- (2) $\Rightarrow$ (3) Suppose that f has n distinct roots $\alpha_1, \dots, \alpha_n$ in F . Since f splits, these are also all roots of f in L . Now, any embedding $\iota : F \hookrightarrow L$ must permute the α_i . Since $F = k(\alpha_1, \dots, \alpha_n)$, this means that

$$\iota(F) = k(\iota(\alpha_1), \dots, \iota(\alpha_n)) = k(\alpha_1, \dots, \alpha_n) = F.$$

- (3) $\Rightarrow$ (1) Let $f \in k[x]$ be irreducible with a root $\alpha \in F$. Write $F = k(\alpha_1, \dots, \alpha_n)$, and let L be the splitting field of $g = fm_{\alpha_1} \cdots m_{\alpha_n}$ over k (so that L automatically contains F). If f does not split over F , then there is some $\beta \in L \setminus F$ that is a root of f . Now, Theorem 5.1.13 gives an isomorphism $\iota : k(\alpha) \rightarrow k(\beta)$ sending α to β . Since L is the splitting field of g over both $k(\alpha)$ and $k(\beta)$, Theorem 5.1.19 gives an isomorphism $L \rightarrow L$ restricting to ι on $k(\alpha)$. But then restricting to F gives an embedding $F \hookrightarrow L$ whose image contains $\iota(\alpha) = \beta$, a contradiction. Thus, f must split over F , as desired. ■

Corollary 5.1.22. Any finite extension is contained in a finite normal extension.

Proof. If F/k is finite, write $F = k(\alpha_1, \dots, \alpha_n)$. Let L be the splitting field of $m_{\alpha_1} \cdots m_{\alpha_n}$ over k , so that $L \supseteq F$ and L/k is normal. ■

5.1.5 Separable Extensions

Definition 5.1.23. If $f \in k[x]$, then $\alpha \in k$ is said to be a *repeated root* for f if $(x - \alpha)^2 \mid f$.

We often choose to work with polynomials without repeated roots. Our intuition tells us that irreducible polynomials never have repeated roots, but this is unfortunately false in general (though true in most familiar cases).

We first present an example of an irreducible polynomial with repeated roots. Consider $\mathbb{F}_p(u)$, the field of rational functions in u with coefficients in $\mathbb{F}_p$. The polynomial $x^p - u$ has no roots in $\mathbb{F}_p(u)$. However, once we adjoin an element s with $s^p = u$, the identity $x^p - s^p = (x - s)^p$ means that s is the only root of this polynomial. Thus, any irreducible factor has repeated roots.

Lemma 5.1.24. Let $f \in k[x]$ be nonzero. The following are equivalent.

- (1) f has no repeated roots in every extension F/k
- (2) There is an extension F/k such that f has $\deg f$ distinct roots in F
- (3) For every extension F/k where f splits, f has $\deg f$ distinct roots in F

If any of these happen, f is said to have *distinct roots*.

Proof.

- (1) $\Rightarrow$ (2) Let F be the splitting field of f over k . Since f contains no repeated roots in F , examining its factorization into linear polynomials shows that it has $\deg f$ distinct roots.

- (2) $\Rightarrow$ (3) Every field in which f splits contains an isomorphic copy of the splitting field of f over k . In particular, F contains such an isomorphic copy, and so we see that the splitting field has $\deg f$ distinct roots of f . Thus, every field in which f splits must also contain $\deg f$ distinct roots of f .
- (3) $\Rightarrow$ (1) Suppose f has a repeated root α in F/k , and let $L \supseteq F$ be a splitting field for f . Then $(x - \alpha)^2$ divides f in $L[t]$, and so examining the factorization of f into linear polynomials shows that it has $\leq \deg f - 1$ roots. $\blacksquare$

Corollary 5.1.25. *Let F/k be a field extension, and let $f \in k[x]$ be nonzero. Then f has distinct roots as an element of $k[x]$ if and only if f has distinct roots as an element of $F[x]$.*

Proof. Let L be the splitting field of f over F . Then f has distinct roots as an element of $k[x]$ if and only if f contains exactly $\deg f$ roots in L if and only if f has distinct roots as an element of $F[x]$. $\blacksquare$

Corollary 5.1.26. *If $f \in k[x]$ has distinct roots and $g \in k[x]$ divides f , then g has distinct roots.*

Proof. If F/k is a field extension with $\alpha \in F$ such that $(x - \alpha)^2 \mid g$ in $F[x]$, then $(x - \alpha)^2 \mid f$ in $F[x]$, and so f does not have distinct roots. $\blacksquare$

Now, requiring that a polynomial has distinct roots is very restrictive. For instance, it means we can't think about the polynomial $(x^2 - 2)^2$, despite the fact that the repeated roots here don't cause any problems for studying the splitting field. As such, we make the following definition.

Definition 5.1.27. A polynomial $f \in k[x]$ is said to be *separable* over k if every irreducible factor of f has distinct roots. In particular, an irreducible factor is separable if and only if it has distinct roots.

Unlike having distinct roots, separability is quite dependent on the base field. For instance, every polynomial is immediately separable over its splitting field, even if it is not separable over some base field. We do not, however, lose separability when extending fields.

Proposition 5.1.28. *Let F/k be a field extension, and let $f \in k[x]$ be separable over k . Then f is separable over F .*

Proof. Let h be an irreducible factor of f in $F[x]$. Then there is some irreducible factor g of f in $k[x]$ such that $h \mid g$ in $F[x]$. Since g has distinct roots, so does h . $\blacksquare$

Separability is captured quite nicely by examining the formal derivative. If $f(x) = \sum_{i=0}^n a_i x^i$, the *formal derivative* of f is given by $f'(x) = \sum_{i=0}^{n-1} (i+1)a_{i+1}x^i$. It is not hard to check that this satisfies the usual product rule from calculus.

If α is a repeated root of f , we can write $f(x) = (x - \alpha)^2 h(x)$. This gives that

$$f'(x) = 2(x - \alpha)h(x) + (x - \alpha)^2 h'(x),$$

so that $f'(\alpha) = 0$. Conversely, if $f(\alpha) = f'(\alpha) = 0$, we may write $f(x) = (x - \alpha)g(x)$, and compute

$$f'(x) = g(x) + (x - \alpha)g'(x),$$

so that $0 = f'(\alpha) = g(\alpha)$, and $(x - \alpha)^2 \mid f(x)$.

Proposition 5.1.29. *Let k be a field and $f \in k[x]$. Then f has distinct roots if and only if f and f' are relatively prime.*

Proof. Suppose that f and f' are relatively prime, and write $1 = gf + hf'$ for $g, h \in k[x]$. Then this identity also holds in all field extensions F/k . In particular, for a field extension F/k and $\alpha \in F$, we have that $1 = g(\alpha)f(\alpha) + h(\alpha)f'(\alpha)$, so we can't have $f(\alpha) = f'(\alpha) = 0$. Since α was arbitrary, f has distinct roots.

Conversely, suppose that f and f' have a common nonconstant factor $g \in k[x]$. If F/k is a splitting field for g and α is a root of g in F , then $f(\alpha) = f'(\alpha) = 0$, so that f does not have distinct roots. $\blacksquare$

Corollary 5.1.30. *Let k be a field and $f \in k[x]$ be irreducible. Then f is separable if and only if $f' \neq 0$.*

Proof. Suppose $f' \neq 0$, and let $g = \gcd(f, f')$. Since f is irreducible, g is either a unit or associate to f . But since $\deg f' < \deg f$, we see that g cannot be associate to f , and so $\gcd(f, f') = 1$, meaning f is separable. Conversely, if $f' = 0$, then $\gcd(f, f') = f \neq 1$, and so f is inseparable. $\blacksquare$

In particular, if $\text{char } k = 0$, then all polynomials are separable, since the only way for $f' = 0$ is if f is constant. Fields of characteristic zero are what are known as *perfect fields*.

Definition 5.1.31. A field k is said to be *perfect* if all polynomials in $k[x]$ are separable.

Definition 5.1.32. Let F/k be a field extension and $\alpha \in F$ be algebraic over k . Then α is said to be *separable* over k if m_α is separable. The extension F/k is said to be separable if every $\alpha \in F$ is separable over k .

5.1.6 Algebraic Closures

Definition 5.1.33. A field k is said to be *algebraically closed* if every polynomial in $k[x]$ splits over k .

If k is algebraically closed and F/k is a field extension, then each $\alpha \in F \setminus k$ must be transcendental over k (since otherwise m_α would be linear). In particular, k has no proper algebraic extensions. We may hope to have a “minimal” extension $\bar{k}/k$ such that $\bar{k}$ is algebraically closed. The following result helps us with that.

Proposition 5.1.34. Let F/k be a field extension with F algebraically closed. Then

$$A = \{\alpha \in F : \alpha \text{ algebraic over } k\}$$

is algebraically closed.

Proof. Clearly A/k is algebraic. If $\alpha \in F$ is algebraic over A , then α is algebraic over k by Corollary 5.1.11. Thus, $\alpha \in A$, and so A is algebraically closed. ■

Note that no proper subfield K of A can be algebraically closed, as A/K would then be a proper algebraic extension, which is impossible. Thus, this field A is a minimal algebraic extension of k . This motivates the following definition.

Definition 5.1.35. Let k be a field. An *algebraic closure* of k is an algebraic extension $\bar{k}/k$ such that $\bar{k}$ is algebraically closed.

We now show that any field indeed has an algebraic closure. By Lemma 5.1.34, it is enough to construct an algebraically closed extension of k .

Lemma 5.1.36. Let k be a field. Then there is an extension F/k such that every nonconstant $f(x) \in k[x]$ has a root in F .

Proof. For each nonconstant monic irreducible $f \in k[x]$, take an indeterminate x_f . Consider the polynomial ring $k[\{x_f\}_f]$, and let $I = \langle f(x_f) : f \in k[x] \text{ monic irreducible} \rangle$.

We claim I is proper. If we had $1 = \sum_{i=1}^n a_i f_i(x_{f_i})$, then we would have this identity in every extension of k . In particular, it would hold in the splitting field of $f_1 \cdots f_n$, but plugging in $x_{f_i} = \alpha_i$ a root of f_i would give $1 = 0$. Thus, I is indeed a proper ideal, and so we can find a maximal ideal $I \subseteq \mathfrak{m} \subseteq k[\{x_f\}_f]$. The field $k[\{x_f\}_f]/\mathfrak{m}$ is then an extension of k , and the coset of x_f is a root of f , as desired. ■

Theorem 5.1.37. Every field admits an algebraic closure.

Proof. Let k be a field, and construct fields

$$k = L_0 \subseteq L_1 \subseteq L_2 \subseteq \cdots,$$

where L_{i+1} is obtained by applying Lemma 5.1.36 to L_i . Let $L = \bigcup_{i=0}^{\infty} L_i$. If $f \in L[x]$ is an irreducible polynomial, then its coefficients lie in L_n for some large enough n . But then f has a root in $L_{n+1} \subseteq L$, and so we see that f must be linear. Thus, L is algebraically closed, and so we are done by Proposition 5.1.34. ■

We would also hope that algebraic closures are unique up to isomorphism. This is true, and we in fact have a better result: any algebraic extension can be realized as a subfield of an algebraic closure.

Theorem 5.1.38. Let K/k be a field extension with K algebraically closed, and let F/k be an algebraic extension. Then there exists an inclusion $\iota : F \hookrightarrow K$ restricting to the identity on k .

Proof. Consider pairs (E, ι_E) , where $k \subseteq E \subseteq F$ is a subfield, and $\iota_E : E \rightarrow K$. Partially order these pairs by $(E, \iota_E) \preceq (E', \iota_{E'})$ if $E \subseteq E'$ and $\iota_{E'}$ restricts to ι_E on E . This poset is nonempty, it contains the pair $(k, k \hookrightarrow K)$.

We apply Zorn's lemma. If we have a chain $\{(E_c, \iota_{E_c}) : c \in C\}$, define $E = \bigcup_{c \in C} E_c$ and ι_E by $\iota_E(\alpha \in E_c) = \iota_{E_c}(\alpha)$. Then (E, ι_E) is clearly an upper bound for the chain, and so this poset has a maximal element (M, ι_M) .

Suppose that $M \subsetneq F$, and let $\alpha \in F \setminus M$. Let $m(x) \in M[x]$ be the minimal polynomial of α over M . Then $\iota_M(m)(x) \in \iota_M(M)[x]$ is irreducible. Since $\iota_M(M) \subseteq K$ and K is algebraically closed, we can find a root β of $\iota_M(m)$ in L . But then Theorem 5.1.13 gives a map $M(\alpha) \rightarrow \iota_M(M)(\beta) \subseteq K$ extending ι_M , contradicting the maximality of (M, ι_M) . Thus, $M = F$, and we have an injection $F \hookrightarrow K$. ■

We note that this map is far from being unique. For instance, one can easily find 3 distinct field embeddings $\mathbb{Q}(\sqrt[3]{2}) \hookrightarrow \mathbb{C}$. We will see that there are at most $[F : k]$ such embeddings, and that we achieve equality when F/k is what is known as *Galois*.

Corollary 5.1.39. *Algebraic closures are unique up to isomorphism.*

Proof. If $\bar{k}$ and $\bar{k}_0$ are algebraic closures for k , then Theorem 5.1.38 gives an embedding $\bar{k} \hookrightarrow \bar{k}_0$. If this is not an isomorphism, then $\bar{k}_0/\bar{k}$ is a proper algebraic extension, which is impossible. Thus, $\bar{k} \cong \bar{k}_0$. ■

We end with a final, somewhat surprising result.

Proposition 5.1.40. *If F/k is algebraic and every nonconstant $f \in k[t]$ splits in F , then F is algebraically closed.*

Proof. Let $g \in F[x]$ be irreducible, and suppose that $g = \sum_{i=0}^n a_i x^i$. Write $K = k(a_1, \dots, a_n)$, and let L be a splitting field for g over K . Then L/K and K/k are both finite, so that L/k is finite. For a root $\alpha \in L$ of g , we can find a minimal polynomial $m_\alpha \in k[x]$, and we must have that $g \mid m_\alpha$ (since g is the minimal polynomial of α over K). Since m_α splits over F , this means that g also splits over F , so that $\deg g = 1$ and F is algebraically closed. ■

5.2 Galois Theory

5.2.1 Automorphisms of Extensions

The key of Galois theory is to not just study field extensions, but to study *automorphisms* of these field extensions.

Definition 5.2.1. If F/k is a field extension, we write $\text{Aut}(F/k)$ for the group of automorphisms $\sigma : F \rightarrow F$ that fix k . If $H \subseteq \text{Aut}(F/k)$, we write $\text{Fix}(H)$ for the set of elements of F that are fixed by each automorphism in H .

At first glance, one would expect there to be many many automorphisms of a field extension, but this is luckily not the case. To get a sense of why, note that all automorphisms $k(\alpha) \rightarrow k(\alpha)$ must send α to another root of m_α , so that there are at most $[k(\alpha) : k]$ automorphisms¹. This generalizes.

Proposition 5.2.2. *Let F/k be a finite extension and let E/k be any extension. Then there are at most $[F : k]$ inclusions $F/k \hookrightarrow E/k$.*

Proof. We induct on $[F : k]$. If $[F : k] = 1$, then $F = k$ and there is exactly one way to include $k/k \hookrightarrow E/k$. Suppose $[F : k] > 1$, and let $\alpha \in F \setminus k$. Then there are at most $\deg m_\alpha = [k(\alpha) : k]$ ways to embed $k(\alpha)/k \hookrightarrow E/k$. By induction, there are at most $[F : k(\alpha)]$ ways to extend this embedding to one $F/k \hookrightarrow E/k$, so there are at most $[F : k(\alpha)][k(\alpha) : k] = [F : k]$ embeddings $F/k \hookrightarrow E/k$. ■

Corollary 5.2.3. *If F/k is a finite field extension, then $|\text{Aut}(F/k)| \leq [F : k]$.*

Proof. Since F/k is finite, embeddings $F/k \hookrightarrow F/k$ are precisely the same as isomorphisms $F/k \rightarrow F/k$. There are at most $[F : k]$ of these. ■

We end with a very useful example of when we can achieve equality.

Proposition 5.2.4. *Let F/k be a field extension, $H \leq \text{Aut}(F/k)$ be finite, and $k' = \text{Fix}(H)$. Then $H = \text{Aut}(F/k')$ and $|H| = [F : k']$.*

¹In fact, there are exactly as many automorphisms as there are roots of m_α in $k(\alpha)$ by Theorem 5.1.13.

Proof. Clearly $H \leq \text{Aut}(F/k')$, so that $|H| \leq |\text{Aut}(F/k')| \leq [F : k']$. If we show that $[F : k'] \leq |H|$, we are done. Let $H = \{\sigma_1, \dots, \sigma_m\}$, and suppose that $a_1, \dots, a_{m+1} \in F$ are k' -linearly independent. If $v_i = (\sigma_1(a_i), \dots, \sigma_m(a_i)) \in F^m$, then the v_i are F -linearly dependent. Take a minimal linear dependence relation $\sum_{i=1}^n \lambda_i v_i = 0$, and scale so that $\lambda_n = 1$.

We in particular have that $\sum_{i=1}^n \lambda_i a_i = 0$ (since H contains the identity), so that some λ_i is not in k' . Say without loss of generality that λ_1 is not fixed by σ_1 . Applying σ_1 to our linear dependence relation, we get $\sum_{i=1}^n \sigma_1(\lambda_i) v_i = 0$ (note that we are using the fact that H is closed under composition here to claim that $\sigma_1(v_i)$ is the same permutation of the elements across all i). But then

$$0 = \sum_{i=1}^n \lambda_i v_i - \sum_{i=1}^n \sigma_1(\lambda_i) v_i = \sum_{i=1}^n (\lambda_i - \sigma_1(\lambda_i)) v_i = \sum_{i=1}^{n-1} (\lambda_i - \sigma_1(\lambda_i)) v_i,$$

where we use the fact that $\lambda_n = 1$. Further, $\lambda_1 - \sigma_1(\lambda_1) \neq 0$, so this is a nontrivial linear dependence relation consisting of $n-1$ vectors. But we picked a minimal linear dependence relation, and so this is a contradiction. Thus, $[F : k'] \leq |H|$, as desired. ■

We may wonder when we get equality in the number of embeddings (without having to pass to some intermediate extension). For this to happen, it would suffice for there to be exactly $\deg m_\alpha$ roots of m_α in E for all $\alpha \in F$. That is, E/k should be normal, and every m_α should have distinct roots, so that E/k should be separable. This is precisely the content of a *Galois* extension.

5.2.2 Galois Extensions

Definition 5.2.5. A field extension F/k is said to be *Galois* if it is normal and separable.

There are many equivalent classifications of Galois extensions

Theorem 5.2.6. Let F/k be a finite extension. The following are equivalent.

- (1) F/k is Galois
- (2) F/k is the splitting field of a separable polynomial
- (3) $|\text{Aut}(F/k)| = [F : k]$
- (4) $k = \text{Fix}(\text{Aut}(F/k))$

Proof.

(1)⇒(2) Since L/k is normal, it is the splitting field of some polynomial. Since L/k is separable, this polynomial is separable.

(2)⇒(3) Suppose F/k is the splitting field of f . We induct on $[F : k]$. If $[F : k] = 1$ then $F = k$ and we are done. Suppose $[F : k] > 1$, and let $\alpha \in F \setminus k$ be a root of f . Then m_α is separable and splits over F (since F is a splitting field and hence normal), so that there are $\deg m_\alpha = [k(\alpha) : k]$ embeddings $k(\alpha)/k \hookrightarrow F/k$.

Fix such an embedding $k(\alpha) \hookrightarrow k(\beta) \subseteq F$. Note that f remains separable over both $k(\alpha)$ and $k(\beta)$, and that L is thus a splitting field for f over both these fields. Thus, we have an isomorphism $\iota_\beta : L \rightarrow L$ with $\iota_\beta(\alpha) = \beta$. Additionally, by induction, $|\text{Aut}(F/k(\alpha))| = [F : k(\alpha)]$. Thus, we have $[F : k(\alpha)] \cdot [k(\alpha) : k] = [F : k]$ automorphisms of F/k given by the compositions $\iota_\beta \circ \sigma$ for $\sigma \in \text{Aut}(F/k(\alpha))$. Each of these compositions are distinct, as if $\iota_\beta \circ \sigma = \iota_\beta \circ \sigma'$, then $\beta = (\iota_\beta \circ \sigma)(\alpha) = (\iota_\beta \circ \sigma')(\alpha) = \beta'$, and so $\sigma = \sigma'$. Thus, $|\text{Aut}(F/k)| \geq [F : k]$, and so we have equality.

(3)⇒(1) If F/k is not normal or not separable, then there is $\alpha \in F$ such that m_α has strictly less than $\deg m_\alpha$ roots in F . Thus, there are strictly less than $[k(\alpha) : k]$ embeddings $k(\alpha) \hookrightarrow F$. Each of these embeddings extends in at most $[F : k(\alpha)]$ ways. Since every automorphism of F/k restricts to an embedding $k(\alpha) \hookrightarrow F$ which it then extends, we see that there are strictly less than $[F : k]$ automorphisms of F/k .

(3)⇒(4) Let $k' = \text{Fix}(\text{Aut}(F/k))$. By Proposition 5.2.4, we have that

$$[F : k] = |\text{Aut}(F/k)| = |\text{Aut}(F/k')| = [F : k'],$$

so that $k = k'$.

(4) $\Rightarrow$ (3) Proposition 5.2.4 shows that $|\text{Aut}(F/k)| = [F : k]$. ■

In particular, Proposition 5.2.4 says that $F/\text{Fix}(H)$ is a Galois extension whenever $H \leq \text{Aut}(F/k)$.

Corollary 5.2.7. *If F/k is a finite Galois extension and $k \subseteq E \subseteq F$ is an intermediate field, then F/E is Galois.*

Proof. If F is the splitting field of the separable polynomial $f \in k[x]$ over k , then F is the splitting field of the (still) separable polynomial $f \in E[x]$ over E . ■

Definition 5.2.8. Let F/k be a Galois extension. The automorphism group $\text{Aut}(F/k)$ is known as the *Galois group* of the extension.

5.2.3 The Galois Correspondence

Suppose we have a field extension F/k . Given an intermediate field $F/E/k$, we can consider the subgroup $\text{Aut}(F/E) \leq \text{Aut}(F/k)$. Conversely, given a subgroup $H \leq \text{Aut}(F/k)$, we can consider the fixed field $\text{Fix}(H) \subseteq F$. This is inclusion-reversing: if $E \subseteq E'$, then $\text{Aut}(F/E) \geq \text{Aut}(F/E')$ and if $H \leq H'$, then $\text{Fix}(H) \supseteq \text{Fix}(H')$. Proposition 5.2.4 tells us that this correspondence is surjective onto subgroups of $\text{Aut}(F/k)$. In the case that F/k is Galois, this correspondence is extremely strong.

Theorem 5.2.9 (Galois Correspondence). *Let F/k be a finite Galois extension.*

(1) *There is an inclusion-reversing bijection*

$$\begin{array}{ccc} \{k \subseteq E \subseteq F\} & \longleftrightarrow & \{H \leq \text{Gal}(F/k)\} \\ E & \longmapsto & \text{Gal}(F/E) \\ \text{Fix}(H) & \longleftarrow & H \end{array}$$

(2) *This bijection is a bijection of lattices. That is, for $k \subseteq E, K \subseteq F$ and $H, N \leq \text{Gal}(F/k)$,*

$$\text{Gal}(F/EK) = \text{Gal}(F/E) \cap \text{Gal}(F/K) \quad \text{Gal}(F/E \cap K) = \langle \text{Gal}(F/E), \text{Gal}(F/K) \rangle$$

$$\text{Fix}(H \cap N) = \text{Fix}(H) \text{Fix}(N)$$

$$\text{Fix}(\langle H, N \rangle) = \text{Fix}(H) \cap \text{Fix}(N)$$

(3) *If $k \subseteq E \subseteq F$ is an intermediate field, then $[F : E] = |\text{Gal}(F/E)|$ and $[E : k] = [\text{Gal}(F/k) : \text{Gal}(F/E)]$. If $H \leq \text{Gal}(F/k)$ is a subgroup, then $[F : \text{Fix}(H)] = |H|$ and $[\text{Fix}(H) : k] = [\text{Gal}(F/k) : H]$.*

(4) *If $k \subseteq E, E' \subseteq F$ are subfields, then $E/k \cong E'/k$ if and only if $E' = \sigma(E)$ for some $\sigma \in \text{Gal}(F/k)$ if and only if $\text{Gal}(F/E)$ and $\text{Gal}(F/E')$ are conjugate. Further,*

$$\sigma \text{Gal}(F/E) \sigma^{-1} = \text{Gal}(F/\sigma(E)).$$

(5) *If $k \subseteq E \subseteq F$ is an intermediate field, then E/k is Galois if and only if $\text{Gal}(F/E) \trianglelefteq \text{Gal}(F/k)$. In this case,*

$$\text{Gal}(E/k) \cong \frac{\text{Gal}(F/k)}{\text{Gal}(F/E)}.$$

Proof.

(1) Proposition 5.2.4 gives us that $H = \text{Gal}(F/\text{Fix}(H))$. Conversely, if $k \subseteq E \subseteq F$ is an intermediate field, then F/E is Galois, so that $E = \text{Fix}(\text{Gal}(F/E))$. Thus, this is indeed a bijection.

(2) We prove the top two statements, as the bottom two follow immediately by applying the bijection. Since $\text{Gal}(F/EK)$ fixes both E and K , we clearly have

$$\text{Gal}(F/EK) \leq \text{Gal}(F/E) \cap \text{Gal}(F/K).$$

Taking fixed fields,

$$EK = \text{Fix}(\text{Gal}(F/EK)) \supseteq \text{Fix}(\text{Gal}(F/E) \cap \text{Gal}(F/K)) \supseteq E, K.$$

Since EK is the smallest field containing both E , and K , we must have equality. Similarly, we clearly have

$$\text{Gal}(F/E \cap K) \geq \langle \text{Gal}(F/E), \text{Gal}(F/K) \rangle.$$

Taking fixed fields,

$$E \cap K = \text{Fix}(\text{Gal}(F/E \cap K)) \subseteq \text{Fix}(\langle \text{Gal}(F/E), \text{Gal}(F/K) \rangle) \subseteq E, K,$$

so we have equality.

- (3) This follows immediately from noting that the extension F/E is Galois and using the fact that $[F : k] = |\text{Gal}(F/k)|$.
- (4) We'll view E/k as an abstract field extension and distinguish the inclusion embedding $\eta : E \subseteq F$. Consider the action of $\text{Gal}(F/k)$ on embeddings $E \hookrightarrow F$ given by composition (i.e. $\sigma \in \text{Gal}(F/k)$ acts on $\iota : E \hookrightarrow F$ to give $\sigma \circ \iota : E \hookrightarrow F$). Note that the stabilizer of ι is precisely $\text{Gal}(F/\iota(E))$.

Now, since F/k is Galois, F is the splitting field of some polynomial over k , and so the splitting field of that same polynomial over $\iota(E)$ for any $\iota : E \hookrightarrow F$. Thus, for any $\iota, \iota' : E \hookrightarrow F$, uniqueness of splitting fields gives us an automorphism $\sigma : F \rightarrow F$ extending $\iota' \circ \iota^{-1} : \iota(E) \rightarrow \iota'(E)$. But then $\sigma \circ \iota = \iota'$, and so we see that our action of $\text{Gal}(F/k)$ on embeddings $E \hookrightarrow F$ is transitive. Recall that stabilizers of elements in the same orbit are conjugate, and so $\text{Gal}(F/\sigma(E)) = \sigma \text{Gal}(F/E) \sigma^{-1}$.

- (5) Suppose first that E/k is Galois, and so normal. Then Theorem 5.1.21 tells us that every automorphism $\sigma \in \text{Gal}(F/k)$ restricts to an automorphism of E . Thus, restriction gives us a homomorphism $\text{Gal}(F/k) \rightarrow \text{Gal}(E/k)$. The kernel of this homomorphism is precisely $\text{Gal}(F/E)$, so $\text{Gal}(F/E) \trianglelefteq \text{Gal}(F/k)$ and $\text{Gal}(E/k) \cong \text{Gal}(F/k)/\text{Gal}(F/E)$.

Conversely, suppose that $\text{Gal}(F/E) \trianglelefteq \text{Gal}(F/k)$. Since the action of $\text{Gal}(F/k)$ on embeddings $E \hookrightarrow F$ is transitive and the stabilizer $\text{Gal}(F/E)$ of the inclusion $E \subseteq F$ is normal, we have that $\text{Gal}(F/E) = \text{Gal}(F/\iota(E))$ for all embeddings $\iota : E \hookrightarrow F$. Taking fixed fields, $E = \iota(E)$ for all $\iota : E \hookrightarrow F$. Thus, Theorem 5.1.21 shows that E/k is normal², and hence Galois (as it is automatically separable).

■

Remark 5.2.10. The content of (5) is that E/k is normal, since it is always separable. Thus, normal extensions correspond to normal subgroups, which is perhaps easier to remember.

We end with how we may deal with Galois extensions when passing to larger fields.

Proposition 5.2.11. *Let F/k be a finite Galois extension and K/k be any extension (where we view $F, K \subseteq \bar{k}$). Then KF/K is Galois, and $\text{Gal}(KF/K) \cong \text{Gal}(F/F \cap K)$. In particular, $[KF : K] = [F : F \cap K]$ and $[KF : F] = [K : K \cap F]$.*

Proof. Since F is the splitting field of some separable polynomial over k , FK is the splitting field of that same polynomial over K . Thus, FK/K is Galois.

Now, we have a map $\text{Gal}(KF/K) \rightarrow \text{Gal}(F/k)$ given by restriction to F . This map is injective: if an element of $\text{Gal}(KF/K)$ fixes all of F , then it must fix KF by the Galois correspondence (as it already fixes K). Now, since the fixed field of $\text{Gal}(KF/K)$ is precisely K , the elements of F that are fixed are precisely those in $F \cap K$, so that this is an isomorphism $\text{Gal}(KF/K) \cong \text{Gal}(F/F \cap K)$. ■

5.3 Applications of Galois Theory

5.3.1 The Fundamental Theorem of Algebra

Theorem 5.3.1. *$\mathbb{C}$ is algebraically closed.*

Proof. Let $f \in \mathbb{C}[x]$ with no roots in $\mathbb{C}$. Then $f(x)\overline{f(x)} \in \mathbb{R}[x]$ has no roots in $\mathbb{C}$, so we may assume that $f(x) \in \mathbb{R}[x]$. Let F be the splitting field of $(x^2 + 1)f(x)$ over $\mathbb{R}$, so that $F/\mathbb{R}$ is Galois, and let $G = \text{Gal}(F/\mathbb{R})$.

Let H be a Sylow 2-subgroup of G . Then $[\text{Fix}(H) : \mathbb{R}] = [G : H]$ is odd. But every polynomial of odd degree has a root in $\mathbb{R}$, so this means $\text{Fix}(H) = \mathbb{R}$ and $G = H$ is a 2-group. Now, $\text{Gal}(F/\mathbb{C})$ is also a 2-group, and so we get a chain of fields $\mathbb{C} = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_n = F$ with $[F_{i+1} : F_i] = 2$ (since there is a corresponding chain of subgroups in any 2-group). But the quadratic formula shows that $\mathbb{C}$ has no irreducible polynomials of degree 2. Thus, $G = 1$ and $F = \mathbb{C}$, as desired. ■

²Technically, we have not shown condition (3) of Theorem 5.1.21. However, F is “large enough” that tracing through the proof that (3) $\Rightarrow$ (1) with $F = E$ and $L = F$ gives us what we need.

5.3.2 The Fundamental Theorem of Symmetric Functions

Consider the polynomial

$$(x - r_1)(x - r_2) \cdots (x - r_n).$$

The coefficient of x^{n-k} is given by

$$s_k(r_1, \dots, r_n) = \sum_{\substack{I \subseteq [n] \\ |I|=k}} \prod_{i \in I} r_i.$$

For instance, $s_1 = r_1 + \cdots + r_n$ and $s_n = r_1 \cdots r_n$. The s_k are called the *elementary symmetric functions* in $r_1, \dots, r_n$. They are called symmetric because $s_k(r_{\sigma(1)}, \dots, r_{\sigma(n)}) = s_k(r_1, \dots, r_n)$ for all $\sigma \in S_n$. More generally, we say an element of $k(r_1, \dots, r_n)$ is *symmetric* if it is invariant under permuting the r_i s. Remarkably, all symmetric functions can be recovered from the elementary symmetric functions.

Theorem 5.3.2 (Fundamental Theorem of Symmetric Functions). *Let k be a field and $\varphi \in k(r_1, \dots, r_n)$. Then φ is symmetric if and only if it is a rational function (with coefficients in k) in the elementary symmetric functions $s_1, \dots, s_n$.*

Proof. Let $F = k(r_1, \dots, r_n)$ and $K = k(s_1, \dots, s_n)$. Since F is the splitting field of the separable polynomial $(x - r_1) \cdots (x - r_n)$ over K , F/K is Galois and $[F : K] \leq n!$.

Now, S_n acts faithfully on F by permuting the r_i , and so we can view it as a subgroup of $\text{Gal}(F/K)$. But then order considerations give $\text{Gal}(F/K) \cong S_n$. In particular, the Galois correspondence gives $K = \text{Fix}(S_n)$, as desired. ■

One can in fact show that a symmetric polynomial is a polynomial in the s_i . The key is to place the lexicographic ordering on monomials. The leading monomial of a symmetric function must then be of the form $r_1^{e_1} \cdots r_n^{e_n}$ with $e_1 \geq \cdots \geq e_n$ (where symmetry allows us to place the order constraint). Since $s_1^{e_1 - e_2} s_2^{e_2 - e_3} \cdots s_n^{e_n}$ has the same leading monomial, subtracting and using induction yields the result.

5.3.3 The Primitive Element Theorem

We can use Galois theory to provide a quick and easy proof of the so-called *Primitive Element Theorem*.

Theorem 5.3.3. *Any finite extension is contained in a finite normal extension. Any finite separable extension is contained in a finite Galois extension.*

Proof. Let F/k be finite, with $F = k(\alpha_1, \dots, \alpha_n)$. Let L be the splitting field of $m_{\alpha_1} \cdots m_{\alpha_n}$. Then L/k contains F , and is a finite normal extension. In the case that F/k is separable, $m_{\alpha_1} \cdots m_{\alpha_n}$ is separable, so L/k is Galois. ■

Theorem 5.3.4 (Primitive Element). *Any finite separable extension is simple.*

Proof. Let F/k be a finite separable extension. By Theorem 5.1.15, it suffices to show that there are finitely many intermediate fields between F and k . Let L/k be a finite Galois extension containing F . Since intermediate fields of L/k are in bijection with $\text{Gal}(L/k)$, there are finitely many such fields, and so also finitely many intermediate fields of F/k . ■

Proofs of the Primitive Element Theorem do exist without using the machinery of Galois theory, though all of them use an argument along the lines of the proof of Theorem 5.1.15.

We prove one more result, taking advantage of our ability to pass to Galois extensions.

Proposition 5.3.5. *A field extension is separable if and only if it is generated by separable elements.*

Proof. Clearly a separable extension is generated by separable elements. Conversely, suppose F is generated by separable elements over k . Since each element of F lies in some finite extension, it suffices to show $k(\alpha_1, \dots, \alpha_n)$ is separable whenever $\alpha_1, \dots, \alpha_n$ are separable. This is contained in the splitting field L of $m_{\alpha_1} \cdots m_{\alpha_n}$, which is separable, and so $k(\alpha_1, \dots, \alpha_n)$ is separable as well. ■

5.3.4 Finite Fields

Note that a finite field must have positive characteristic. The key to our study of finite fields is the following result.

Proposition 5.3.6. *Let k be a field of characteristic $p > 0$. The map $t \mapsto t^p$ is a field endomorphism. If k is finite, the map is a field automorphism.*

Proof. The content here is that our map is an additive homomorphism. The binomial theorem gives

$$(a + b)^p = \sum_{i=0}^p \binom{p}{i} a^i b^{p-i} = a^p + b^p + \sum_{i=1}^{p-1} \binom{p}{i} a^i b^{p-i}.$$

Since p is prime, $p \mid \binom{p}{i}$ for $1 \leq i \leq p-1$. Since $\text{char } k = p$, this means $(a + b)^p = a^p + b^p$. We note that this map is always injective, as $a = 0$ if and only if $a^p = 0$. Thus, if k is finite, this map is bijective. ■

Definition 5.3.7. Let k be a field of characteristic p . The map $\phi : t \mapsto t^p$ is called the *Frobenius endomorphism*. When ϕ is surjective, it is called the *Frobenius automorphism*.

The fact that the Frobenius automorphism is indeed an automorphism is quite useful.

Proposition 5.3.8. *Let k be a field of characteristic p . Then k is perfect if and only if the map $t \mapsto t^p$ is surjective. In particular, finite fields are perfect.*

Proof. Suppose first that the map $t \mapsto t^p$ is surjective, and let $f \in k[x]$ be irreducible. If $f' = 0$, then we can write $f(x) = g(x^p)$ for some $g \in k[x]$ (since the coefficient of x^n for $p \nmid n$ must be 0). But every coefficient of g is a p th power, so we can write $g(x^p) = h(x)^p$ for some $h \in k[x]$. This contradicts the fact that f is irreducible, and so all irreducible polynomials are separable.

Conversely, suppose that $\alpha \in k$ is not a p th power, and let $f(x) = x^p - \alpha$. Since f has no roots in k , no irreducible factor of f is monic. Thus, any irreducible factor of f has repeated roots, since f can have at most one root (if $\beta^p = \alpha$ then $f(x) = (x - \beta)^p$), and so k is not perfect. ■

Now, we know that $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ is a field of order p . Since every finite field of characteristic p is a finite-dimensional vector space over $\mathbb{F}_p$, its order must be a power of p . Now, if F is a finite field of order p^n , then $a^{p^n-1} = 1$ for all $a \in F^\times$ by Lagrange's Theorem, so that $a^{p^n} = a$ for all $a \in F$. In particular, F is a splitting field of $x^{p^n} - x$, and so we see that all fields of order p^n are isomorphic. Further, since $x^{p^n} - x$ has p^n distinct roots in F , we see that it is separable, and so the extension $F/\mathbb{F}_p$ is Galois.

Conversely, suppose we have any $n \geq 1$, and let F be the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$. Since $(a + b)^{p^n} = a^{p^n} + b^{p^n}$, we see that the roots of $x^{p^n} - x$ form a subfield of F . But since F is generated by these roots, F must consist of exactly the roots of $x^{p^n} - x$, and so in particular $|F| = p^n$. We have thus deduced the following.

Theorem 5.3.9. *Let p be a prime and $n \geq 1$. Then there exists a unique up to isomorphism field $\mathbb{F}_{p^n}$ of order p^n . Further, the extension $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois.*

As it turns out, the Galois group $\text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ has quite nice structure.

Theorem 5.3.10. *The Galois group $\text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ is cyclic and generated by the Frobenius automorphism ϕ .*

Proof. Since $|\text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)| = [\mathbb{F}_{p^n} : \mathbb{F}_p] = n$, it suffices to show that ϕ has order n . For any $i < n$, we have that

$$\text{Fix}(\phi^i) = \{a \in \mathbb{F}_{p^n} : a^{p^i} - a = 0\},$$

which has at most $p^i < p^n$ elements. Thus, $\phi^i \neq \text{id}$, and so ϕ has order n . ■

Recall that the cyclic group $\mathbb{Z}/n\mathbb{Z}$ has a subgroup of order m if and only if $m \mid n$, and that such a subgroup is unique. Applying the Galois correspondence gives us the following.

Theorem 5.3.11. *The field $\mathbb{F}_{p^n}$ has a subfield of order p^m if and only if $m \mid n$. Further, there is at most one subfield of order p^m , and it is given by $\{a \in \mathbb{F}_{p^n} : a^{p^m} = a\}$ if it exists.*

In particular, for any prime power q and $n \geq 1$, we can study the field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$. The same arguments show that $\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ is cyclic of order n , and that there is an intermediate field $\mathbb{F}_{q^n}/E/\mathbb{F}_q$ of order q^m if and only if $m \mid n$.

Our knowledge of how finite fields embed into one another will allow us to construct the algebraic closure of the finite field $\mathbb{F}_q$. We construct a field

$$\overline{\mathbb{F}_q} = \bigcup_{n \geq 1} \mathbb{F}_{q^n},$$

where we identify $\mathbb{F}_{q^m} \subseteq \mathbb{F}_{q^n}$ whenever $m \mid n$ ³. Since we can always embed $\mathbb{F}_{q^n}, \mathbb{F}_{q^m} \subseteq \mathbb{F}_{q^{nm}}$, we can add and multiply any two elements of $\overline{\mathbb{F}_q}$, and it is not hard to see that these operations make $\overline{\mathbb{F}_q}$ into a field. Now, the extension $\overline{\mathbb{F}_q}/\mathbb{F}_q$ is algebraic: any element of $\overline{\mathbb{F}_q}$ lies in some $\mathbb{F}_{q^n}$, and is thus algebraic over $\mathbb{F}_q$. Further, $\overline{\mathbb{F}_q}$ is algebraically closed: if f is a polynomial of degree n , and m is large enough that all coefficients of f lie in $\mathbb{F}_{q^m}$, then the splitting field of f has (finite) dimension $\leq n!$ over $\mathbb{F}_{q^m}$, and so is some finite field $\mathbb{F}_{q^k}$. This means that f splits over $\overline{\mathbb{F}_q}$, as desired.

We end with a cute application to number theory, using an entirely oversized hammer.

Proposition 5.3.12. *Let q be a prime power. Then $m \mid n$ if and only if $q^m - 1 \mid q^n - 1$.*

Proof. If $m \mid n$, say $n = md$, then

$$q^n - 1 = (q^m)^d - 1 = (q^m - 1)(q^{m(d-1)} + \cdots + q^m + 1).$$

Conversely, suppose that $q^m - 1 \mid q^n - 1$. Since $\mathbb{F}_{q^n}^\times$ is cyclic, it contains a subgroup of order $q^m - 1$, and every element of this subgroup satisfies $a^{q^m - 1} = 1$. Thus, this subgroup, together with 0, forms the set of roots to $x^{q^m} - x$. In particular, this is a subfield of $\mathbb{F}_{q^n}$ of order q^m , and so $m \mid n$. ■

5.3.5 Cyclotomy

Throughout this section, we will use ζ_n to refer to the primitive n th root of unity $e^{2\pi i/n} \in \mathbb{C}$.

Definition 5.3.13. The n th cyclotomic polynomial $\Phi_n(t)$ is

$$\Phi_n(t) = \prod_{\substack{\gcd(k,n)=1 \\ 1 \leq k \leq n}} (t - \zeta_n^k).$$

Clearly $\deg \Phi_n = \varphi(n)$, where φ denotes Euler's totient function. Since every n th root of unity is a primitive d th root of unity for some $d \mid n$, we have that

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x).$$

For instance, $\Phi_1(x) = x - 1$, and so when p is prime,

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + \cdots + x + 1.$$

While it is not at all clear that Φ_n should have rational coefficients, this formula shows us that it does. Indeed, assume inductively that $\Phi_d(x)$ has integer coefficients for $d < n$. Since each $\Phi_d(x)$ is monic, we can use polynomial division in $\mathbb{Z}[x]$ to compute $q(x), r(x) \in \mathbb{Z}[x]$ such that $x^n - 1 = q(x) \prod_{\substack{d \mid n \\ d < n}} \Phi_d(x) + r(x)$ with $\deg r < n - \varphi(n)$. But since $\prod_{\substack{d \mid n \\ d < n}} \Phi_d(x)$ divides $x^n - 1$ in $\mathbb{C}[x]$, degree considerations show $r(x) = 0$ and $q(x) = \Phi_n(x)$.

We can compute Φ_n for some small values. For instance, $\Phi_4(t) = (t^4 - 1)/\Phi_1(t)\Phi_2(t) = t^2 + 1$, and $\Phi_6(t) = (t^6 - 1)/\Phi_1(t)\Phi_2(t)\Phi_3(t) = t^2 - t + 1$. We may hope that Φ_n has nonzero coefficients ± 1 , but this turns out to be false, though the smallest n for which it fails is $n = 105$. The reason n must get so large is due to Migotti's theorem, which we will show later this section.

Proposition 5.3.14. *The cyclotomic polynomial $\Phi_n(x)$ is irreducible in $\mathbb{Q}[x]$ for all $n \geq 1$.*

³We are actually constructing the colimit $\varinjlim \mathbb{F}_{q^n}$

Proof. Since Φ_n is monic with integer coefficients, it suffices to show Φ_n is irreducible in $\mathbb{Z}[x]$ by Gauss' Lemma. Suppose we can write $\Phi_n = fg$, where $f(x) \in \mathbb{Z}[x]$ is monic irreducible, and $g(x) \in \mathbb{Z}[x]$ with $\deg g > 0$.

The primitive n th roots of unity are split among f and g . In particular, for a root η of f and a root ξ of g , we can write $\xi = \eta^k$ for some k relatively prime to n . Pick η and ξ such that k is minimized. If $p \mid k$ is prime, then $\xi = (\eta^p)^{k/p}$. Since $k/p < k$, the minimality of k means that η^p is not a root of f , and thus a root of g . But then minimality again means we must have $k = p$ is prime.

Now, η is a root of $g(x^p)$. Since f is irreducible with η as a root, $f = m_\eta$, so that $f(x) \mid g(x^p)$ in $\mathbb{Q}[x]$ (and thus $\mathbb{Z}[x]$ since f is monic). Write $g(x^p) = f(x)h(x)$.

We now work mod p to get $\overline{g(x^p)} = \overline{g(x)^p} = \overline{f(x)h(x)^p}$. But this means that $\overline{f(x)}$ and $\overline{g(x)}$ share an irreducible common factor in $\mathbb{F}_p[x]$. But since $\overline{f}\overline{g} = \overline{\Phi_n}$ divides $\overline{x^n - 1}$, this means $\overline{x^n - 1}$ does not have distinct roots. But the only root of $(\overline{x^n - 1})' = \overline{nx^{n-1}}$ is 0, which is not a root of $\overline{x^n - 1}$, and so $\overline{x^n - 1}$ does have distinct roots, a contradiction. Thus, Φ_n is irreducible, as desired. ■

In particular, Φ_n is the minimal polynomial of ζ_n , and $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$.

Theorem 5.3.15. $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

Proof. For every k with $\gcd(k, n) = 1$, we have an automorphism of $\mathbb{Q}(\zeta_n)$ given by $\zeta_n \mapsto \zeta_n^k$ (by Theorem 5.1.13). This is clearly an injective homomorphism $(\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$. Since both sides have size $\varphi(n)$, this is an isomorphism. ■

Theorem 5.3.16. Let $\ell = \text{lcm}(n, m)$ and $d = \gcd(n, m)$. Then $\mathbb{Q}(\zeta_n, \zeta_m) = \mathbb{Q}(\zeta_\ell)$ and $\mathbb{Q}(\zeta_n) \cap \mathbb{Q}(\zeta_m) = \mathbb{Q}(\zeta_d)$.

Proof. Clearly $\mathbb{Q}(\zeta_n, \zeta_m) \subseteq \mathbb{Q}(\zeta_\ell)$. If $an + bm = d$, then

$$\zeta_n^b \zeta_m^a = e^{2\pi i \cdot \frac{b}{n}} e^{2\pi i \cdot \frac{a}{m}} = e^{2\pi i \cdot \frac{an+bm}{nm}} = e^{2\pi i \cdot \frac{d}{nm}} = e^{2\pi i/\ell} = \zeta_\ell,$$

so we have equality.

Similarly, we clearly have $\mathbb{Q}(\zeta_n) \cap \mathbb{Q}(\zeta_m) \supseteq \mathbb{Q}(\zeta_d)$. Now, Proposition 5.2.11 gives that $[\mathbb{Q}(\zeta_\ell) : \mathbb{Q}(\zeta_m)] = [\mathbb{Q}(\zeta_n) : \mathbb{Q}(\zeta_n) \cap \mathbb{Q}(\zeta_m)]$. Since $[\mathbb{Q}(\zeta_\ell) : \mathbb{Q}(\zeta_m)] = \varphi(\ell)/\varphi(m)$ and $[\mathbb{Q}(\zeta_n) : \mathbb{Q}(\zeta_d)] = \varphi(n)/\varphi(d) = \varphi(\ell)/\varphi(m)$, we have equality. ■

We now work towards showing why coefficients of Φ_n lie in $\{-1, 0, 1\}$ for small n .

Lemma 5.3.17.

(1) If p is a prime and $n \geq 1$, then

$$\Phi_{pn}(x) = \begin{cases} \Phi_n(x^p) & p \mid n \\ \Phi_n(x^p)/\Phi_n(x) & p \nmid n \end{cases}.$$

(2) If $2 \nmid n$, then $\Phi_{2n}(x) = \Phi_n(-x)$.

Proof.

(1) We have that

$$\Phi_n(x^p) = \prod_{\substack{\gcd(k, n)=1 \\ 1 \leq k \leq n}} (x^p - \zeta_n^k) = \prod_{\substack{\gcd(k, n)=1 \\ 1 \leq k \leq n}} \prod_{i=0}^{p-1} (x - \zeta_{pn}^{k+ni}).$$

If $p \mid n$, then $\gcd(k+ni, pn) = 1$ for all k and i in the product. Further, for every $1 \leq \ell \leq pn$ with $\gcd(\ell, pn) = 1$, we can uniquely write $\ell = k + ni$ for $0 \leq i \leq p-1$, and we get $\gcd(k, n) = 1$, so the right hand side is equal to $\Phi_{pn}(x)$.

Now, if $p \nmid n$, then for each k , there is exactly one i_k such that $p \mid k + ni_k$, and $\gcd(k + ni, pn) = 1$ for $i \neq i_k$. Thus, $k + ni_k \equiv pk \pmod{pn}$, and so we have a factor of $x - \zeta_n^k$ for each k . Every other term in the product contributes $x - \zeta_{pn}^\ell$ for some $\gcd(\ell, pn) = 1$. Again, we see that we can obtain every primitive pn th root of unity in the product, and so the right hand side is equal to $\Phi_n(x)\Phi_{pn}(x)$.

(2) We have that

$$\Phi_{2n}(x) = \Phi_n(x^2)/\Phi_n(x) = \prod_{\substack{\gcd(k, n)=1 \\ 1 \leq k \leq n}} (x + \zeta_n^k) = (-1)^{\varphi(n)} \prod_{\substack{\gcd(k, n)=1 \\ 1 \leq k \leq n}} (-x - \zeta_n^k) = \Phi_n(-x).$$

■

Theorem 5.3.18 (Migotti). *Let $n = 2^a p^b q^c$ for primes p and q . Then the coefficients of $\Phi_n(x)$ lie in $\{-1, 0, 1\}$.*

Proof. If any of a, b, c are > 1 , we may repeatedly apply Lemma 5.3.17 (1) (for instance, if $b > 1$, then we have $\Phi_n(x) = \Phi_{2^a p q^c}(x^{p^{b-1}})$). Thus, we reduce to the case that $a, b, c \leq 1$. By applying Lemma 5.3.17 (2), we reduce to the case that $a = 0$ and $b, c \leq 1$. If either of b or c is 0, we are done (since we know that $\Phi_p(x) = x^{p-1} + \cdots + x + 1$), so suppose that $b = c = 1$.

Since p and q are distinct primes, the multiples $0, p, \dots, (q-1)p$ run through all remainders modulo q , and so we can pick exponents e_i such that

$$\Phi_q(x^p) = x^{p(q-1)} + \cdots + x^p + 1 = \sum_{i=0}^{q-1} (t^q)^{e_i} \cdot x^i.$$

Thus, we have that

$$\Phi_q(x^p)(x-1) = \sum_{i=0}^{q-1} ((x^q)^{e_{i-1}} - (x^q)^{e_i}) \cdot t^i,$$

where $e_{-1} = e_{q-1} + 1$. But for $a \geq b$, we have $(x^q)^a - (x^q)^b = (x^q)^b(x^q - 1)((x^q)^{a-b-1} + \cdots + 1)$ and $(x^q)^b - (x^q)^a = -(x^q)^b(x^q - 1)((x^q)^{a-b-1} + \cdots + 1)$, so that $\Phi_q(x^p)(x-1)/(x^q - 1)$ is a polynomial. Further, the coefficients of $\Phi_q(x^p)(x-1)/(x^q - 1)$ lie in $\{-1, 0, 1\}$ (since the coefficients of each term in the sum lie in $\{-1, 0, 1\}$, and no exponent appears in two different terms in the sum, as each term corresponds to a unique remainder modulo q). But $\Phi_q(x^p)(x-1)/(x^q - 1) = \Phi_q(x^p)/\Phi_q(x)$, which is equal to $\Phi_{pq}(x)$ by Lemma 5.3.17, and so we are done. ■

Note that 105 is the smallest integer divisible by 3 distinct odd primes, and indeed the coefficients of Φ_{105} do not lie in $\{-1, 0, 1\}$.

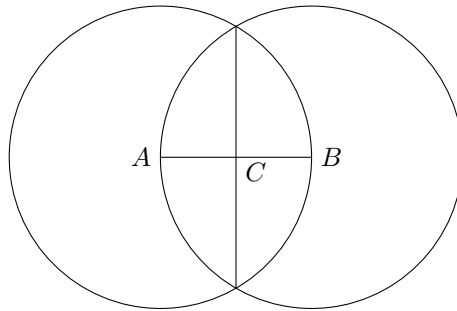
5.3.6 Straightedge and Compass Constructions

We discuss what we can construct using a straightedge and compass. We allow the following actions:

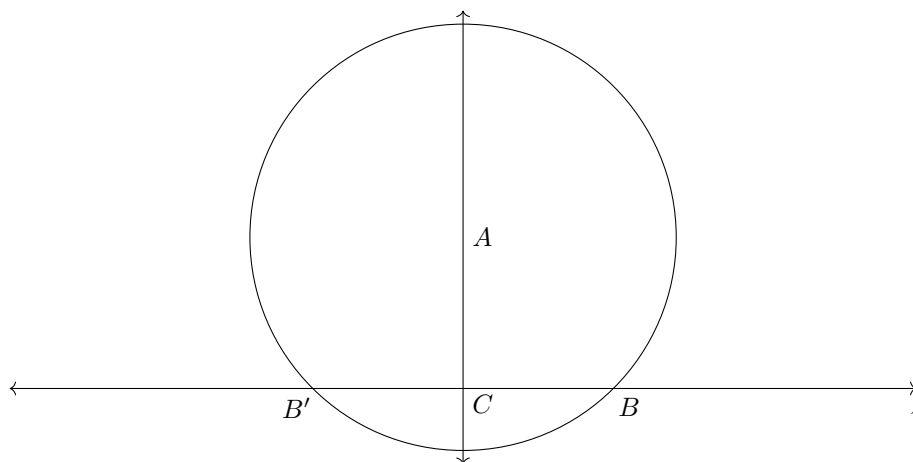
- We may mark the intersection points of any two lines and circles that have been drawn.
- We may draw a line through any two marked points
- We may draw a circle centered at a given marked point and passing through another given marked point

We'll say that a point is *constructible* if it can eventually become one of our marked points.

If we have constructed two points A and B , we can construct a point C bisecting the segment, as follows:



Thus, if we have constructed a point A and a line ℓ not containing A (but necessarily containing another constructed point B), we can construct a line through A perpendicular to ℓ :



where C bisects the segment from B to B' (if $B = B'$, then the line from A to B is perpendicular to ℓ). Iterating this construction, we may also construct a line through A parallel to ℓ ⁴.

Thus, if we have any constructed segment AB and any point O , we may translate AB to have an endpoint at O by examining the intersection of the line through O parallel to AB and the line through B parallel to OA ⁵.

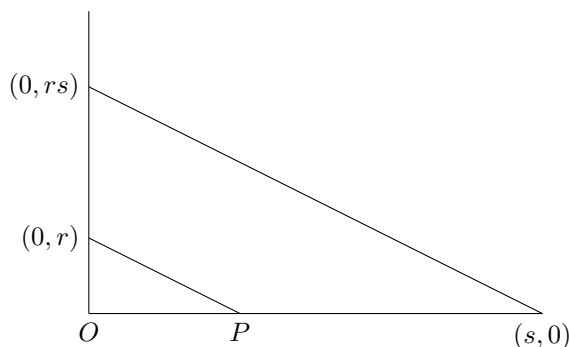
From now on, we'll assume we start with points $O = (0, 0)$ and $P = (1, 0)$, and construct everything else from there.

Definition 5.3.19. A real number r is *constructible* if the point $(r, 0)$ is constructible. A complex number $r + is$ is *constructible* if the point (r, s) is constructible. We write $\mathcal{C}_{\mathbb{R}}$ and $\mathcal{C}_{\mathbb{C}}$ for the set of constructible real and complex numbers respectively.

Our previous discussions have shown that, $r + is$ is constructible if and only if r and s are constructible. The reader likely expects that the constructible numbers form a field (why else would this discussion be in the field theory chapter?), and they would be correct.

Proposition 5.3.20. $\mathcal{C}_{\mathbb{R}}$ and $\mathcal{C}_{\mathbb{C}}$ form a field.

Proof. It clearly suffices to show that $\mathcal{C}_{\mathbb{R}}$ is a field. Let $r, s \in \mathcal{C}_{\mathbb{R}}$. Translating the segment $(0, 0) - (0, s)$ to begin at $(0, r)$ and drawing appropriate circles shows that $r \pm s \in \mathcal{C}_{\mathbb{R}}$. The following diagram shows that rs is constructible (where we construct the line parallel to $P - (0, r)$ passing through $(s, 0)$):

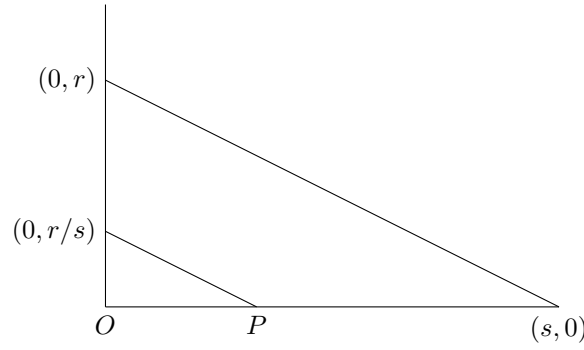


Similarly, the following diagram shows that r/s is constructible⁶:

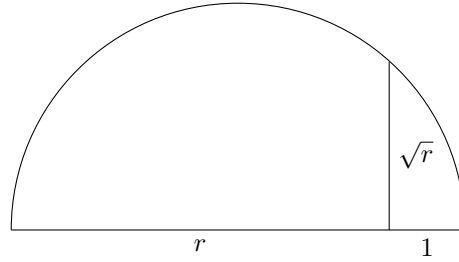
⁴this requires constructing a point not on ℓ or the line from A perpendicular to ℓ , but this is not too hard after drawing enough circles

⁵We need O , A , and B to not be collinear, but this problem can be solved by first translating AB to some point not on the common line, and then translating that translate to O

⁶We've been assuming $s > 1$, but the reader can verify a similar diagram for $s < 1$



We finally remark that $\mathcal{C}_{\mathbb{R}}$ is closed under square roots:



This leads to our key result about constructibility.

Theorem 5.3.21. *A real number γ is constructible if and only if there are real fields $\mathbb{Q} = F_0 \subseteq \cdots \subseteq F_m$ with $\gamma \in F_m$ and $[F_i : F_{i-1}] = 2$ for $1 \leq i \leq m$.*

Proof. Suppose first that we have such a chain of fields. We induct on m , with $m = 0$ being trivial. By induction, every element of F_{m-1} is constructible, and γ is the root of some quadratic polynomial with coefficients in F_{m-1} . But we can construct square roots, and so the quadratic formula shows that γ (and indeed, every element of F_m) is constructible.

Conversely, suppose γ is constructible. Then each step of the construction gives a field F_i generated over $\mathbb{Q}$ by all numbers that have been constructed so far. It suffices to show that $[F_i : F_{i-1}] \leq 2$. But computing the intersection of two lines does not require leaving the field, and computing the intersection of a line with a circle or two circles only requires passing to a quadratic extension, which is precisely what we need. ■

Corollary 5.3.22. *If $\gamma \in \mathcal{C}_{\mathbb{C}}$, then $[\mathbb{Q}(\gamma) : \mathbb{Q}]$ is a power of 2.*

Proof. We have a chain of fields

$$\mathbb{Q} = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_m \subseteq F_m(i) \ni \gamma.$$

We can apply this theorem to determine what numbers are and are not constructible. First, note that constructing an angle θ is equivalent to constructing the complex number $e^{i\theta}$, and that constructing a regular n -gon is equivalent to constructing the angle $2\pi/n$. There are three famous impossibilities that we get (almost)⁷ for free:

- **Trisecting the Angle:** Given any angle, is it possible to use a straightedge and compass to construct it? We will show that one cannot construct a $\pi/9$ angle (which in particular means that one cannot trisect a $\pi/3$ angle). Indeed, ζ_{18} is a root of the irreducible polynomial $\Phi_{18}(t)$, which has degree $\varphi(18) = 6$.
- **Doubling the Cube:** Given a cube, is it possible to construct a cube of double the volume? Essentially, we are asking if $\sqrt[3]{2}$ is constructible, which is false since it is a root of the irreducible polynomial $t^3 - 2$.
- **Squaring the Circle:** Given a circle, is it possible to construct a square of the same area? If so, then $\sqrt{\pi}$ would be constructible, which is impossible, as it is not even algebraic over $\mathbb{Q}$.

⁷We use that π is transcendental

We end with a result about when exactly we can construct a regular n -gon.

Proposition 5.3.23. *A regular n -gon is constructible if and only if $\varphi(n)$ is a power of 2.*

Proof. If ζ_n is constructible, then $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \deg \Phi_n = \varphi(n)$ is a power of 2. Conversely, suppose that $\varphi(n) = 2^r$, so that $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is a Galois extension whose degree is a power of 2, and let $G = \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$. Taking a series of subgroups $1 = G_r \leq \cdots \leq G_0 = G$ such that $|G_i| = 2^{r-i}$, we get a chain of fields $\mathbb{Q} = F_0 \subseteq \cdots \subseteq F_m \ni \zeta_n$ with $[F_i : F_{i-1}] = 2$. Taking intersections with $\mathbb{R}$ shows that $\cos(2\pi i/n)$ and $\sin(2\pi i/n)$ are constructible. ■

Since we know how φ behaves on prime factorizations, it is not hard to see that a regular n -gon is constructible if and only if $n = 2^a p_1 \cdots p_k$, where $p_1, \dots, p_k$ are distinct odd primes with $p_i = 2^{n_i} + 1$ ⁸.

5.3.7 Solvability of Polynomials

It is well known that the roots to the equation $ax^2 + bx + c$ are given by $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. While this may not exist in our ground field (for instance, if we are working over $\mathbb{R}$ and $b^2 - 4ac < 0$), we can pass to an extension that contains the appropriate square root. We will examine more generally when we can come up with such formulas.

Definition 5.3.24. An extension F/k is said to be *radical* if we can find

$$k \subseteq k(\alpha_1) \subseteq k(\alpha_1, \alpha_2) \subseteq \cdots \subseteq k(\alpha_1, \dots, \alpha_r) = F$$

and $n_1, \dots, n_r \in \mathbb{N}$ so that $\alpha_i^{n_i} \in k(\alpha_1, \dots, \alpha_{i-1})$. We call an extension $k(\alpha)/k$ where $\alpha^n \in k$ for some n a *pure radical extension*⁹.

We will work to classify precisely how the automorphism groups of each intermediate extension look. First, we need some technical results (though we remark that these results are of independent interest).

Lemma 5.3.25. *Let G be any abelian group and k any field. Then the set of homomorphisms $G \rightarrow k^\times$ is a linearly independent subset of k^G , the vector space of (set) functions $G \rightarrow k$.*

Proof. Suppose otherwise, and take a minimal linear dependence relation $\sum_{i=1}^n a_i \chi_i = 0$ for $a_i \in k^\times$ and $\chi_i : G \rightarrow k^\times$. Since the χ_i are nonzero, we must have that $n > 1$. Take $g \in G$ so that $\chi_1(g) \neq \chi_n(g)$. Now, for all $h \in G$,

$$0 = \sum_{i=1}^n a_i \chi_i(gh) = \sum_{i=1}^n (a_i \chi_i(g)) \chi_i(h).$$

Also, multiplying our original linear dependence relation by $\chi_n(g)$, $0 = \sum_{i=1}^n (a_i \chi_n(g)) \chi_i(h)$ for all $h \in H$. Subtracting, we see that

$$0 = \sum_{i=1}^n (a_i \chi_i(g) - a_i \chi_n(g)) \chi_i = \sum_{i=1}^{n-1} a_i (\chi_i(g) - \chi_n(g)) \chi_i.$$

By construction, the coefficient of χ_1 is nonzero. But we picked a minimal linear dependence, and so this is a contradiction. ■

Lemma 5.3.26. *Let $G \leq \text{Aut}(k)$ be finite. Suppose that, for each $\sigma \in G$, we had $x_\sigma \in k^\times$ such that $x_{\sigma\tau} = x_\sigma \cdot \sigma(x_\tau)$. Then there is $\alpha \in k^\times$ such that $x_\sigma = \alpha/\sigma(\alpha)$.*

Proof. Consider the sum $\sum_{\tau \in G} x_\tau \tau$. Since G is a linearly independent subset of functions $k \rightarrow k$, this sum is nonzero. Take $\beta \in k$ so that $\sum_{\tau \in G} x_\tau \tau(\beta) = \alpha \neq 0$. Then,

$$x_\sigma \cdot \sigma(\alpha) = x_\sigma \sum_{\tau \in G} \sigma(x_\tau) \cdot \sigma\tau(\beta) = \sum_{\tau \in G} x_{\sigma\tau} \cdot \sigma\tau(\beta) = \alpha,$$

as desired. ■

We're now ready to begin tackling radical extensions.

Theorem 5.3.27 (Kummer). *Suppose that k contains a primitive n th root of unity. If F/k is Galois with $\text{Gal}(F/k) \cong \mathbb{Z}/n\mathbb{Z}$, then there is $\alpha \in F$ with $\alpha^n \in k$ and $F = k(\alpha)$. Conversely, if $F = k(\alpha)$ for such an α , then F/k is Galois and $\text{Gal}(F/k)$ is cyclic of order dividing n .*

⁸Primes of this form are called *Fermat primes*. It's not hard to check that $2^n + 1$ is a prime only if n is a power of 2.

⁹"Pure radical" is not standard terminology

Proof. Let $\zeta \in k$ be a primitive n th root of unity. Suppose first that F/k is Galois and $\text{Gal}(F/k) = \langle \sigma \rangle$ is cyclic of order n . If we let $x_{\sigma^i} = \zeta^{-i}$, then

$$x_{\sigma^i} \cdot \sigma^i(x_{\sigma^j}) = \zeta^{-i} \cdot \sigma^i(\zeta^{-j}) = \zeta^{-i-j} = x_{\sigma^{i+j}},$$

where we use the fact that $\zeta \in k$ is fixed by σ . Thus, there is $\alpha \in F$ with $\zeta^{-i} = \alpha/\sigma^i(\alpha)$, which is to say that $\sigma(\alpha) = \zeta\alpha$. Thus, $\sigma(\alpha^n) = \zeta^n\alpha^n = \alpha^n$, so α^n is fixed by $\text{Gal}(F/k)$. Thus, $\alpha^n \in k$. Moreover, $\sigma^i(\alpha) = \zeta^i\alpha$ is a root of m_α for all i , so that $[k(\alpha) : k] = \deg m_\alpha \geq n$, meaning $F = k(\alpha)$.

Conversely, suppose that $F = k(\alpha)$ for $\alpha^n \in k$. Then the polynomial $t^n - \alpha^n$ has n distinct roots in F (namely, $\zeta^i\alpha$ for $0 \leq i \leq n-1$), so F is the splitting field of a separable polynomial over k . Thus, F/k is Galois. Now, we have a homomorphism $\text{Gal}(F/k) \rightarrow \langle \zeta \rangle$ given by $\sigma \mapsto \sigma(\alpha)/\alpha$ (since σ must send α to some other root of $t^n - \alpha^n$). Further, this homomorphism is injective (as an element of $\text{Gal}(F/k)$ is determined how it acts on α), meaning $\text{Gal}(F/k)$ is cyclic of order dividing n . ■

Now, the following result will allow us to take advantage of our newfound understanding of radical extensions.

Lemma 5.3.28. *Suppose $k = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_r = L$ where F_i/F_{i-1} is Galois with abelian Galois group for $1 \leq i \leq r$. If $k \subseteq E \subseteq L$ with E/k Galois, then $\text{Gal}(E/k)$ is solvable.*

Proof. If $r = 0$, then $E = L = k$ and $\text{Gal}(E/k)$ is trivial. Thus, suppose $r > 0$ and proceed by induction on r .

Since $\text{Gal}(F_1/k)$ is abelian, $\text{Gal}(F_1/E \cap F_1) \trianglelefteq \text{Gal}(F_1/k)$, so that $E \cap F_1/k$ is Galois. Further, $\text{Gal}(E \cap F_1/k)$ is a quotient of $\text{Gal}(F_1/k)$, and thus abelian. Now, $\text{Gal}(E/E \cap F_1) \trianglelefteq \text{Gal}(E/k)$, and the quotient is $\text{Gal}(E \cap F_1/k)$, which is abelian.

Proposition 5.2.11 tells us that EF_1/F_1 is Galois. Since we have a tower $F_1 \subseteq \cdots \subseteq F_r = L$ of length $r-1$, induction tells us that $\text{Gal}(EF_1/F_1)$ is solvable. By Proposition 5.2.11, $\text{Gal}(E/E \cap F_1) \cong \text{Gal}(EF_1/F_1)$ is solvable.

Since $\text{Gal}(E/k)$ has a normal solvable subgroup whose quotient group is solvable, $\text{Gal}(E/k)$ is solvable. ■

We are now ready to prove the most important result of this section.

Theorem 5.3.29 (Galois). *Suppose $\text{char } k = 0$ and F/k is Galois. Then F is contained in a radical extension of k if and only if $\text{Gal}(F/k)$ is solvable.*

Proof. Suppose first that $\text{Gal}(F/k)$ is solvable, and let $[F : k] = n$. Since we are in characteristic 0, we may take an extension $F^* = F(\zeta)$, where ζ is a primitive n th root of unity. Write $k^* = k(\zeta) \subseteq F^*$, and note $k^*F = F^*$. Since F/k is Galois, Proposition 5.2.11 tells us that $F^* = k^*F/k^*$ is Galois. Further, $\text{Gal}(F^*/k^*)$ is isomorphic to a subgroup of $\text{Gal}(F/k)$, and is thus solvable.

Let $G = \text{Gal}(F^*/k^*)$ and take a composition series $1 = G_0 \trianglelefteq \cdots \trianglelefteq G_r = G$ for G . Then each composition is cyclic of prime order, say $|G_i/G_{i-1}| = p_i$. Taking $K_i = \text{Fix}(G_i)$, we have

$$k^* = K_r \subseteq \cdots \subseteq K_0 = F^*,$$

with K_{i-1}/K_i Galois and $\text{Gal}(K_{i-1}/K_i) \cong G_i/G_{i-1}$ cyclic of order p_i .

Note that p_i divides $|G|$, which divides $[F : k] = n$, so that k contains primitive p_i th roots of unity for all i . Thus, Theorem 5.3.27, each extension K_{i-1}/K_i is a pure radical extension. Since $k^* = k(\zeta)/k$ is also a pure radical extension, we see that $F^* \supseteq F$ is a radical extension of k , as desired.

Conversely, suppose that F is contained in some radical extension L of k . Write $k = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_r = L$, where $L_i = L_{i-1}(\alpha_i)$ and $\alpha_i^{n_i} \in L_{i-1}$. Let n be a common multiple of the n_i , and let ζ denote a primitive n th root of unity. Write $L^* = L(\zeta)$, and define $K_{-1} = k$, $K_0 = k(\zeta)$, and $K_i = K_{i-1}(\alpha_i)$ for $1 \leq i \leq r$. We then have

$$k = K_{-1} \subseteq K_0 \subseteq K_1 \subseteq \cdots \subseteq K_r = L^*.$$

The extension $K_0/K_{-1} = k(\zeta)/k$ is Galois (as it is the splitting field of $x^n - 1$) with an abelian Galois group (since the Galois group injects into $(\mathbb{Z}/n\mathbb{Z})^\times$). For $i > 0$, $K_i = K_{i-1}(\alpha_i)$, so Theorem 5.3.27 tells us that K_i/K_{i-1} is Galois with cyclic Galois group. Thus, by Lemma 5.3.28, $\text{Gal}(F/k)$ is solvable, as desired. ■

Tracing through the proof in fact gives us the following.

Corollary 5.3.30. *Let $\text{char } k = 0$ and F/k be Galois with $[F : k] = n$, and suppose that k contains a primitive n th root of unity. Then $\text{Gal}(F/k)$ is solvable if and only if F is a radical extension of k .*

Now, the reader should wonder about the solvability of actual polynomials. We say that a polynomial is *solvable by radicals* if its splitting field is contained in some radical extension. We therefore see that a polynomial is solvable by radicals if and only if the Galois group of its splitting field is solvable.

We may wonder if it is possible for *some* root of a given polynomial to be contained in a radical extension, even if its whole splitting field is not. Unfortunately, this turns out not to be the case.

Proposition 5.3.31. *If $\text{char } k = 0$, any radical extension of k is contained in a Galois radical extension of k .*

Proof. Let L/k be a radical extension, and write $k = L_0 \subseteq \cdots \subseteq L_r$ with $L_i = L_{i-1}(\alpha_i)$, where $\alpha_i^{n_i} \in L_{i-1}$. We induct on r . If $r = 1$, then $L = k(\alpha)$ for some α with $\alpha^n \in k$. If $L^+ = L(\zeta)$, where ζ is a primitive n th root of unity, then L^+/k is radical and the splitting field of $x^n - \alpha^n$, and thus Galois.

Suppose now that we can construct a Galois radical extension L_{r-1}^+ containing L_{r-1} , and let $L_r = L_{r-1}(\alpha)$, where $\alpha^n \in L_{r-1}$. Define

$$h(x) = \prod_{\sigma \in \text{Gal}(L_{r-1}^+/k)} (x^n - \sigma(\alpha^n)).$$

The coefficients of h lie in $\text{Fix}(\text{Gal}(L_{r-1}^+/k)) = k$. Define L_r^+ to be the splitting field of h over L_{r-1}^+ , so that L_r^+ is Galois. Since α is a root of h , we can identify L_r in L_r^+ . Further, L_r^+ is generated over L_{r-1}^+ by roots of h , all of whose n th powers lie in L_{r-1}^+ , meaning that L_r^+ is radical. Thus, L_r^+ is a Galois radical extension containing L_r , as desired. ■

This means that, if we have a formula for one root of a polynomial in terms of radicals, we can pass to a Galois radical extension (forcing the polynomial to split) and get a formula for all roots in terms of radicals.

Corollary 5.3.32 (Insolvability of the Quintic). *There is no formula using radicals and the coefficients of a quintic equation to determine its solutions.*

Proof. If there was such a formula, then we could view $r_1 \in k(r_1, \dots, r_5)$ in a radical extension of $k(s_1, \dots, s_5)$. By passing to a Galois radical extension, the whole field $k(r_1, \dots, r_5)$ is contained in a radical extension of $l(s_1, \dots, s_5)$. But $\text{Gal}(k(r_1, \dots, r_5)/k(s_1, \dots, s_5)) \cong S_5$ is not solvable. ■

Of course, we are yet to produce an actual polynomial whose roots cannot be expressed in terms of radicals. The following proposition will help.

Proposition 5.3.33. *Let $f \in \mathbb{Q}[x]$ be irreducible and have prime degree p . Suppose that f contains $p - 2$ real roots and 2 complex roots, and let E be a splitting field for f over $\mathbb{Q}$. Then $\text{Gal}(E/\mathbb{Q}) \cong S_p$.*

Proof. Embed $E \subseteq \mathbb{C}$, and write $G = \text{Gal}(E/\mathbb{Q})$. Then G acts faithfully on the roots of f , so we have $G \hookrightarrow S_p$. Further, since f is irreducible, G acts transitively on the roots of f , so that in particular $p \mid |G|$ (as the size of any orbit must divide $|G|$). Thus, $\text{im } G$ contains a p -cycle. Further, complex conjugation is an element of G that swaps only two roots, so $\text{im } G$ contains a transposition. But this means $\text{im } G = S_p$. ■

For instance, $x^5 - 10x + 5$ is irreducible by Eisenstein's criterion, and has exactly 3 real roots.

5.3.8 Natural Irrationalities

Proposition 5.3.34 (Natural Irrationalities). *Suppose that $u \in \mathbb{C}(r_1, \dots, r_n)$ lies in a radical extension of $\mathbb{C}(s_1, \dots, s_n)$. Then there is a radical extension $L/\mathbb{C}(s_1, \dots, s_n)$ such that $u \in L \subseteq \mathbb{C}(r_1, \dots, r_n)$.*

Proof. Notice that

$$f(x) = \prod_{\sigma \in S_n} (x - \sigma(u))$$

has coefficients in $k = \mathbb{C}(s_1, \dots, s_n)$ (by the fundamental theorem of symmetric functions). If F is the splitting field of f over k , then F/k is Galois and contained in a radical extension of k (since we may extend to a Galois radical extension, over which f must split). Thus, $\text{Gal}(F/k)$ is solvable. But $\mathbb{C}$ contains primitive m th roots of unity for all m , and so F is a radical extension of k . Noting that f splits over $\mathbb{C}(r_1, \dots, r_n)$, we see that $F \subseteq \mathbb{C}(r_1, \dots, r_n)$, as desired. ■

Natural irrationalities is actually quite surprising. It says that never need to leave the “larger” field $\mathbb{C}(r_1, \dots, r_n)$ when solving polynomials by radicals (while this may feel obvious when working with indeterminate over $\mathbb{C}$, it is far less obvious when working with elements of $\overline{\mathbb{Q}}$ over $\mathbb{Q}(\zeta_m)$, where the result still holds for an appropriate choice of m). For instance, although $(\sqrt{s_1} - 1)(\sqrt{s_1} + 1)$ lies in $\mathbb{C}(r_1, \dots, r_n)$, we really didn’t need to take $\sqrt{s_1}$ in the first place¹⁰.

We’ll investigate now what happens if we don’t have all the necessary roots of unity, by examining the case where we are working in $\mathbb{R}$.

Definition 5.3.35. An extension $F/\mathbb{Q}$ is said to be *real* if we have an embedding $F \hookrightarrow \mathbb{R}$.

Lemma 5.3.36. Let $F/E/k$ be field extensions with $F = k(\alpha)$ and $m = [F : E]$.

- (1) If $\alpha^m \in E$ then $E = k(\alpha^m)$
- (2) If $\alpha^n \in k$ and k contains a primitive n th root of unity, then $\alpha^m \in E$.

Proof.

- (1) Let $L = k(\alpha^m) \subseteq E$. Then $[F : L] \leq m$ (as α is a root of $x^m - \alpha^m \in L[x]$) and $L \subseteq E$, so that $L = E$.
- (2) By Kummer’s Theorem 5.3.27, F/k is Galois and $\text{Gal}(F/k)$ is cyclic. Thus, F/E is Galois, and $\text{Gal}(F/E)$ is cyclic, say $\text{Gal}(F/E) = \langle \sigma \rangle$ where σ has order m . Since $\alpha^n \in F$, we see that $\sigma(\alpha)^n = \sigma(\alpha^n) = \alpha^n$, so that $\sigma(\alpha) = \alpha\zeta$ for some n th root of unity ζ . Since $\zeta \in k$, $\sigma^i(\alpha) = \alpha\zeta^i$, and in particular $\zeta^m = 1$ since σ has order m . Thus, $\sigma(\alpha^m) = \sigma(\alpha)^m = (\alpha\zeta)^m = \alpha^m$, and so $\alpha^m \in \text{Fix}(\text{Gal}(F/E)) = E$. ■

The following lemma is key to our study of real extensions.

Lemma 5.3.37. Let k be a field of characteristic 0, and let $R/L/F/k$ be a tower of fields, where F/k is Galois, L/k is purely radical, and R contains $p < \infty$ roots of unity. Then $[F : k] \leq p$. In particular, when $R = \mathbb{R}$, $[F : k] \leq 2$.

Proof. Write $L = k(\alpha)$, where $\alpha^n \in k$, and let ζ denote a primitive n th root of unity. Write $k^* = k(\zeta)$, $F^* = F(\zeta)$, $L^* = L(\zeta)$, and let $m = [L^* : F^*]$.

Now, F^*/F is Galois (as the splitting field of $x^n - 1$), and so Proposition 5.2.11 gives that $[L : L \cap F^*] = [L^* : F^*] = m$ (since $F^*L = L^*$). Now, since $L^* = k^*(\alpha)$ and $\alpha^n \in k^*$, Lemma 5.3.36 gives $\alpha^m \in F^*$. Thus, $\alpha^m \in F^* \cap L$, and $[L : L \cap F^*] = m$, so Lemma 5.3.36 means that $L \cap F^* = k(\alpha^m)$.

Now, we clearly have $(L \cap k^*)F \subseteq L \cap F^*$, and we show equality. Since F/k is Galois,

$$[(L \cap k^*)F : L \cap k^*] = [F : F \cap L \cap k^*] = [F : F \cap k^*]$$

and

$$[F^* : k^*] = [Fk^* : k^*] = [F : F \cap k^*].$$

Also, $k^*(L \cap F^*) = F^*$ and $k^* \cap (L \cap k^*) = L^* \cap k^*$. Thus, since k^*/k is Galois,

$$[F^* : k^*] = [k^*(L \cap F^*) : k^*] = [L \cap F^* : k^* \cap (L \cap F^*)] = [L \cap F^* : L \cap k^*],$$

and so we see that $L \cap F^* = (L \cap k^*)F$.

Since $\text{Gal}(k^*/k)$ is abelian, $L \cap k^*/k$ is Galois, and so $(L \cap k^*)F/k$ is Galois, which is to say $L \cap F^*/k$ is Galois. Write $\beta = \alpha^m$, and recall that $L \cap F^* = k(\beta)$. Since some power of β lies in k , we must have that $\sigma(\beta)/\beta$ is a root of unity for all $\sigma \in \text{Gal}(L \cap F^*/k)$. Since elements of $\text{Gal}(L \cap F^*/k)$ are determined by how they act on β , and since R contains p roots of unity, we have that

$$[F : k] \leq [L \cap F^* : k] = |\text{Gal}(L \cap F^*/k)| \leq p. \quad \blacksquare$$

Now, since radical extensions are built out of pure radical extensions, it is not too surprising that we have the following.

Theorem 5.3.38. Let F/k be Galois and L/k be a radical extension with $L \subseteq \mathbb{R}$. Then $[F \cap L : k]$ is a power of 2.

¹⁰Of course, this is quite a silly example, but natural irrationalities shows that all examples are like this

Proof. We induct on $[F : k]$, with $[F : k] = 1$ being trivial. We'll write $E = F \cap L$.

Suppose first we can find an intermediate $k \subsetneq K \subsetneq F$ with K/k Galois. Then induction means that $[E \cap K : k] = [L \cap K : k]$ is a power of 2. Also, since KL is a radical extension of K and F/K is Galois, $[F \cap KL : K]$ is a power of 2 by induction. Since $EK \subseteq F \cap KL$, this means $[EK : K]$ is a power of 2. Now, since K/k is Galois, Proposition 5.2.11 gives that $[E : E \cap K] = [EK : K]$ is a power of 2, and so $[E : k] = [E : E \cap K][E \cap K : k]$ is a power of 2.

Thus, suppose there exists no such intermediate field, so that $\text{Gal}(F/k)$ is simple. Suppose $E \supsetneq k$ (since we are otherwise done), and let $\alpha \in E \setminus k$. Since F/k is Galois, m_α splits over F , and so F is a splitting field of m_α over k (since there are no intermediate Galois extensions). But since $\alpha \in L$ is a root of m_α , this means that m_α is solvable by radicals, and so $\text{Gal}(F/k)$ is solvable. Thus, $\text{Gal}(F/k)$ is cyclic of prime order, and so $[F : k]$ is prime. In particular, there are no proper subfields $k \subsetneq K \subsetneq F$.

Since $k \subsetneq E \subseteq F$, we have that $E = F \subseteq L$. Without loss of generality, suppose that L is a minimal radical extension containing F , and let $k \subseteq M \subsetneq L$ such that L/M is purely radical and M/k is radical. Then $F \not\subseteq M$, and so $F \cap M = k$. Since F/k is Galois, Proposition 5.2.11 shows that FM/M is Galois, and FM is contained in the purely radical extension L/k . Thus, Lemma 5.3.37 shows that $[FM : M] \leq 2$, and so $[F : k] = [F : F \cap M] = [FM : M] \leq 2$, as desired. ■

We may apply this result to polynomials with real roots lying in radical extensions of $\mathbb{Q}$.

Theorem 5.3.39. *Let $k \subseteq \mathbb{R}$, and suppose $f \in k[x]$ is irreducible and splits over $\mathbb{R}$ with splitting field F . If f has a root in a real radical extension of k , then $[F : k]$ is a power of 2. In particular, $\deg f$ is a power of 2.*

Proof. Let $k \subseteq L \subseteq \mathbb{R}$ be a real radical extension containing a root α of f , and set $E = F \cap L$. By Theorem 5.3.38, $[E : k]$ is a power of 2. More generally, for any intermediate field $k \subseteq K \subseteq F$, KL/K is a radical extension and F/K is a Galois extension, so $[F \cap KL : K]$ is a power of 2. Since $EK \subseteq F \cap KL$, this means $[EK : K]$ is a power of 2.

Write $G = \text{Gal}(F/k)$ and $H = \text{Gal}(F/E)$. If $U \leq G$ and $K = \text{Fix}(U)$, then $\text{Fix}(U \cap H) = EK$, so that $[U : U \cap H] = [EK : K]$ is a power of 2. If p is an odd prime, let P be a Sylow p -subgroup of G , so that $[P : P \cap H]$ is a power of 2. Since $p \neq 2$, this means $P \cap H = P$, so $P \subseteq H$. But P was arbitrary, and so H contains the subgroup N generated by all Sylow p -subgroups of G .

Set $K = \text{Fix}(N) \supseteq E$, so that K contains α . Since $N \trianglelefteq G$, K/k is Galois, f splits over K . Thus, $K = F$, so $N = 1$. But this means that G has no nontrivial Sylow p -subgroups for all odd primes p , so G is a 2-group. ■

If we are dealing with a cubic with 3 real roots, then this means that it is impossible to write any of its roots only in terms of n th roots of real numbers: the only way to express roots of f using radicals involves passing to the complex numbers. This is a quite surprising result, as one would expect the real numbers to be sufficient to express real roots of a polynomial.

Remark 5.3.40. Recall that if a real number γ is constructible, $[\mathbb{Q}(\gamma) : \mathbb{Q}]$ is a power of 2, which is to say that $\deg m_\gamma$ is a power of 2. We see now that this is no coincidence, as constructibility in particular means γ can be expressed using radicals.

5.4 More on Fields

5.4.1 The Inverse Galois Problem for Abelian Groups

The inverse Galois problem consists of looking for what groups can be expressed as the Galois group of some extension over $\mathbb{Q}$. Recall that

$$\text{Gal}\left(\frac{\mathbb{C}(r_1, \dots, r_n)}{\mathbb{C}(s_1, \dots, s_n)}\right) \cong S_n.$$

Since any finite group G can be realized as a subgroup of S_n for some n , this means that there is some Galois extension F/k such that $\text{Gal}(F/k) \cong G$. However, we do not know whether we can take $k = \mathbb{Q}$ (which is significantly more restrictive, note that the relevant fields are quite large). In this section, we will prove that any finite *abelian* group can be realized as the Galois group of some extension over $\mathbb{Q}$.

Lemma 5.4.1. *For any $n \geq 1$, there are infinitely many primes $p \equiv 1 \pmod{n}$.*

Proof. Let p be a prime divisor of $\Phi_m(m)$. Then p divides $m^m - 1$, and so in particular $p \nmid m$. Let d be the order of m modulo p , so that $d \mid m$. Write $m = de$.

If $e > 1$, then we can write $x^m - 1 = \Phi_m(x)(x^d - 1)g(x)$ for some polynomial $g(x)$. Dividing by $x^d - 1$ and plugging in $x = m$,

$$0 \equiv \Phi_m(m)g(m) = 1 + m^d + \dots + (m^d)^{e-1} \equiv e \pmod{p},$$

so that $p \mid de = m$, a contradiction. Thus, $d = m$, and $m = d \mid p - 1$, so $p \equiv 1 \pmod{m}$ and $p > m$.

Now, for $m \geq 2$, we have that $|\Phi_m(m)| > m - 1$ by examining the location of m and the primitive m th roots of unity on the complex plane. Thus, for $k \geq 2$, $|\Phi_{nk}(nk)| > 1$ has a prime divisor, which is necessarily $\geq nk$, and so we have found infinitely many primes congruent to 1 (mod n). ■

Remark 5.4.2. This is a special case of Dirichlet's Theorem, which states that there are infinitely many primes congruent to $k \pmod{n}$ whenever $\gcd(n, k) = 1$.

Remark 5.4.3. The above proof actually shows that, if $p \mid \Phi_n(m)$, then $p \nmid m$ and either $p \mid n$ or $p \equiv 1 \pmod{n}$.

Since $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p - 1$, which we can make divisible by any integer we want, we get a hint towards how we can obtain our desired result.

Theorem 5.4.4. *Let G be a finite abelian group. Then there is a Galois extension $F/\mathbb{Q}$ such that $\text{Gal}(F/\mathbb{Q}) \cong G$.*

Proof. Write

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z},$$

and let $p_1, \dots, p_r$ be distinct primes with $p_i \equiv 1 \pmod{n_i}$. Set $n = p_1 \cdots p_r$. Then,

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_r\mathbb{Z})^\times \cong \mathbb{Z}/(p_1 - 1)\mathbb{Z} \times \cdots \times \mathbb{Z}/(p_r - 1)\mathbb{Z}.$$

Since $n_i \mid p_i - 1$, we can find $N \leq \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ such that $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})/N$ is isomorphic to $\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z} \cong G$. Taking $F = \text{Fix}(N)$ gives the result. ■

5.4.2 More on Finite Fields

We provide some interesting formulas.

Proposition 5.4.5. *Let $\text{GL}_n(\mathbb{F}_p)$ be the group of invertible $n \times n$ matrices with entries in $\mathbb{F}_p$. Then*

$$|\text{GL}_n(\mathbb{F}_p)| = p^{\binom{n}{2}} \prod_{i=1}^n (p^i - 1).$$

Proof. Write $G = \text{GL}_n(\mathbb{F}_p)$. Let

$$\mathcal{F} = \{0 = V_0 \leq V_1 \leq \cdots \leq V_n = \mathbb{F}_p^n : \dim V_i = i\},$$

and let G act on $\mathcal{F}$ by multiplication on the left. Given a basis $v_1, \dots, v_n$ of $\mathbb{F}_p^n$, we'll write

$$[v_1, \dots, v_n] = 0 \leq \langle v_1 \rangle \leq \langle v_1, v_2 \rangle \leq \cdots \leq \langle v_1, \dots, v_n \rangle = \mathbb{F}_p^n \in \mathcal{F},$$

and note that every element of $\mathcal{F}$ arises in this way. This action is transitive: if $e_1, \dots, e_n$ is the standard basis of $\mathbb{F}_p^n$ and $v_1, \dots, v_n$ is some basis, then the matrix $(v_1 \ \cdots \ v_n)$ sends $[e_1, \dots, e_n]$ to $[v_1, \dots, v_n]$.

Now, the stabilizer of $[e_1, \dots, e_n]$ is precisely the set B of upper triangular matrices, which has order $p^{\binom{n}{2}}(p - 1)^n$. We can also count $|\mathcal{F}|$: there are $\frac{p^n - 1}{p - 1}$ ways to pick a 1-dimensional subspace V_1 of $\mathbb{F}_p^n$, then $\frac{p^{n-1} - 1}{p - 1}$ ways to pick a 1-dimensional subspace V_2/V_1 of $\mathbb{F}_p^n/V_1$, and so on. Thus, orbit-stabilizer yields

$$|G| = |B| \cdot |\mathcal{F}| = p^{\binom{n}{2}}(p - 1)^n \prod_{i=1}^n \frac{p^i - 1}{p - 1} = p^{\binom{n}{2}} \prod_{i=1}^n (p^i - 1). \quad \blacksquare$$

We can in fact use this technique to study the Sylow p -subgroups of G . For any $V_0 \leq \cdots \leq V_n \in \mathcal{F}$, let

$$U(V_0 \leq \cdots \leq V_n) = \{M \in G : MV_i = V_i, M \text{ acts trivially on } V_{i+1}/V_i\}.$$

Now, for any $g \in G$, we have that $h \in U(F)$ if and only if $ghg^{-1} \in U(gF)$ for all $F \in \mathcal{F}$. Thus, we have a G -equivariant map $\mathcal{F} \rightarrow \{H \leq G\}$, where G acts on its subgroups by conjugation. Let $U = U([e_1, \dots, e_n])$ be the set of upper triangular matrices with 1 along the diagonal. Since the action of G on $\mathcal{F}$ is transitive, we see that this restricts to a G -equivariant surjection $\mathcal{F} \rightarrow \{gUg^{-1} : g \in G\}$.

We claim that we in fact have a G -equivariant bijection. If we have distinct $[e_1, \dots, e_n]$ and $[v_1, \dots, v_n]$, let i be minimal so that $v_i \notin \langle e_1, \dots, e_i \rangle$. Then there is some $j > i$ such that the j th coordinate of v_i is nonzero. If $g \in U$ is the matrix with 1s along the diagonal and a 1 in the (i, j) entry, then $gv_i - v_i \in \langle e_i \rangle$ is nonzero. If g acted trivially on $\langle v_1, \dots, v_i \rangle / \langle v_1, \dots, v_{i-1} \rangle$, then we would have $gv_i - v_i \in \langle v_1, \dots, v_{i-1} \rangle = \langle e_1, \dots, e_{i-1} \rangle$, which is impossible. Thus, there is some element in U not in $U([v_1, \dots, v_n])$, and so we indeed have a bijection.

In particular, the stabilizer of U under conjugation, which is to say $N_G(U)$, is the stabilizer of $[e_1, \dots, e_n]$, which is precisely B . Since $B/U \cong (\mathbb{F}_p^\times)^n$ is not divisible by p , this means that U is a Sylow p -subgroup of G (since $[G : U] \equiv [B : U] \pmod{p}$). Further, U has precisely $|\mathcal{F}| = \prod_{i=1}^n \frac{p^i - 1}{p - 1}$ conjugates in G .

Now, for a prime power q , we know that the field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ has degree n . Since $\mathbb{F}_{q^n}^\times$ is cyclic, we can write $\mathbb{F}_{q^n} = \mathbb{F}_q(\alpha)$ for some generator α . The minimal polynomial of α over $\mathbb{F}_q$ must then have degree exactly n , and so there are irreducible polynomials of every degree over $\mathbb{F}_q$. We provide a precise count.

Proposition 5.4.6. *There are exactly $\frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}$ monic irreducible polynomials of degree n over $\mathbb{F}_q$, where*

$$\mu(d) = \begin{cases} (-1)^k & d = p_1 \cdots p_k \\ 0 & d \text{ not squarefree} \end{cases}$$

is the Möbius function.

Proof. Note that every irreducible monic polynomial of degree n over $\mathbb{F}_q$ must split over $\mathbb{F}_{q^n}$ (adjoining one root gives a field of order q^n , and the extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is normal). Further, any root of an irreducible monic polynomial of degree n must generate $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$, and any generator is necessarily a root of an irreducible monic polynomial of degree n . Thus, if $T = \{\alpha \in \mathbb{F}_{q^n} : \mathbb{F}_{q^n} = \mathbb{F}_q(\alpha)\}$, then we wish to determine $|T|/n$.

Any element of $\mathbb{F}_{q^n}$ not lying in T lies in some proper subfield of $\mathbb{F}_{q^n}$, and thus some maximal proper subfield, which has order $q^{n/r}$ for some prime $r \mid n$. If we write K_r for this subfield, then we see that

$$\left| \bigcap_{i=1}^m K_{r_i} \right| = q^{n/r_1 \cdots r_m}.$$

Thus, inclusion-exclusion gives

$$\left| \bigcup_{\substack{r|n \\ r \text{ prime}}} K_r \right| = \sum_{\substack{1 < s|n \\ s \text{ squarefree}}} (-\mu(s)) q^{n/s},$$

so that

$$|T| = q^n - \left| \bigcup_{\substack{r|n \\ r \text{ prime}}} K_r \right| = \sum_{\substack{s|n \\ s \text{ squarefree}}} \mu(s) q^{n/s} = \sum_{d|n} \mu(d) q^{n/d}.$$

■

5.4.3 Wedderburn's Theorem

We provide a proof of Wedderburn's little theorem.

Theorem 5.4.7 (Wedderburn). *A finite division ring is a field.*

Proof. Let D be a finite division ring, and let $k = Z(D)$ be its center. Then D is a finite-dimensional algebra over k , say of dimension n . Let $|k| = q > 1$, so that $|D| = q^n$.

For each $x \in D$ not contained in k , its centralizer Z_x has order q^d for some $d < n$. Letting $D^\times$ act on itself by conjugation, we get the class equation

$$q^n - 1 = |D^\times| = |k^\times| + \sum_{C \in \mathcal{C}} |C| = q - 1 + \sum_{\substack{C \in \mathcal{C} \\ \dim_k Z_x = d}} \frac{q^n - 1}{q^d - 1},$$

where the sum is taken over nontrivial conjugacy classes C (and x is a representative of C). In particular, we see that $q^d - 1 \mid q^n - 1$, and so $d \mid n$ for every d appearing in the sum. Writing $q^n - 1$ and $q^d - 1$ as a product of

cyclotomic polynomials, we see that $\Phi_n(q)$ divides each term in the sum on the right (since $d < n$ and $d \mid n$ for each term in the sum), and divides the left hand side. Thus, $\Phi_n(q)$ divides $q - 1$, and so $|\Phi_n(q)| \leq q - 1$.

Now, suppose that $n > 1$, and let ζ be a primitive n th root of unity. We in particular have that $\zeta \neq 1$. Now, for $\gcd(k, n) = 1$, we have that $|q - \zeta^k| > q - 1$ (by considering their positions in the complex plane), and so

$$|\Phi_n(q)| = \prod_{\gcd(k, n)=1} |q - \zeta^k| > q - 1,$$

a contradiction. Thus, $n = 1$, and so $D = k$ is a field. ■

In fact, this means that a finite domain is a field, as a finite domain is a division ring (multiplication by an element is injective, and thus surjective, yielding an inverse).

5.4.4 (In)Separable Extensions

We recall the following key result about when polynomials are separable.

Proposition 5.4.8. *Let k be a field.*

- (1) $f \in k[x]$ has distinct roots if and only if f and f' are relatively prime
- (2) If $f \in k[x]$ is irreducible, then f is separable if and only if $f' \neq 0$

In particular, if $f \in k[x]$ is irreducible and inseparable, then $\text{char } k = p > 0$ and $f(x) = g(x^p)$ for some $g \in k[x]$.

Corollary 5.4.9. *Let $\text{char } k = p > 0$, and let $f \in k[x]$ be irreducible. Then $f(x) = g(x^{p^n})$ for some $n \geq 0$ and some separable irreducible $g \in k[x]$.*

Proof. We induct on $\deg f$ ($\deg f = 1$ is trivial, since f is automatically separable). If f is separable, we are done. Otherwise, let $f(x) = h(x^p)$ for some irreducible $h \in k[x]$. By induction, we can write $h(x) = g(x^{p^n})$ for some separable irreducible $g \in k[x]$, and so $f(x) = g(x^{p^{n+1}})$. ■

Recall that an extension is separable if m_α is separable for every α . We'll examine what happens when the exact opposite of this is true.

Definition 5.4.10. A field extension F/k is said to be *purely inseparable* if m_α is not separable for all $\alpha \in F \setminus k$.

Remark 5.4.11. The extension k/k is trivially purely inseparable. A nontrivial extension cannot be both separable and purely inseparable, so we will need to turn to imperfect fields to find examples.

The following result allows us to characterize purely inseparable extensions.

Theorem 5.4.12. *Let F/k be algebraic and $\text{char } k = p > 0$. The following are equivalent.*

- (1) F/k is purely inseparable
- (2) For every $\alpha \in F$, there is $n \geq 0$ such that $\alpha^{p^n} \in k$
- (3) Every element of F has minimal polynomial of the form $x^{p^n} - a$ for $n \geq 0$ and $a \in k$

Proof.

- (1) $\Rightarrow$ (2) Let $\alpha \in F$. Then we can write $m_\alpha(x) = f(x^{p^n})$ for some $n \geq 0$ and $f \in k[x]$ separable. But this means α^{p^n} is separable over k , and so $\alpha^{p^n} \in k$.
- (2) $\Rightarrow$ (3) Let $\alpha \in F$ with $\alpha^{p^n} \in k$. We know that m_α divides $x^{p^n} - \alpha^{p^n} = (x - \alpha)^{p^n}$, and so is of the form $(x - \alpha)^r$ for some $r \mid p^n$. But this means $r = p^m$ for some m , and $m_\alpha(x) = x^{p^m} - \alpha^{p^m}$ (in particular, $\alpha^{p^m} \in k$).
- (3) $\Rightarrow$ (1) Suppose that $\alpha \in F$ is separable over k , and write $m_\alpha(x) = x^{p^n} - a = (x - \alpha)^{p^n}$. Since m_α is separable, it has no repeated roots, and so $p^n = 1$, which is to say $m_\alpha = x - \alpha$. In particular, $\alpha \in k$, so F/k is purely inseparable. ■

Corollary 5.4.13. *If $\text{char } k = p > 0$ and $F = k(\alpha)$ where $\alpha^{p^n} \in k$ for some $n \geq 0$, then F/k is purely inseparable.*

Proof. An arbitrary element of F is of the form $\sum_i a_i \alpha^i$, where the a_i lie in k . Now,

$$\left(\sum_i a_i \alpha^i \right)^{p^n} = \sum_i a_i^{p^n} (\alpha^{p^n})^i \in k,$$

so F/k is purely inseparable. ■

Corollary 5.4.14. *Let $\text{char } k = p > 0$ and $F/E/k$ be a tower of field extensions. Then F/k is purely inseparable if and only if F/E and E/k are purely inseparable.*

Proof. Suppose that F/k is purely inseparable. For $\alpha \in F$, we can find some n such that $\alpha^{p^n} \in k \subseteq E$, so F/E is purely inseparable. Similarly, for $\alpha \in E \subseteq F$, we can find n such that $\alpha^{p^n} \in k$, so E/k is purely inseparable.

Conversely, suppose that F/E and E/k are purely inseparable. For $\alpha \in F$, we can find n such that $\alpha^{p^n} \in E$. Now, we can find m such that $\alpha p^{n+m} = (\alpha p^n)^{p^m} \in k$, so F/k is purely inseparable. ■

Corollary 5.4.15. *Let $\text{char } k = p > 0$ and let F/k be a finite purely inseparable extension. Then $[F : k]$ is a power of p .*

Proof. We induct on $[F : k]$, with $[F : k] = 1$ being trivial. If $[F : k] > 1$, let $\alpha \in F \setminus k$. Then $k(\alpha)/k$ is purely inseparable, and $[k(\alpha) : k] = \deg m_\alpha$ is a power of p . By induction, $[F : k(\alpha)]$ is also a power of p , and so $[F : k]$ is a power of p . ■

One may hope that we can separate any extension into a separable part and a purely inseparable part. Indeed, this turns out to be true.

Theorem 5.4.16. *Let F/k be an algebraic extension and*

$$S = \{\alpha \in F : \alpha \text{ separable over } k\}.$$

Then S/k is a separable extension, F/S is a purely inseparable extension, and S is unique with respect to this property.

Proof. We may assume $\text{char } k = p > 0$, as otherwise k is perfect and there is nothing to prove. We first note that S is indeed a field: if $\alpha, \beta \in S$, then $k(\alpha, \beta)$ is separable (as it is generated by separable elements), and so $\alpha \pm \beta, \alpha\beta^{\pm 1} \in S$. Clearly S/k is separable. Now, for any $\alpha \in F$, we can write $m_\alpha(x) = f(x^{p^n})$ for some separable $f \in k[t]$. This means that α^{p^n} is separable over k , so that $\alpha^{p^n} \in S$ and F/S is purely inseparable.

Suppose now we have some other intermediate field $F/T/k$ such that F/T is purely inseparable and T/k is separable. Clearly we have that $T \subseteq S$. But since S/k is separable, S/T is separable as well (since polynomials remain separable when passing to larger fields). Since F/T is purely inseparable, S/T is purely inseparable, and so we must have $S = T$. ■

Corollary 5.4.17. *F/k is purely inseparable if and only if no elements of $F \setminus k$ are separable over k .*

Proof. These are both equivalent to S (as defined above) being equal to k . ■

Definition 5.4.18. Let F/k be an algebraic extension, and let S be as in Theorem 5.4.16. The field S is called the *separable part* of F , and we write $F_{\text{sep}} = S$. The *separable degree* of F/k is

$$[F : k]_s = [S : k].$$

Proposition 5.4.19. *Let $F/E/k$ be a tower of field extensions. Then F/k is separable if and only if F/E and E/k are separable.*

Proof. If F/k is separable, then clearly E/k is separable. Additionally, F/E is separable, as m_α^k is separable for all $\alpha \in F$, and so $m_\alpha^E \mid m_\alpha^k$ is separable as well.

Conversely, suppose that F/E and E/k are separable. Then $E \subseteq F_{k\text{-sep}} \subseteq F$ and $F/F_{k\text{-sep}}$ is separable (since F/E is separable). But since $F/F_{k\text{-sep}}$ is purely inseparable, we have that $F = F_{k\text{-sep}}$. ■

In the case of finite extensions, we see that F/k is separable if and only if $[F : k] = [F : k]_s$.

Now, if F/k is finite and separable, then there are exactly $[F : k]$ ways to embed F into $\bar{k}$ (this can be proved by induction, noting that there are exactly $\deg m_\alpha = [k(\alpha) : k]$ ways to embed $k(\alpha) \hookrightarrow \bar{k}$ when α is separable)¹¹. On the

¹¹If F/k is infinite and separable, there are infinitely many embeddings $F \hookrightarrow \bar{k}$.

other hand, if F/k is purely inseparable, then there is exactly one way to embed $F \hookrightarrow \bar{k}$, since $m_\alpha(x) = x^{p^n} - \alpha^{p^n} = (x - \alpha)^{p^n}$ has exactly one root for all $\alpha \in F$. Thus, in general, there are $[F : k]_s$ ways to embed $F \hookrightarrow \bar{k}$, since we can first embed $F_{\text{sep}} \hookrightarrow \bar{k}$ in $[F : k]_s$ ways, and the rest of F must be embedded uniquely. In particular, $[k(\alpha) : k]_s$ is the number of distinct roots of m_α . Further, in the case that F/k is normal, all embeddings $F \hookrightarrow \bar{k}$ have the same image, and so $|\text{Aut}(F/k)| = [F : k]_s$. This new definition allows us to examine how separable degrees interact with each other.

Proposition 5.4.20. *Let $F/E/k$ be a tower of field extensions. Then $[F : k]_s$ is finite if and only if $[F : E]_s$ and $[E : k]_s$ are, and in this case*

$$[F : k]_s = [F : E]_s [E : k]_s.$$

Proof. Since each embedding $E/k \hookrightarrow \bar{k}$ and $F/E \hookrightarrow \bar{k}$ extends uniquely to an embedding $F/k \hookrightarrow \bar{k}$, we see that $[F : k]_s$ is infinite if either of $[F : E]_s$ or $[E : k]_s$ are. Conversely, since every embedding arises in this manner, we see that $[F : k]_s = [F : E]_s [E : k]_s$ when both sides are finite¹². ■

We may hope to alternatively find some intermediate field $F/E/k$ where F/E is separable and E/k is purely inseparable. This isn't possible in general, but is possible in the finite case.

Proposition 5.4.21. *Let F/k be a finite normal extension and $E = \text{Fix}(\text{Aut}(F/k))$. Then F/E is separable and E/k is purely inseparable.*

Proof. Proposition 5.2.4 shows that F/E is Galois (and in particular separable). Now, suppose that $\alpha \in E$ is separable over k . Since F/k is normal, m_α splits in F . Now, if $\alpha \notin k$, then m_α has at least one other root β . Since F/k is normal (and so a splitting field), we can extend the map $k(\alpha) \rightarrow k(\beta)$ to an automorphism of F (by uniqueness of splitting fields). But α is fixed by $\text{Aut}(F/k)$, a contradiction. Thus, $\alpha \in k$, and so E/k is purely inseparable. ■

Now, we would hope again that $[F : k]_s = [F : E]$, and gladly this also turns out to be true.

Lemma 5.4.22. *If F/k is purely inseparable and $f \in k[x]$ is separable and irreducible, then $f \in F[x]$ is irreducible.*

Proof. Let g be an irreducible divisor of f in $F[x]$. Let L be a splitting field of f over F , so that g splits over L into roots that are separable over k . This means $g \in L_{\text{sep}}[x]$, and so the coefficients of g lie in $L_{\text{sep}} \cap F$. Since F/k is purely inseparable, this means $g \in k[x]$, and so we must have $g = f$ (up to units), since f is irreducible. ■

Proposition 5.4.23. *Let F/k be a finite extension, and let S and I be intermediate fields such that S/k and F/I are separable and F/S and I/k are inseparable. Then $[S : k] = [F : I]$.*

Proof. If $\alpha \in S$, then $m_\alpha \in k[x]$ remains irreducible in $I[x]$, and so is the minimal polynomial of α over I . Thus, $[k(\alpha) : k] = [I(\alpha) : I]$. Since S/k is separable, the Primitive Element Theorem tells us we can pick α so that $S = k(\alpha)$, and so

$$[S : k] = [k(\alpha) : k] = [I(\alpha) : I] \leq [F : I].$$

Conversely, since F/I is separable, we can write $F = I(\beta)$. Since E/S is purely inseparable, we can find some $n \geq 0$ with $\beta^{p^n} \in S$ (where $p = \text{char } k > 0$, since there is nothing to prove in characteristic 0). Now, $F = I(\beta)$ is purely inseparable over $I(\beta^{p^n})$ by Corollary 5.4.13. But F/I is separable, and so $F/I(\beta^{p^n})$ is separable as well. Thus, $F = I(\beta^{p^n})$. Since $\beta^{p^n} \in L$, we may use the result from the last paragraph to get

$$[F : I] = [I(\beta^{p^n}) : I] = [k(\beta^{p^n}) : k] \leq [S : k].$$

■

We can use this fact to give a proof that separable degrees are multiplicative without appealing to algebraic closures¹³. Suppose we have a finite tower $k \subseteq E \subseteq F$. We can find E_{sep} and F_{sep} such that E_{sep}/k and F_{sep}/E are separable while E/E_{sep} and F/F_{sep} are purely inseparable¹⁴. Clearly $E_{\text{sep}} \subseteq F_{\text{sep}}$, and so we can find an intermediate field K with K/E_{sep} separable and F_{sep}/K purely inseparable. Since $K/E_{\text{sep}}/k$ is separable and $F/F_{\text{sep}}/K$ is purely inseparable, we have that $[F : k]_s = [K : k]$.

Now, $[K : k] = [K : E_{\text{sep}}][E_{\text{sep}} : k] = [K : E_{\text{sep}}][E : k]_s$. Since K is an intermediate field of $F_{\text{sep}}/E_{\text{sep}}$ with K/E_{sep} separable and F_{sep}/K purely inseparable, while E is an intermediate field with F_{sep}/E separable and E/E_{sep} purely inseparable, the above theorem gives $[K : E_{\text{sep}}] = [F_{\text{sep}} : E] = [F : E]_s$. Thus, $[F : k]_s = [F : E]_s [E : k]_s$, as desired.

We end with a nice application of the theory we have developed.

¹²We are sweeping under the rug the fact that $\bar{k} = \bar{E}$, which follows immediately from the fact that E/k is algebraic.

¹³Many people, including the author, prefer to avoid using algebraic closures when not necessary. Since a sufficiently large splitting field is usually enough, this pedagogy is not actively harmful.

¹⁴Note that F_{sep} is taken with the base field E , not k

Theorem 5.4.24. *Let F/k be algebraic, and suppose every nonconstant polynomial $f \in k[x]$ has a root in F . Then F is algebraically closed.*

Proof. Suppose first that F/k is separable. Since every nonzero monic irreducible $f \in k[x]$ has a root in F , we can write $f = m_\alpha$ for some α . In particular, since F/k is separable, f is separable. Let L be a splitting field for f over k , so that L/k is Galois, and write $L = k(\beta)$ by the Primitive Element Theorem. Then m_β has a root $\gamma \in F$ by assumption, and so we can find a map $L = k(\beta) \hookrightarrow k(\alpha) \subseteq F$. Thus, f splits in $F[t]$, and so F is algebraically closed by Proposition 5.1.40.

We now drop the assumption that F/k is separable, and let $\text{char } k = p > 0$. Let

$$E = \{\alpha \in F : \alpha^{p^n} \in k \text{ for some } n \geq 0\}.$$

For any $\alpha, \beta \in E$, we can find a common $n \geq 0$ so that $\alpha^{p^n}, \beta^{p^n} \in k$, showing that $(\alpha \pm \beta)^{p^n}, (\alpha\beta^{\pm 1})^{p^n} \in k$. Thus, E is a field. Further, for any $\alpha \in E$ with $\alpha^{p^n} \in k$, the polynomial $t^{p^{n+1}} - \alpha^{p^n} \in k[t]$ has a root $\beta \in F$. This means $0 = \beta^{p^{n+1}} - \alpha^{p^n} = (\beta^p - \alpha)^{p^n}$, so that $\alpha = \beta^p$ and $\beta^{p^{n+1}} \in k$. Thus, α is a p th power in E , so the Frobenius endomorphism is surjective. Thus, E is perfect by Proposition 5.3.8, so that F/E is separable.

By the first paragraph, it now suffices to show that every nonconstant polynomial in $E[x]$ has a root in F . Let $g(x) = \sum_{i=0}^n a_i x^i \in E[t]$, and take m large enough that $a_i^{p^m} \in k$ for $1 \leq i \leq n$. Then,

$$g(x)^{p^m} = \left(\sum_{i=0}^n a_i x^i \right)^{p^m} = \sum_{i=0}^n a_i^{p^m} x^{ip^m} \in k[x],$$

so $g(x)^{p^m}$ has a root α in F . But since $g(\alpha)^{p^m} = 0$, we must have that $g(\alpha) = 0$, so g has a root in F , as desired. ■

5.4.5 Transcendental Extensions

So far, we've exclusively been working with algebraic extensions. We now turn to examining transcendental extensions. Recall that $\alpha \in F/k$ is transcendental over k if it is not algebraic over k .

Definition 5.4.25. We say an extension F/k is *totally transcendental* if every $\alpha \in F \setminus k$ is transcendental over k .

Proposition 5.4.26. *The extension $k(\alpha)/k$ is totally transcendental, where α is transcendental over k . Further, $k(\alpha)/k(\beta)$ is finite for all $\beta \in k(\alpha) \setminus k$.*

Proof. Write $\beta = f(\alpha)/g(\alpha)$, and consider $h \in k(\beta)[x]$ given by $h(x) = f(x) - \beta g(x)$, so that $h(\alpha) = 0$. Since $\beta g(x)$ does not have coefficients in k , h is nonzero, so α is algebraic over $k(\beta)$. Now, β must be transcendental over k , as α would otherwise also be algebraic over k . ■

Our main approach to studying transcendental extensions will be to develop an analog of linear independence that allows us to derive analogous results from linear algebra.

Definition 5.4.27. If F/k is a field extension, we say $S \subseteq F$ is *algebraically independent over k* if $f(\alpha_1, \dots, \alpha_n) \neq 0$ for all $\alpha_1, \dots, \alpha_n \in S$ and nonzero $f \in k[x_1, \dots, x_n]$.

Example. The elements $x_1, \dots, x_n \in k(x_1, \dots, x_n)$ (the field of rational functions in $x_1, \dots, x_n$) are algebraically independent.

Just as we can uniquely determine the structure of the span of S by $|S|$ when S is linearly independent, we can determine the structure of $k(S)$ by $|S|$ when S is algebraically independent.

Proposition 5.4.28. *Let F/k be an extension, and $S \subseteq F$ be algebraically independent. Then $k(S)$ is isomorphic to the rational function field in $|S|$ variables.*

Proof. For each $\alpha \in S$, let x_α be an indeterminate. Then evaluating α for x_α gives a map $k[x_\alpha]_{\alpha \in S}$, which is injective since S is algebraically independent. Thus, we have a field inclusion $k(x_\alpha)_{\alpha \in S} \rightarrow k(S)$, which is surjective because $k(S)$ is generated by elements of S . ■

Definition 5.4.29. A field extension F/k is said to be *purely transcendental* if $F = k(S)$ for some algebraically independent $S \subseteq F$.

Recall that $\{\alpha\}$ is linearly independent over k if and only if $\alpha = 0$, and that if S is linearly independent, then $\{\alpha\} \cup S$ is linearly dependent if and only if $\alpha \in \langle S \rangle$. Note that $\{\alpha\}$ is transcendently independent over k if and only if α is transcendental over k . By analogy, the corresponding statement for algebraic extensions should therefore be that if S is algebraically independent, then $\{\alpha\} \cup S$ is algebraically dependent if and only if α is algebraic over $k(S)$. This is indeed true.

Proposition 5.4.30. *Let F/k , and suppose that S and T are disjoint subsets of F with S algebraically independent over k . Then $S \cup T$ is algebraically independent over k if and only if T is algebraically independent over $k(S)$.*

Proof. Suppose first that $S \cup T$ is algebraically dependent over k , say $f(s_1, \dots, s_n, t_1, \dots, t_m) = 0$. Viewing $f(s_1, \dots, s_n, x_1, \dots, x_m) \in k(s_1, \dots, s_n)[x_1, \dots, x_m]$, we see that T is algebraically independent over $k(S)$. Conversely, suppose that T is algebraically independent over $k(S)$, say $f \in k(S)[x_1, \dots, x_m]$ is nonzero with $f(t_1, \dots, t_m) = 0$. Clearing the denominators of the coefficients of f then shows that $S \cup T$ is algebraically dependent (to verify that clearing denominators doesn't result in 0, note that each term after clearing denominators will have a different $x_1^{e_1} \cdots x_m^{e_m}$ factor, and so they can't possibly cancel). ■

Corollary 5.4.31. *Purely transcendental extensions are totally transcendental.*

Proof. Let F/k be purely transcendental with $F = k(S)$, and suppose $\beta \in F$ is algebraic over k . Since β is the quotient of two polynomials in S , we may without loss of generality assume that S is finite.

We induct on $|S|$. If $|S| = 0$, then $F = k$ and we are done. For $|S| > 0$, let $\alpha \in S$ and write $S' = S \setminus \{\alpha\}$. Set $E = k(S')$, so that $F = E(\alpha)$ and α is transcendental over E , since $S' \cup \{\alpha\} = S$ is algebraically independent over k . Since $E(\alpha)/E$ is totally transcendental and β is algebraic over $k \subseteq E$, we see that $\beta \in E$. By induction, $\beta \in k$, as desired. ■

We'll now turn to finding the analog of a basis for an extension.

Proposition 5.4.32. *If F/k is a field extension, there exists an intermediate field E such that F/E is algebraic and E/k is purely transcendental. Further, for any set S with $F = k(S)$, we can pick $E = k(S')$, where $S' \subseteq S$ is maximally algebraically independent.*

Proof. Let $F = k(S)$, and use Zorn's lemma to find $S' \subseteq S$ that is maximal among algebraically independent subsets. Set $E = k(S')$, so we clearly have that E/k is transcendental. Since $S \setminus S'$ generates F over E , it suffices to show that each of these elements are algebraic over E . But for $\alpha \in S \setminus S'$, $S' \cup \{\alpha\}$ is algebraically dependent, so α is algebraic over $k(S') = E$. ■

Remark 5.4.33. If we can find a finite generating set S , we do not need to use Zorn's lemma to find S' .

Definition 5.4.34. For a field extension F/k , a *transcendence basis* for F over k is an algebraically independent set S such that $F/k(S)$ is algebraic.

Thus, we have established that *every field extension has a transcendence basis*, and that in particular every finitely generated extension has a finite transcendence basis. The following result now establishes that every transcendence basis has the same size (in the finite case).

Theorem 5.4.35. *Let F/k be a field extension, and suppose that $F/k(S)$ is algebraic for some finite $S \subseteq F$. Then every $T \subseteq F$ with $|T| > |S|$ is algebraically dependent.*

Proof. We induct on $|S|$, with $|S| = 0$ being trivial. Suppose $|S| > 0$, and let $\alpha \in T$ with α transcendental over k (as we are otherwise done).

If $\alpha \in S$, let $S' = S \setminus \{\alpha\}$, $T' = T \setminus \{\alpha\}$, and $K = k(\alpha)$. Since F is algebraic over $k(S) = K(S')$ and $|T'| > |S'|$, induction gives that T' is algebraically dependent over $K = k(\alpha)$, meaning that $T' \cup \{\alpha\} = T$ is algebraically dependent over k .

Now, suppose that $\alpha \notin S$. Let $\{\alpha\} \subseteq R \subseteq S \cup \{\alpha\}$ be a maximal algebraically independent set. Since α is algebraic over $k(S)$, we see that R is a proper subset of $S \cup \{\alpha\}$, and so $|R| \leq |S|$. Further, $k(S \cup \{\alpha\})/k(R)$ is algebraic, and since $F/k(S)$ is algebraic, we have that $F/k(R)$ is algebraic. Thus, we may now apply the previous paragraph to R . ■

Corollary 5.4.36. *If F/k has a finite transcendence basis, then all transcendence bases have the same cardinality.*

It is in fact always true that all transcendence bases must have the same cardinality, even when we are not working in the finite case, though this doesn't really matter to us. This fact allows us to define the analog of dimension.

Definition 5.4.37. If F/k is a field extension and S is a transcendence basis for F over k , the *transcendence degree* of F/k is $\text{trdeg}_k F = |S|$.

As we would expect, transcendence degree is additive.

Proposition 5.4.38. Let $F/E/k$ be a tower of field extensions. Then $\text{trdeg}_k F$ is finite if and only if $\text{trdeg}_E F$ and $\text{trdeg}_k E$ are finite. In this case, $\text{trdeg}_k F = \text{trdeg}_E F + \text{trdeg}_k E$.

Proof. If $\text{trdeg}_k F$ is finite, then $\text{trdeg}_k E$ is finite (since elements of E that are algebraically independent over k are in particular elements of F that are algebraically independent over k) and $\text{trdeg}_E F$ is finite (since elements of F that are algebraically independent over E are in particular algebraically independent over k).

Conversely, suppose $\text{trdeg}_k E$ and $\text{trdeg}_E F$ are finite, and let S and T be transcendence bases of E/k and F/E respectively. Note that no element of T lies in E , so S and T are disjoint. Thus, since T is algebraically independent over $E \supseteq k(S)$, we have that $S \cup T$ is algebraically independent over k .

Now, we want to show that $F/k(S \cup T)$ is algebraic. Let A consist of all elements of F that are algebraic over $k(S \cup T)$, and note that $E \subseteq A$, since $E/k(S)$ is algebraic. This means that $E(T) \subseteq A$ as well. Since $F/E(T)$ is algebraic, this means F/A is algebraic, and so $F/k(S \cup T)$ is algebraic, since $A/k(S \cup T)$ is algebraic by definition. ■

We can use this fact about transcendence degrees to construct a totally transcendental extension that is not purely transcendental. Consider $\mathbb{Q}(x, \sqrt{-(x^2+1)})/\mathbb{Q}$, and suppose that $\frac{f(x)}{g(x)} + \frac{h(x)}{r(x)}\sqrt{-(x^2+1)}$ is algebraic over $\mathbb{Q}$. Then $\frac{f(x)}{g(x)} - \frac{h(x)}{r(x)}\sqrt{-(x^2+1)}$ is also algebraic over $\mathbb{Q}$, so that $\frac{h(x)}{r(x)}\sqrt{-(x^2+1)}$ is algebraic over $\mathbb{Q}$, and so $\frac{h(x)^2}{r(x)^2}(x^2+1) \in \mathbb{Q}(x)$ is algebraic over $\mathbb{Q}$. Since $\mathbb{Q}(x)/\mathbb{Q}$ is totally transcendental, this means that $\frac{h(x)^2}{r(x)^2}(x^2+1) \in \mathbb{Q}$, and so $h(x)^2(x^2+1) = cr(x)^2$ for some $c \in \mathbb{Q}$. But examining the factorization into irreducibles on both sides yields a contradiction, as the left hand side has an odd number of x^2+1 factors, while the right hand side has an even number.

Now, $\mathbb{Q}(x, \sqrt{-(x^2+1)})/\mathbb{Q}$ has transcendence degree 1, and so if it were purely transcendental, we could write $\mathbb{Q}(x, \sqrt{-(x^2+1)}) = \mathbb{Q}(\gamma)$ for some γ , necessarily transcendental over $\mathbb{Q}$. Writing $x = p/q$ and $\sqrt{-(x^2+1)} = u/v$ for $p, q, u, v \in \mathbb{Q}[\gamma]$ and using the fact that $x^2 + \sqrt{-(x^2+1)}^2 + 1 = 0$, we have that $p^2v^2 + u^2q^2 + v^2q^2 = 0$. Since γ is transcendental over $\mathbb{Q}$, we can view this as a polynomial identity, and note that $v, q \neq 0$. Thus, taking $s \in \mathbb{Q}$ that is not a root of q or v , we have that

$$0 = p(s)^2v(s)^2 + u(s)^2q(s)^2 + v(s)^2q(s)^2 \geq v(s)^2q(s)^2 > 0,$$

a contradiction. This means $\mathbb{Q}(x, \sqrt{-(x^2+1)})/\mathbb{Q}$ is not purely transcendental.

Proposition 5.4.39. Let F/k be a field extension, $S \subseteq F$ be algebraically independent over k , and $F/A/k$ be an intermediate field such that A/k is algebraic.

- (1) S is algebraically independent over A
- (2) A consists of precisely the elements of $A(S)$ that are algebraic over k
- (3) If $F/k(S)$ or A/k is finite, then $[A : k] = [A(S) : k(S)] \leq [F : k(S)]$

Proof.

- (1) Clearly it suffices to prove this in the case S is finite, say $|S| = n$. Let m be the size of the largest subset of S that is algebraically independent over A . Then $\text{trdeg}_A A(S) = m$ and, since $\text{trdeg}_k A = 0$, we have that $\text{trdeg}_k A(S) = m$. But

$$m = \text{trdeg}_k A(S) = \text{trdeg}_{k(S)} A(S) + \text{trdeg}_k k(S) \geq \text{trdeg}_k k(S) = n,$$

so $m = n$.

- (2) Since $A(S)/A$ is purely transcendental, it is in particular totally transcendental. Since elements of $A(S)$ that are algebraic over k are in particular algebraic over A , the result follows.
- (3) If $F/k(S)$ is finite, we first show A/k is finite. Otherwise, find a chain $k \subsetneq A_1 \subsetneq A_2 \subsetneq \cdots$ of subfields of A (for example, by successively adjoining an element of A at each step). Set $L_i = A_i(S)$, and notice that part (2) applied to L_i gives $L_i \cap A = A_i$. This yields a chain $k(S) \subsetneq L_1 \subsetneq L_2 \subsetneq \cdots$, contradicting the fact that $F/k(S)$ is finite. Thus, A/k is finite, say $[A : k] = m$.

Write $K = K(S)$. We induct on m , with $m = 1$ being trivial. If $m > 1$, let $\alpha \in A \setminus k$, so that part (1) means that S is algebraically independent over $k(\alpha)$. Since $k(\alpha)(S) = K(\alpha)$ and $[A : k(\alpha)] < m$, induction applied to $F/k(\alpha)$ tells us that $[A : k(\alpha)] = [A(S) : K(\alpha)]$. Thus, it suffices to show $[k(\alpha) : k] = [K(\alpha) : K]$.

We have that $m_\alpha^K \mid m_\alpha^k$. Working in a splitting field, the roots of m_α^K are roots of m_α^k , and are thus algebraic over k . Since the coefficients of m_α^K are elementary symmetric functions in these roots, they must also be algebraic over k . Since K/k is purely (and thus totally) transcendental, this means that the coefficients of m_α^K lie in k , and so $m_\alpha^K = m_\alpha^k$. Thus,

$$[k(\alpha) : k] = \deg m_\alpha^k = \deg m_\alpha^K = [K(\alpha) : K],$$

as desired. ■

We can use the theory we have developed so far, to prove the following remarkable theorem.

Theorem 5.4.40. *Let F/k be finitely generated, and $F/E/k$ be a subfield. Then E/k is finitely generated.*

Proof. Since F/k is finitely generated, $\text{trdeg}_k F$ is finite, and so $\text{trdeg}_k E$ is finite as well. If S is a (necessarily finite) transcendence basis of E over k and $K = k(S)$, it suffices to show E/K is finite.

Let T be a transcendence basis for F over K . Since F/k is finitely generated, $F/K(T)$ is finitely generated. Since $F/K(T)$ is also algebraic, it is finite. Since E/K is algebraic, Proposition 5.4.39 yields $[E : K] \leq [F : K(T)] < \infty$. ■

This result should be quite surprising, as the analogous result does not hold for groups, modules, or algebras.

We now present another application of the theory we have developed so far, applied to linearly disjoint subfields.

Definition 5.4.41. Let $k \subseteq K, L \subseteq F$ be fields. We say that K and L are *linearly disjoint* over k if every k -linearly independent subset of L is also K -linearly independent and if every k -linearly independent subset of K is also L -linearly independent.

Lemma 5.4.42. *Let $k \subseteq K, L \subseteq F$ be fields. If every k -linearly independent subset of L is also K -linearly independent, then every k -linearly independent subset of K is also L -linearly independent.*

Proof. Suppose that $\beta_1, \dots, \beta_n \in K$ are k -linearly independent, and that $\sum_{i=1}^n \alpha_i \beta_i = 0$, where $\alpha_1, \dots, \alpha_n \in L$. Pick a k -basis $\{\gamma_j\}$ for L , and write $\alpha_i = \sum_{j=1}^m a_{i,j} \gamma_j$, where $a_{i,j} \in k$. Then,

$$0 = \sum_{i=1}^n \alpha_i \beta_i = \sum_{i=1}^n \left(\sum_{j=1}^m a_{i,j} \gamma_j \right) \beta_i = \sum_{j=1}^m \left(\sum_{i=1}^n a_{i,j} \beta_i \right) \gamma_j.$$

Since the γ_j are k -linearly independent, they are also K -linearly independent, and so $\sum_{i=1}^n a_{i,j} \beta_i = 0$ for all j . But the β_i are k -linearly independent, so $a_{i,j} = 0$ for all i, j . Thus, the β_i are L -linearly independent. ■

Lemma 5.4.43. *Suppose we have $k \subseteq A, E \subseteq F$, where A/k is algebraic. If $[B : k] \leq [BE : E]$ for every $k \subseteq B \subseteq A$ with B/k finite, then E and A are linearly disjoint over k .*

Proof. Let $\alpha_1, \dots, \alpha_n \in A$ be k -linearly independent, and set $B = k(\alpha_1, \dots, \alpha_n)$. Then B/k is finite, and we can without loss of generality assume that the α_i span B as a k -vector space (by adding on additional α_i s if need be). Let R denote the E -span of the α_i , so that $B \subseteq R \subseteq BE$. Since the products $\alpha_i \alpha_j$ lie in $B \subseteq R$, this means R forms a ring, and so $BE = E[\alpha_1, \dots, \alpha_n] \subseteq R$. Thus, $R = BE$, and the α_i E -span BE . Since $[BE : E] \geq [B : k] = n$, we see that the α_i must therefore be E -linearly independent, as desired. ■

Proposition 5.4.44. *Let $k \subseteq A, E \subseteq F$, with A/k algebraic and E/k purely transcendental. Then E and A are linearly disjoint over k .*

Proof. It suffices to show $[B : k] \leq [BE : E]$ for all $A/B/k$ with B/k finite, which follows from Proposition 5.4.39. ■

5.4.6 Lüroth's Theorem

In this section, we will show that, if E/k is a field extension with $\text{trdeg}_k E = 1$ such that E is contained in a purely transcendental extension of k , then E is itself purely transcendental (the case when E is contained in a purely transcendental extension of transcendence degree 1 is Lüroth's theorem). We will first deal with the case of Lüroth's theorem, though we first need a lemma about polynomials.

Lemma 5.4.45. *Let $p, q \in k[x]$ be relatively prime and $u, v \in k[y]$ be relatively prime, and suppose that u and v are not both constants. Set $f(x, y) = u(y)p(x) - v(y)q(x)$, and suppose we can write $f = gh$ for $g \in k[x, y]$ and $h \in k[x]$. Then h is a constant.*

Proof. We will show $h \mid p, q$ in $k[x]$. Suppose without loss of generality that u is not constant, and let F/k be an extension containing a root α of u . Note that $v(\alpha)$ is nonzero, as m_α would otherwise divide both u and v . Now,

$$h(x)g(x, \alpha) = u(\alpha)p(x) - v(\alpha)q(x) = -v(\alpha)q(x),$$

and so h divides q in $F[x]$, and thus in $k[x]$. This also means that h divides $u(y)p(x)$ in $k[x, y]$, so that h divides p in $k(y)[x]$, and thus in $k[x]$. Thus, $h \mid p, q$ is constant. ■

Corollary 5.4.46. *Suppose $p, q \in k[x]$ are relatively prime, and $F = k(\beta)$, where β is transcendental over k . If $f(x) = p(x) - \beta q(x) \in F[x]$, then $\deg f = \max(\deg p, \deg q)$ and f is irreducible when $\deg f > 0$.*

Proof. Since $\beta \notin k$, none of the coefficients of $p(x)$ equal the coefficients of $\beta q(x)$, and so $\deg f = \max(\deg p, \deg q)$. Assume $\deg f > 0$.

Since $k(\beta) \cong k(y)$ for an indeterminate y , it suffices to show $f(x, y) = p(x) - yq(x)$ is irreducible in $k(y)[x]$. If $f(x, y)$ factored, we could use Gauss' lemma to write $f = gh$ for $g, h \in k[x, y]$. Since f has y -degree 1, we see that (without loss of generality) h must not involve y , and so lie in $k[x]$. But then the lemma applies with $u = 1$ and $v = y$, so h is constant and f is not reducible after all. ■

Theorem 5.4.47 (Lüroth). *Let $F/E/k$ be a tower of fields, where $F = k(\alpha)$ for some α transcendental over k . Then $E = k(\beta)$ for some element β .*

Proof. Suppose $E \supsetneq k$, and let $\beta \in E \setminus k$ be arbitrary. Write $\beta = p(\alpha)/q(\alpha)$ for relatively prime $p, q \in k[x]$, and set $d = \max(\deg p, \deg q)$. Then the polynomial $p(x) - \beta q(x) \in k(\beta)[x]$ is irreducible of degree d and has α as root. Thus, $[F : k(\beta)] = d$. In particular, F/E is finite, and so $[E : k(\beta)] = d/n$, where $n = [F : E]$. It therefore suffices to find an element β such that $d = n$.

Let g be the minimal polynomial of α over E . Since α is transcendental over k , some coefficient β of g must lie in $E \setminus k$. Writing $\beta = p(\alpha)/q(\alpha)$, we again have that α is a root of $p(x) - \beta q(x) \in E[x]$, and so $g \mid p(x) - \beta q(x)$ in $E[x]$. Thus, writing $\beta = p(\alpha)/q(\alpha)$, there is $h \in F[x]$ such that

$$q(\alpha)p(x) - p(\alpha)q(x) = g(x)h(x).$$

Replacing α with an indeterminate y and applying Gauss' lemma to clear denominators, we can write

$$q(y)p(x) - q(x)p(y) = g_0(x, y)h_0(x, y)$$

for $g_0, h_0 \in F[x, y]$ with $g_0 = \gamma g$ for some $\gamma \in k(y)$. Since g is monic, we must have that $\gamma \in k[y]$. Now, β is a coefficient of g , and so $\beta\gamma$ is a coefficient of $g_0 \in k[y][x]$, meaning $\beta\gamma \in k[y]$. If $\gamma = r(y)$ and $\beta\gamma = s(y)$, then $pr = qs$ and $p \mid s, q \mid r$. Thus,

$$d = \max(\deg p, \deg q) \leq \max(\deg s, \deg r).$$

Since $s(y)$ and $r(y)$ are both coefficients of g_0 , the y -degree of g_0 is at least $\max(\deg s, \deg r) \geq d$. But the y -degree of $q(y)p(x) - p(y)q(x)$ is at most d , and so g_0 must have y -degree d . This forces h_0 to have y -degree 0, which means that h_0 is constant. But this means $g_0(x, y) = -g_0(y, x)$, and so g_0 has x -degree d . But $\deg_x g_0 = \deg_x g = n$, so $d = n$, as desired. ■

Corollary 5.4.48. *If $k \subseteq E \subseteq k(\alpha)$, then $E = k(\beta)$ for some β .*

Proof. If α is algebraic over k , then Artin's Theorem 5.1.15 tells us that there are finitely many intermediate fields between k and $k(\alpha)$, and so finitely many intermediate fields between k and E , meaning E/k is simple. If α is transcendental over k , then this follows from Lüroth's theorem. ■

We'll now work to tackle the case when $\text{trdeg}_k F = 2$, which will then allow us to take on the general case.

Lemma 5.4.49. *Suppose that $f, g \in k[x, y]$ are k -linearly independent. Then $f(x, x^n)$ and $g(x, x^n)$ are k -linearly independent for all but finitely many $n \geq 0$.*

Proof. Write

$$\begin{aligned} f(x, y) &= p_r(x)y^r + p_{r-1}(x)y^{r-1} + \cdots + p_1(x)y + p_0(x) \\ g(x, y) &= q_s(x)y^s + q_{s-1}(x)y^{s-1} + \cdots + q_1(x)y + q_0(x) \end{aligned}$$

for $p_i, q_j \in k[x]$ and $p_r, q_s \neq 0$. Let $\deg p_r = u$ and $\deg q_s = v$, and write α and β for the leading coefficients of p_r and q_s respectively. Take

$$m = \max \left(\max_i \deg p_i, \max_j \deg q_j \right).$$

Then, for any $n > m$, the leading term of $f(x, x^n)$ is αx^{u+rn} and the leading term of $g(x, x^n)$ is βx^{v+sn} . If $f(x, x^n)$ and $g(x, x^n)$ are linearly dependent, this relation must therefore be $\beta f(x, x^n) - \alpha g(x, x^n)$. But $\beta f(x, y) - \alpha g(x, y)$ is nonzero, and so only has finitely many roots in $k(x)[y]$. Since x^n is a root when $n > m$ and $f(x, x^n)$ and $g(x, x^n)$ are linearly dependent, there can only be finitely many such n . ■

Lemma 5.4.50. *Let $\gamma \in k \subseteq k(x)$. Then there is a subring $R \subseteq k(x)$ and a homomorphism $\sigma : R \rightarrow k$ such that*

- $k[x] \subseteq R$
- $\sigma(x) = \gamma$
- $\sigma|_k = \text{id}$
- If $r \in R$ with $\sigma(r) \neq 0$, then $r^{-1} \in R$ and $\sigma(r^{-1}) = \sigma(r)^{-1}$

Proof. Let R consist of all elements of $k(x)$ that can be written in the form $p(x)/q(x)$ with $q(\gamma) \neq 0$. It is not hard to check that this is a ring, and clearly $k[x] \subseteq R$. Let σ be given by evaluation at γ , so that $\sigma(\gamma) = \gamma$ and $\sigma|_k = \text{id}$. If $r = p(x)/q(x) \in R$ with $\sigma(r) \neq 0$, then $p(\gamma) \neq 0$, so $r^{-1} = q(x)/p(x) \in R$ and $\sigma(r^{-1}) = \sigma(r)^{-1}$. ■

We're now ready to prove our generalization of Lüroth's theorem.

Theorem 5.4.51. *If $k \subseteq E \subseteq F = k(x, y)$, where x and y are indeterminates, with $\text{trdeg}_k E = 1$, then $E = k(\beta)$ for some $\beta \in E$.*

Proof. Since F/k is finitely generated, E/k is as well. Pick $\alpha \in E \setminus k$, so that α is transcendental over k . Since $\text{trdeg}_k E = 1$, this means $E/k(\alpha)$ is algebraic. Since $E/k(\alpha)$ is also finitely generated, we can write $E = k(\alpha)[\beta_1, \dots, \beta_r]$.

Write $\alpha = f(x, y)/g(x, y)$, and note that f and g are k -linearly independent since $\alpha \notin k$. Similarly, write $\beta_i = p_i(x, y)/q_i(x, y)$. Since $q_i \in k(x)[y]$ has finitely many roots, we can find n such that $q_i(x, x^n) \neq 0$ for all i . Further, we can pick n such that $f(x, x^n)$ and $g(x, x^n)$ are linearly independent over k . Set $\gamma = x^n$.

Now, use the above lemma to find a subring $k(x)[y] \subseteq R \subseteq k(x, y)$ and a homomorphism $\sigma : R \rightarrow k(x)$. This homomorphism then satisfies $\sigma(q(x, y)) = q(x, \gamma)$. In particular, $\sigma(f)$ and $\sigma(g)$ are linearly independent over k , and so $\sigma(\alpha) = \sigma(f)/\sigma(g) \notin k$ is transcendental over k . Now, for any nonzero polynomial h over k , $\sigma(h(\alpha)) = h(\sigma(\alpha)) \neq 0$, since $\sigma(\alpha)$ is transcendental over k , so that $h(\alpha)^{-1} \in R$. Thus, $k(\alpha) \subseteq R$. Additionally, $\sigma(q_i) = q_i(x, \gamma) \neq 0$, so that $\beta_i = p_i/q_i \in R$ for all i . This means that R contains $k(\alpha)[\beta_1, \dots, \beta_r] = E$. Now, restricting σ gives a field homomorphism $E \rightarrow k(x)$, and so Lüroth's theorem applied to $\sigma(E)$ shows that $E = k(\beta)$ for some β . ■

Theorem 5.4.52. *Let $k \subseteq E \subseteq F$, where F/k is purely transcendental and $\text{trdeg}_k E = 1$. Then $E = k(\beta)$ for some $\beta \in E$.*

Proof. Write $F = k(S)$, where S is algebraically independent over k . We first show that we can restrict to the case when S is finite. Let $T \subseteq S$ be a finite subset such that $E \cap k(T) \supsetneq k$, and note that such a T must exist (since $E \supsetneq k$ and each element is expressed using finitely many elements of S). If $\beta \in (E \cap k(T)) \setminus k$, then β is transcendental over k , and so $E/k(\beta)$ is algebraic, since $\text{trdeg}_k E = 1$. Thus, elements of E are algebraic over $k(T)$. However, $S \setminus T$ is algebraically independent over $k(T)$, and so $L = k(T)(S \setminus T)$ is purely transcendental over $k(T)$. Thus, $E \subseteq k(T)$, and so we may restrict to the case when S is finite.

We induct on $|S|$ (the base cases of 1 and 2 are covered by previous results). Let $\alpha \in S$, and write $k^* = k(\alpha)$ and $E^* = E(\alpha)$. If $k^* \cap E \supsetneq k$, then the result of the previous paragraph applies, and we may use Lüroth's theorem to obtain the result. Thus, assume $k^* \cap E = k$, so that $k^* \subsetneq E^*$.

Now,

$$\text{trdeg}_k k^* + \text{trdeg}_{k^*} E^* = \text{trdeg}_k E^* = \text{trdeg}_k E + \text{trdeg}_E E^*.$$

Since $\text{trdeg}_k k^* = \text{trdeg}_k E = 1$, we have that $\text{trdeg}_{k^*} E^* = \text{trdeg}_E E^*$. Since L/k^* is purely transcendental, $\text{trdeg}_{k^*} E^* \geq 1$. But since E^* is generated by a single element over E , $\text{trdeg}_E E^* \leq 1$, and so $\text{trdeg}_{k^*} E^* = 1$.

Now, L/k^* is purely transcendental, and generated by $S \setminus \{\alpha\}$. Thus, by induction, $E^* = k^*(\beta)$ for some β transcendental over $k^* = k(\alpha)$. Then α and β are algebraically independent over k , and $k \subseteq E \subseteq E^* \subseteq k(\alpha, \beta)$. Thus, the case when $\text{trdeg}_k F = 2$ applies, and so we are done. ■

5.4.7 The Artin-Schreier Theorem

In this section, we will investigate what happens when we have a finite proper extension L/k with L algebraically closed. As it will turn out, it is very rare that this happens, and in the case it does, k “resembles” $\mathbb{R}$ and L/k “resembles” $\mathbb{C}/\mathbb{R}$.

Suppose that L is algebraically closed and that $[L : k] = n$ is finite. If $f \in k[x]$ is irreducible, then it has a root α in L , and we see that

$$n = [L : k] \geq [k(\alpha) : k] = \deg f.$$

The following result will then allow us to rule out the possibility of k being imperfect, which will open the door to applying Galois theory.

Proposition 5.4.53. *Let $\text{char } k = p > 0$, and suppose that k is not perfect. Then there is an irreducible polynomial in $k[x]$ of degree p^e for all $e \geq 1$.*

Proof. Let $a \in k$ not be a p th power, and define $f(x) = x^{p^e} - a$. We claim that f is irreducible. Let α be a root of f in some splitting field, and let $g = m_\alpha$.

Suppose f factorizes into irreducibles as $f = g_1 \cdots g_m$. Since $f(x) = x^{p^e} - \alpha^{p^e} = (x - \alpha)^{p^e}$, each g_i must have α as a root, and so $g_i = g$ and $f = g^m$. In particular, $p^e = \deg f = m \deg g$, so m is a power of p . But $-a = f(0) = g(0)^m$, so that $a = (-g(0))^m$ is an m th power. Since a is not a p th power by construction, we must have that $m = 1$ and $f = g$ is irreducible. ■

We now use this fact to reduce to studying extensions of prime degree.

Lemma 5.4.54. *Let L/k with L algebraically closed and $1 < [L : k] < \infty$. Then either $[L : k] = 2$ and $L = k(\sqrt{-1})$, or we can find an intermediate field $L/F/k$ such that*

- L/F is Galois
- $[L : F] = p$ a prime
- -1 is a square in F

Proof. Since we know k must be perfect, L/k is separable. Since L is algebraically closed, L/k is normal, and thus Galois. Let $i \in L$ be a root of $x^2 + 1$. If $L = k(i)$, then $[L : k] = 2$ and we are done. If $L \supsetneq k(i)$, let $G = \text{Gal}(L/k(i))$, so that $|G| > 1$. We can then find a subgroup $H \leq G$ of prime order, and taking $F = \text{Fix}(H)$ suffices. ■

Now, if $[L : F] = p$, we will split into two cases: $\text{char } k \neq p$ and $\text{char } k = p$. We first deal with the case when $\text{char } k \neq p$.

Lemma 5.4.55. *If L is algebraically closed, $[L : F] = p$ a prime, and $\text{char } F \neq p$, then F contains a primitive p th root of unity.*

Proof. Since L is algebraically closed and has characteristic $\neq p$, it contains a primitive p th root of unity ζ . Then ζ is a root of $\frac{x^p - 1}{x - 1}$, and so $[F(\zeta) : F] = \deg m_\zeta \leq p - 1 < [L : F]$. Since $[L : F]$ is prime, there are no proper intermediate fields, and so $F = F(\zeta)$ and $\zeta \in F$. ■

Proposition 5.4.56. *Let L/F be a Galois extension of prime degree p , and assume that F contains a primitive p th root of unity. If $p = 2$, assume also that -1 is a square in F . Then there is $\alpha \in L$ such that $x^p - \alpha$ is irreducible. In particular, L is not algebraically closed.*

Proof. By Kummer's theorem 5.3.27, we can write $L = F(\alpha)$ for some α with $\alpha^p \in F$. To show $x^p - \alpha$ is irreducible, let β be a root in some extension of L , and note that it suffices to show $[L(\beta) : L] = p$. Kummer's theorem tells us that $[L(\beta) : L]$ divides p , and so suppose towards a contradiction that $[L(\beta) : L] = 1$, so that $\beta \in L$.

Write $\text{Gal}(L/F) = \langle \sigma \rangle$. Since $\beta^{p^2} = \alpha^p \in F$, $\delta = \sigma(\beta)/\beta$ satisfies $\delta^{p^2} = 1$. Thus, δ^p is a p th root of unity, and thus lies in F . If $\zeta = \sigma(\delta)/\delta$, then ζ is again a p th root of unity, and so lies in F . Now, a quick induction on i gives

$$\sigma^i(\beta) = \beta \delta^i \zeta^{i(i-1)/2}.$$

In particular, since $\sigma^p = \text{id}$, we have that $\delta^p \zeta^{p(p-1)/2} = 1$. If p is odd, this means that $\delta^p = 1$. If $p = 2$, then $\delta^2 = \pm 1$. Since we are assuming -1 is a square in F in this case, $\delta \in F$. Thus, $\zeta = \sigma(\delta)/\delta = 1$ and so $\delta^p = \delta^2 = 1$ in this case as well. Now,

$$\sigma(\alpha) = \sigma(\beta^p) = \sigma(\beta)^p = \beta^p \delta^p = \alpha,$$

so σ is the identity, contradicting the fact that σ generates G . Thus, $\beta \notin L$, as desired. $\blacksquare$

To deal with the case when $\text{char } k = p$, we first need to develop an analog of Kummer's theorem.

Lemma 5.4.57. *Let F/k be a finite Galois extension. Suppose that, for each $\sigma \in G$, we have $x_\sigma \in k$ such that $x_{\sigma\tau} = x_\sigma + \sigma(x_\tau)$. Then there is $\alpha \in k$ such that $x_\sigma = \sigma(\alpha) - \alpha$.*

Proof. Since the sum $\sum_{\sigma \in \text{Gal}(F/k)} \sigma$ is nonzero, we can pick β such that $\sum_{\sigma \in \text{Gal}(F/k)} \sigma(\beta) \neq 0$. Noticing that this sum lies in $\text{Fix}(\text{Gal}(F/k)) = k$, we may scale β to assume this sum is equal to -1 . Let $\alpha = \sum_{\sigma \in \text{Gal}(F/k)} x_\sigma \sigma(\beta)$. Now,

$$\sigma(\alpha) = \sigma \left(\sum_{\tau \in \text{Gal}(F/k)} x_\tau \tau(\beta) \right) = \sum_{\tau \in \text{Gal}(F/k)} \sigma(x_\tau) \sigma\tau(\beta) = \sum_{\tau \in \text{Gal}(F/k)} x_{\sigma\tau} \sigma\tau(\beta) - x_\sigma \sum_{\tau \in \text{Gal}(F/k)} \sigma\tau(\beta) = \alpha + x_\sigma.$$

Theorem 5.4.58. *Let F/k be a proper extension with $\text{char } k = p > 0$. Then F/k is Galois of degree p if and only if we can write $F = k(\alpha)$ for some α satisfying $\alpha^p - \alpha \in k$.*

Proof. Suppose first that F/k is Galois of degree p , and write $G = \text{Gal}(F/k)$. We can then find a homomorphism θ from G into the prime subfield of k (as they both have order p). Writing $x_\sigma = \theta(\sigma)$, we have that

$$x_{\sigma\tau} = x_\sigma + x_\tau = x_\sigma + \sigma(x_\tau),$$

since $x_\tau \in k$ is fixed by G . Thus, we can write $\theta(\sigma) = \sigma(\alpha) - \alpha$. In particular, since θ is injective, $\sigma(\alpha) \neq \alpha$ when $\sigma \neq \text{id}$, and so $\alpha \notin k$. Since $[F : k]$ is prime, this means $F = k(\alpha)$, so it remains to show $\alpha^p - \alpha \in k$. Let σ be the generator of G with $\theta(\sigma) = 1$. Then $\sigma(\alpha) = \alpha + 1$, and so

$$\sigma(\alpha^p - \alpha) = \sigma(\alpha)^p - \sigma(\alpha) = (\alpha + 1)^p - (\alpha + 1) = \alpha^p - \alpha.$$

This means $\alpha^p - \alpha$ is fixed by G , and thus lies in k .

Conversely, suppose $F = k(\alpha)$, where $a = \alpha^p - \alpha \in k$. If $f(x) = x^p - x - a$, then we see that $\alpha, \alpha + 1, \dots, \alpha + p - 1$ are distinct roots of f (since we are in characteristic p), and so f is separable and F is a splitting field for f over k . Thus, F/k is Galois. Since each $\alpha + n$ generates F over k , it follows that each irreducible factor of f has the same degree (namely, $[F : k]$). But since $\deg f = p$ is prime and doesn't split over k , this means that the only irreducible factor of f has degree p , and $[F : k] = p$. $\blacksquare$

We can now state the analog statement when $\text{char } k = p$.

Proposition 5.4.59. *Let L/F be a Galois extension of prime degree p , and suppose $\text{char } F = p$. Then there exists $\alpha \in L$ such that $x^p - x - \alpha \in L[x]$ is irreducible. In particular, L is not algebraically closed.*

Proof. Write $L = F(\beta)$ for some β with $b = \beta^p - \beta \in F$. Set $\alpha = \beta^{p-1}b$ and $f(x) = x^p - x - \alpha$.

Let γ be a root of f in some extension field of L . If $L(\gamma) \supsetneq L$, then the analog of Kummer's theorem tells us that $[L(\gamma) : L] = p$, and so in particular f is irreducible. Thus, assume towards a contradiction that $\gamma \in L$. Write $\gamma = g(\beta)$, where $g \in F[x]$ has degree $\leq p - 1$. Take $h \in F[x]$ such that $g(x)^p = h(x^p)$ (that is, the coefficients of h are the p th power of the corresponding coefficients of g). Then,

$$b\beta^{p-1} = \gamma^p - \gamma = g(\beta)^p - g(\beta) = h(\beta^p) - g(\beta).$$

Substituting $\beta^p = b + \beta$,

$$b\beta^{p-1} = h(b + \beta) - g(\beta).$$

We compare the coefficients of β^{p-1} . If $a \in F$ is the coefficient of x^{p-1} in $g(x)$, then a^p is the coefficient of x^{p-1} in $h(x)$, and thus in $h(x + b)$ (since $\deg g = \deg h \leq p - 1$, so x^{p-1} would be the leading term). Thus, $b = a^p - a$, and so a is a root of $x^p - x - b$ in F . But since $[F(\beta) : F] = p$, the minimal polynomial of β over F must be $x^p - x - b$, and so this is in particular irreducible over F , a contradiction. Thus, $\gamma \notin L$, as desired. ■

Putting this all together, we have shown the following.

Theorem 5.4.60. *Let L/k , where L is algebraically closed and $1 < [L : k] < \infty$. Then $[L : k] = 2$ and $L = k(\sqrt{-1})$.*

Proof. If this is not the case, then Lemma 5.4.54 allows us to find an intermediate field F such that L/F is Galois of prime degree and -1 is a square in F . Propositions 5.4.56 and 5.4.59 together yield a contradiction. ■

Corollary 5.4.61. *If L is algebraically closed, then $\text{Aut}(L)$ has no finite subgroup of order exceeding 2.*

Proof. If $G \leq \text{Aut}(L)$ is finite, then $L/\text{Fix}(G)$ has degree $|G|$. Thus, $|G| \leq 2$. ■

It is clear now how L/k must “resemble” $\mathbb{C}/\mathbb{R}$, since we are adjoining only a square root of -1 . We now turn to studying how k “resembles” $\mathbb{R}$.

Definition 5.4.62. A field is said to be *formally real* if -1 is not a sum of squares. A field is said to be *realclosed* if it is formally real and no proper algebraic extension is formally real.

Intuitively, a realclosed field is the same as $\mathbb{R}$, and a formally real field is a subfield of $\mathbb{R}$.

Lemma 5.4.63. *Let k be formally real.*

- (1) *No sum of nonzero squares in k is zero*
- (2) *$\text{char } k = 0$*

If k is realclosed and S is the set of nonzero squares in k , then

- (3) *S is closed under addition*
- (4) *$k = -S \sqcup \{0\} \sqcup S$*

Proof.

- (1) If $0 = a_1^2 + \cdots + a_n^2$ with $a_i \neq 0$ (so that $n \geq 2$ in particular), then $-1 = (a_2/a_1)^2 + \cdots + (a_n/a_1)^2$, a contradiction.
- (2) For any $n \geq 0$, we must have

$$\underbrace{1 + \cdots + 1}_n = \underbrace{1^2 + \cdots + 1^2}_n \neq 0.$$

- (3) Suppose $t = r^2 + s^2 \in k$ is not a square, and let α be a root of $x^2 - t$ in some extension field. Then $k(\alpha) \supsetneq k$, and so $k(\alpha)$ is not formally real. Write

$$-1 = \sum_{i=1}^n (a_i + b_i \alpha)^2 = \sum_{i=1}^n a_i^2 + t \sum_{i=1}^n b_i^2 + 2\alpha \sum_{i=1}^n a_i b_i.$$

Since $\alpha \notin k$, we must have

$$-1 = \sum_{i=1}^n a_i^2 + t \sum_{i=1}^n b_i^2 = \sum_{i=1}^n a_i^2 + (rb_i)^2 + (sb_i)^2$$

is a sum of squares, a contradiction. Thus, S is closed under addition.

- (4) If t is a nonsquare, the same argument as in (3) allows us to write $-1 = u + tv$, where u and v are sums of squares, and thus themselves squares. We necessarily have $v \neq 0$. Thus, $-t = 1/v + u/v$ is the sum of two squares, and so a square, meaning we have $k = -S \cup \{0\} \cup S$. Since -1 isn't a square in k , this union is disjoint. ■

Theorem 5.4.64 (Artin-Schreier). *Let L/k where L is algebraically closed and $1 < [L : k] < \infty$. Then $[L : k] = 2$, $L = k(\sqrt{-1})$, and k is realclosed.*

Proof. We have already seen that $[L : k] = 2$ and $L = k(\sqrt{-1})$. Write $i^2 = -1$ and S for the set of squares in k . We show S is closed under addition.

Let $r, s \in k$ be nonzero, and let $a + bi \in L$ be a square root for $r + si$, since L is algebraically closed. Then

$$r^2 + s^2 = (r + si)(r - si) = (a + bi)^2(a - bi)^2 = ((a + bi)(a - bi))^2 = (a^2 + b^2)^2,$$

so S is indeed closed under addition. In particular, since $i \notin k$, we see that $-1 \notin S$, and so k is formally real. Since $[L : k] = 2$ and L is algebraically closed, there are no proper algebraic extensions of k except for L , and so k is realclosed (since $i^2 = -1$ in L). ■

We can take this correspondence further: if k is realclosed, it turns out $k(\sqrt{-1})$ is algebraically closed. To prove this, we note that our proof of the Fundamental Theorem of Algebra in §5.3.1 only requires that $\mathbb{C}$ is closed under square roots and that any odd polynomial has a root in $\mathbb{R}$. We can prove these facts for *any* realclosed field.

Lemma 5.4.65. *If k is realclosed and $L = k(\sqrt{-1})$, then every element of L is a square in L .*

Proof. Write $i = \sqrt{-1}$. Note first that every element of k is a square in L : if $t \in k$ is not a square in k , then $-t = s^2$ for some $s \in k$, and so $t = (is)^2$. Now, let $a + bi \in L$ be arbitrary.

It suffices to find $x, y \in L$ with $(x + iy)^2 = a + bi$, so we want $x^2 - y^2 = a$ and $2xy = b$. Set $a^2 + b^2 = r^2$ for $r \in k$ (since squares in k are closed under addition), and let $x \in L$ be such that $x^2 = \frac{a+r}{2}$. If $x = 0$, then $r = -a$ and $b = 0$, so we are done. Thus, assume $x \neq 0$, and set $y = b/2x$, so we automatically have $2xy = b$. Now,

$$4x^2(x^2 - y^2) = (2x^2)^2 - (2xy)^2 = (a + r)^2 - b^2,$$

so that

$$x^2 - y^2 = \frac{(a + r)^2 + b^2}{4x^2} = \frac{2a^2 + 2ar}{2(a + r)} = a,$$

as desired. ■

Lemma 5.4.66. *Let k be formally real, and let F/k be finite, where $[F : k]$ is odd. Then F is formally real. In particular, if k is realclosed, then every odd-degree polynomial has a root in k .*

Proof. We induct on $n = [F : k]$, with $n = 1$ being trivial. Suppose $n > 1$, and let $\alpha \in F \setminus k$. If $[F : k(\alpha)], [k(\alpha) : k] < n$, then induction means that $k(\alpha)$ is formally real, and so induction again means that F is formally real. Thus, assume $F = k(\alpha)$.

Suppose that F is not formally real, and write $-1 = \sum g_i(\alpha)^2$ where $\deg g_i < n$. Let $h(x) = 1 + \sum g_i(x)^2$, so that $h(\alpha) = 0$ and $m_\alpha \mid h$. If $m = \max_i \deg g_i$, then $\deg h = 2m$, as the x^{2m} coefficient is a sum of squares, and so nonzero. Thus, $\deg(h/m_\alpha) = 2m - n$ is odd and $< n$. If $f \mid h/m_\alpha$ is an irreducible factor of odd degree, and β is a root of f in some extension of k , then induction applied to $k(\beta)/k$ shows that $k(\beta)$ is formally real. But $f \mid h$, so $h(\beta) = 0$ and -1 is a sum of squares in $k(\beta)$, a contradiction. ■

Thus, replicating our proof of the Fundamental Theorem of Algebra gives the following.

Theorem 5.4.67. *If k is realclosed, then $k(\sqrt{-1})$ is algebraically closed.*

Theorem 5.4.68. *A field k is realclosed if and only if there exists a proper finite extension L/k with L algebraically closed.*

5.5 Infinite Galois Theory

5.5.1 Infinite Galois Extensions

So far, we have developed Galois theory for *finite* extensions. We will now generalize the theory to possibly infinite algebraic extensions. We'll typically reserve F for finite extensions, and use L for general extensions.

In the past, whenever we wanted polynomials $f_1, \dots, f_n$ to split, we would pass to the splitting field of their product $f_1 \cdots f_n$. When dealing with possibly infinitely many polynomials, this won't work, and so we'll need to develop a slightly stronger notion of splitting fields.

Definition 5.5.1. A field extension L/k is said to be a splitting field of a (possibly infinite) set of polynomials $S \subseteq k[x]$ if every $f \in S$ splits and L is generated over k by the roots of elements of S .

The uniqueness of splitting fields 5.1.19 was one of our most powerful tools in developing Galois theory, and thankfully it still holds in the infinite case.

Theorem 5.5.2. Let $\iota : k_1 \rightarrow k_2$ be an isomorphism of fields. Let $S_i \subseteq k_i[x]$, and suppose that ι yields a bijection between S_1 and S_2 . If L_i/k_i is a splitting field of S_i , then there is an isomorphism $L_1 \cong L_2$ restricting to ι on k_1 .

Proof. We apply Zorn's lemma. Consider the poset

$$\{(F, \eta) : F \subseteq L_1, \eta : F \rightarrow L_2, \eta|_{k_1} = \iota\}$$

partially ordered by declaring $(F, \eta) \preceq (F', \eta')$ if $F \subseteq F'$ and $\eta'|_F = \eta$. We claim that every chain has an upper bound: indeed if $\{(F_i, \eta_i)\}_{i \in I}$ is a chain, set $F = \bigcup_{i \in I} F_i$ and $\eta(\alpha \in F_i) = \eta_i(\alpha)$. It is not hard to check that η is well-defined and that (F, η) is an upper bound for the chain. Thus, by Zorn's lemma, we can find a maximal element (M, σ) .

We first show that $M = L_1$. Suppose otherwise that $M \subsetneq L_1$, and let $\alpha \in L_1 \setminus M$ be a root of $f \in S_1$ (since M cannot contain all roots of members of S_1). Since f does not split over M , $\sigma(f) = \iota(f)$ does not split over $\sigma(M)$, and so we can find $\beta \in L_2 \setminus \sigma(M)$ that is a root of $\iota(f)$. But we can then extend to a map $M(\alpha) \rightarrow \sigma(M)(\beta)$ by sending $\alpha \rightarrow \beta$, contradicting the maximality of M . Thus, $M = L_1$.

Now, we show that σ is indeed an isomorphism¹⁵. It suffices to show that $\sigma(L_1)$ contains all roots of all elements of S_2 . For any $g \in S_2$ of degree n , we can write $g = \iota(f)$ for $f \in S_1$, and we know that f splits over L_1 . Thus, the n roots of f must be sent to n (necessarily distinct) roots of g , and so $\sigma(L_1)$ contains all roots of g . Thus, σ is surjective, and so an isomorphism. ■

The remarkable thing is that we can in fact salvage Theorem 5.2.6 for infinite extensions.

Theorem 5.5.3. Let L/k be algebraic. The following are equivalent.

- (1) L/k is Galois
- (2) We can write $L = \bigcup L_i$, where L_i/k is a finite Galois extension for each i
- (3) L is the splitting field of some collection of polynomials in $k[x]$
- (4) $\text{Fix}(\text{Aut}(L/k)) = k$

Proof.

- (1) $\Rightarrow$ (2) For each $\alpha \in L$, let L_α be the subfield generated by the roots of m_α . Since L/k is normal, m_α splits in L , so L_α is a splitting field of m_α over k . Since L/k is separable, m_α is separable, so L_α/k is Galois (and finite, since $[L_\alpha : k] \leq (\deg m_\alpha)!$). Clearly $L = \bigcup_{\alpha \in L} L_\alpha$.
- (2) $\Rightarrow$ (3) Let L_i/k be the splitting field of the separable polynomial f_i , and let $S \subseteq k[x]$ consist of all the f_i . Then every element of S splits in L , and L is generated by the roots of elements of S (since L_i is generated by the roots of elements of f_i), so L/k is a splitting field for S .
- (3) $\Rightarrow$ (4) Let L be the splitting field of $S \subseteq k[x]$ over k , and suppose $\alpha \in \text{Fix}(\text{Aut}(L/k))$. We only need finitely many roots of elements of S to express α , so we can find a finite $S' \subseteq S$ such that α is contained in the splitting field L' of S' over k . Then L'/k is the splitting field of the separable polynomial $\prod_{f \in S'} f$, and so is finite and Galois. Since L is the splitting field of S over k , it is also the splitting field of S over L' , and so uniqueness of splitting fields tells us that every element of $\text{Aut}(L'/k)$ extends to an element of $\text{Aut}(L/k)$. Thus, $\alpha \in \text{Fix}(\text{Aut}(L'/k)) = k$.
- (4) $\Rightarrow$ (1) Fix $\alpha \in L$, and note that $A = \{\sigma(\alpha) : \sigma \in \text{Aut}(L/k)\}$ must be finite (as it is a subset of the roots of m_α). Let $f(x) = \prod_{\beta \in A} (x - \beta)$, and note that the coefficients of f are elementary symmetric polynomials in the elements of A . Since every element of $\text{Aut}(L/k)$ must permute A , the coefficients of f lie in $\text{Fix}(\text{Aut}(L/k)) = k$. In particular, $m_\alpha \mid f$, and so $f = m_\alpha$, since every root of f is a root of m_α . Thus, m_α splits with distinct roots in L , so L/k is normal and separable. ■

¹⁵This was never a problem before because we had a finite dimensionality hypothesis, so the injectivity of σ forced it to be an isomorphism

In particular, we see that if $F/E/k$ is an intermediate field of a Galois extension F/k , then F/E is Galois, as F is still the splitting field of the relevant set of polynomials over E . We remark that Proposition 5.2.4 shows that $\text{Gal}(F/k)$ must be infinite for infinite Galois extensions F/k . The proof that (4) $\Rightarrow$ (1) also shows that $\{\sigma(\alpha) : \sigma \in \text{Gal}(F/k)\}$ consists of all roots of m_α .

Example. The extension $\overline{\mathbb{Q}}/\mathbb{Q}$ is Galois, as it is the splitting field of $\mathbb{Q}[x]$. Indeed, $\overline{k}/k$ is Galois for any perfect field k (in particular, $\overline{\mathbb{F}_q}/\mathbb{F}_q$ is Galois for all prime powers q). An element of $\text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$ is determined by how it acts the copy of $\mathbb{F}_{q^n}$ for each n . Since $\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) \cong \mathbb{Z}/n\mathbb{Z}$, we can view elements as determined by a sequence $(a_1, a_2, a_3, \dots)$, where a_i is considered mod n . Since $\mathbb{F}_{q^n} \subseteq \mathbb{F}_{q^m}$ when $n \mid m$, we must have that $a_n \equiv a_m \pmod{n}$. Every sequence satisfying these compatibility relations in turn gives rise to an element of $\text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$. Note that we are taking the inverse limit $\varprojlim \mathbb{Z}/n\mathbb{Z}$.

Example. We'll define $\mathbb{Q}_\sqrt{}/\mathbb{Q}$ to be the splitting field of $\{x^2 - r : r \in \mathbb{Q}\}$. Note that we can equivalently describe

$$\mathbb{Q}_\sqrt{} = \mathbb{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \dots).$$

This gives us a concrete way to describe $\text{Gal}(\mathbb{Q}_\sqrt{}/\mathbb{Q})$ by determining how we map $\sqrt{p} \rightarrow \pm\sqrt{p}$. Thus, $\text{Gal}(\mathbb{Q}_\sqrt{}/\mathbb{Q}) \cong \{\pm 1\}^{\mathbb{N}}$.

Example. Let $\mathbb{Q}(\zeta_{p^\infty}) = \bigcup_n \mathbb{Q}(\zeta_{p^n})$ for a prime p . We can describe an element $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q})$ by describing how it acts on each $\mathbb{Q}(\zeta_{p^n})$. We must have that $\sigma(\zeta_{p^n}) = \zeta_{p^n}^{a_n}$ for some $a_n \in (\mathbb{Z}/p^n\mathbb{Z})^\times$, and σ is uniquely determined by the sequence $(a_1, a_2, \dots)$. However, we must also have

$$\zeta_{p^n}^{a_n} = \sigma(\zeta_{p^n}) = \sigma(\zeta_{p^{n+1}}^p) = \sigma(\zeta_{p^{n+1}})^p = \zeta_{p^{n+1}}^{a_{n+1}p} = \zeta_{p^n}^{a_{n+1}},$$

so that $a_n \equiv a_{n+1} \pmod{p^n}$. It's not hard to see that every sequence satisfying these compatibility relations gives rise to an element of $\text{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q})$. We are taking the limit $\varprojlim (\mathbb{Z}/p^n\mathbb{Z})^\times$.¹⁶

Unfortunately, we can see already that the Galois correspondence must break down. Let $H \leq \text{Gal}(\mathbb{Q}_\sqrt{}/\mathbb{Q})$ consist of the automorphisms σ such that $\sigma(\sqrt{p}) = \sqrt{p}$ for all but finitely many $p \in \{-1, 2, 3, 5, \dots\}$. Then $H \leq \text{Gal}(\mathbb{Q}_\sqrt{}/\mathbb{Q})$, but we do have that $\text{Fix}(H) = \mathbb{Q}$. The solution to this will be to put a topology on the Galois group, and then to only think about closed subsets.

5.5.2 The Krull Topology

Intuitively, we want to declare σ and τ to be “close” if $\sigma|_E = \tau|_E$ for some intermediate field E . We can in fact define these sets fairly intrinsically.

Lemma 5.5.4. *Let L/k be a Galois extension.*

(1) *For $\sigma \in \text{Gal}(L/k)$ and an intermediate field E ,*

$$\sigma \text{Gal}(L/E) = \{\tau \in \text{Gal}(L/k) : \sigma|_E = \tau|_E\}.$$

(2) *If F is an intermediate field with F/k finite, then*

$$[\text{Gal}(L/k) : \text{Gal}(L/F)] = [F : k].$$

Proof.

- (1) Since $\text{Gal}(L/E)$ is precisely the set of automorphisms that restrict to the identity on E , $\sigma \text{Gal}(L/E)$ is precisely the set of automorphisms that restrict to σ on E .
- (2) $[\text{Gal}(L/k) : \text{Gal}(L/F)]$ is the number of left cosets of $\text{Gal}(L/F)$. By (1), this is the number of distinct embeddings $F \hookrightarrow L$. Since F/k is separable, there are exactly $[F : k]$ such embeddings.

■

Definition 5.5.5. The *Krull Topology* on $\text{Gal}(L/k)$ is the topology whose base open sets are given by $\sigma \text{Gal}(L/F)$ for intermediate fields $L/F/k$ with F/k finite.

¹⁶This is the group of units of the p -adics.

We should verify that these open sets indeed do form a basis, and so generate a valid topology. If $\sigma \text{Gal}(L/F)$ and $\sigma' \text{Gal}(L/F')$ intersect nontrivially, we can pick τ in their intersection, so that

$$\tau \text{Gal}(L/FF') \subseteq \tau \text{Gal}(L/F) \cap \tau \text{Gal}(L/F') = \sigma \text{Gal}(L/F) \cap \sigma' \text{Gal}(L/F').$$

Note that if $F \subseteq F'$, then $\sigma \text{Gal}(L/F) \supseteq \sigma \text{Gal}(L/F')$, so “smaller” open sets correspond to agreeing on “larger” extensions of k .

Example. The Krull topology on $\text{Gal}(\mathbb{Q}_{\sqrt{}}/\mathbb{Q}) \cong \{\pm 1\}^{\mathbb{N}}$ is the product topology: a base open set around σ consists of all τ that agree with σ on a given finite set of coordinates, so that any open set only has freedom in finitely many coordinates.

Example. The Krull topology on $\text{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q})$ declares σ and τ to be close if the corresponding sequences $(a_1, a_2, \dots)$ and $(b_1, b_2, \dots)$ agree on the first few coordinates.

Theorem 5.5.6. *Let L/k be Galois.*

- (1) *The Krull topology makes $\text{Gal}(L/k)$ into a topological group.*
- (2) *If L/k is finite, the Krull topology is discrete.*
- (3) *The Krull topology is Hausdorff.*
- (4) *$\text{Gal}(L/E)$ is closed for all intermediate fields $L/E/k$.*
- (5) *If $H \leq \text{Gal}(L/k)$, then the closure of H is given by $\overline{H} = \text{Gal}(L/\text{Fix}(H))$.*

Proof.

- (1) We will first show that multiplication is continuous at $(\sigma, \tau) \in \text{Gal}(L/k) \times \text{Gal}(L/k)$. Pick a base open set $\sigma\tau \text{Gal}(L/F)$. Note we may shrink $\sigma\tau \text{Gal}(L/F)$ (i.e. make F larger) to assume that F/k is Galois. Now, we claim the base open set $\sigma \text{Gal}(L/F) \times \tau \text{Gal}(L/F)$ lies in the preimage of $\sigma\tau \text{Gal}(L/F)$ under multiplication. Indeed, if $f|_F = \sigma$ and $g|_F = \tau$, then

$$fg(\alpha) = f(g(\alpha)) = f(\tau(\alpha)) = \sigma(\tau(\alpha)) = \sigma\tau(\alpha)$$

for all $\alpha \in F$, where we are using the fact that $\tau(\alpha) \in F$ since F is Galois. Thus, multiplication is continuous.

We'll now show that inversion is continuous at $\sigma \in \text{Gal}(L/k)$. Take a base open set $\sigma^{-1} \text{Gal}(L/F)$, and once again shrink to assume F is Galois. If $f \in \sigma \text{Gal}(L/F)$, then $f|_F = \sigma|_F$. Since F is Galois, f and σ restrict to automorphisms of F , and so we also have $f^{-1}|_F = \sigma^{-1}|_F$. Thus, $\sigma \text{Gal}(L/F)$ lies in the preimage of $\sigma^{-1} \text{Gal}(L/F)$, and so inversion is continuous.

- (2) If L/k is finite, then $\sigma \text{Gal}(L/L) = \{\sigma\}$ is open for all σ .
- (3) Suppose that $\sigma \neq \tau$, and let $\sigma(\alpha) \neq \tau(\alpha)$. Then σ and τ disagree on $k(\alpha)$, and so $\sigma \text{Gal}(L/k(\alpha))$ and $\tau \text{Gal}(L/k(\alpha))$ are disjoint open sets containing σ and τ respectively.
- (4) We will show that the complement is open. If $\sigma \notin \text{Gal}(L/E)$, then σ does not restrict to the identity on E , say $\sigma(\alpha) \neq \alpha$. Then no element of the base open set $\sigma \text{Gal}(L/k(\alpha))$ fixes α , and so $\sigma \text{Gal}(L/k(\alpha)) \subseteq \text{Gal}(L/E)^c$. Thus, $\text{Gal}(L/E)^c$ is open, so $\text{Gal}(L/E)$ is closed.
- (5) Since $\text{Gal}(L/\text{Fix}(H))$ is closed and contains H , we clearly have $\overline{H} \subseteq \text{Gal}(L/\text{Fix}(H))$. Suppose now that $\sigma \in \text{Gal}(L/\text{Fix}(H))$, and take a base open set $\sigma \text{Gal}(L/F)$. By shrinking the set if necessary, assume F/k is a finite Galois extension.

Since F/k is Galois, automorphisms of L restrict to automorphisms of F , and so we have $H|_F \leq \text{Gal}(F/k)$. Since $\sigma|_F$ fixes $\text{Fix}(H|_F)$, we see that $\sigma|_F \in \text{Gal}(F/\text{Fix}(H|_F)) = H|_F$ by finite Galois theory, so there is $h \in H$ with $\sigma|_F = h|_F$. In particular, $\sigma \text{Gal}(L/F) \cap H \ni h$ is nonempty. Since $\sigma \text{Gal}(L/F)$ was an arbitrary base open set, this means $\sigma \in \overline{H}$. ■

We end with some results about how the Krull topology behaves across field extensions.

Theorem 5.5.7. *Let L/k be Galois, and let E be an intermediate field.*

- (1) The subspace topology on $\text{Gal}(L/E)$ equals the Krull topology on $\text{Gal}(L/E)$.
- (2) If E/k is Galois, the restriction map $\text{Gal}(L/k) \rightarrow \text{Gal}(E/k)$ is a continuous homomorphism.

Proof.

- (1) Let $\sigma \text{Gal}(L/F) \subseteq \text{Gal}(L/E)$ be a base open set in the Krull topology, with F/E finite. If we write $F = E(\alpha_1, \dots, \alpha_n)$ and $F^* = k(\alpha_1, \dots, \alpha_n)$, then F^*/k is finite. Now, $\sigma \text{Gal}(L/F^*)$ is open in the Krull topology on $\text{Gal}(L/k)$, and consists of all τ that agree with σ on F^* . Thus, $\sigma \text{Gal}(L/F^*) \cap \text{Gal}(L/E)$ consists of all $\tau \in \text{Gal}(L/E)$ that agree with σ on F^* , and thus on F (since they agree with σ on $\alpha_1, \dots, \alpha_n$). That is,

$$\sigma \text{Gal}(L/F) = \sigma \text{Gal}(L/F^*) \cap \text{Gal}(L/E)$$

is open in the subspace topology.

Conversely, suppose that $\sigma \text{Gal}(L/F)$ is a base open set of the Krull topology on $\text{Gal}(L/k)$. We want to show $\sigma \text{Gal}(L/F) \cap \text{Gal}(L/E)$ is open in the Krull topology on $\text{Gal}(L/E)$. Note EF/E is finite. Since $\sigma \text{Gal}(L/F) \cap \text{Gal}(L/E)$ consists precisely of the elements of $\text{Gal}(L/E)$ that agree with σ on F , it consists precisely of the elements of $\text{Gal}(L/E)$ that agree with σ on EF . Thus,

$$\sigma \text{Gal}(L/F) \cap \text{Gal}(L/E) = \sigma \text{Gal}(L/EF)$$

is open in the Krull topology on $\text{Gal}(L/E)$.

- (2) Let $\sigma \text{Gal}(E/F)$ be a base open set of the Krull topology on $\text{Gal}(E/F)$, where F/k is finite. Then its preimage consists of all elements of $\text{Gal}(L/k)$ agreeing with σ on F , which is to say that its preimage is $\sigma \text{Gal}(L/F)$, which is open. ■

5.5.3 The Infinite Galois Correspondence

We're now ready to state and prove the Galois correspondence for infinite extensions.

Theorem 5.5.8. *Let L/k be a Galois extension.*

- (1) *There is an inclusion-reversing bijection*

$$\begin{array}{ccc} \{k \subseteq E \subseteq L\} & \longleftrightarrow & \left\{ \begin{array}{c} H \leq \text{Gal}(L/k) \\ \text{closed} \end{array} \right\} \\ E & \longmapsto & \text{Gal}(L/E) \\ \text{Fix}(H) & \longleftarrow & H \end{array}$$

- (2) *This bijection is a bijection of lattices. That is, for $k \subseteq E, K \subseteq L$ and closed subgroups $H, N \leq \text{Gal}(L/k)$,*

$$\text{Gal}(L/EK) = \text{Gal}(L/E) \cap \text{Gal}(L/K) \quad \text{Gal}(L/E \cap K) = \overline{\langle \text{Gal}(L/E), \text{Gal}(L/K) \rangle}$$

$$\text{Fix}(H \cap N) = \text{Fix}(H) \text{Fix}(N) \quad \text{Fix}(\overline{\langle H, N \rangle}) = \text{Fix}(H) \cap \text{Fix}(N)$$

- (3) *For $H \leq \text{Gal}(L/k)$, $\overline{H} = \text{Gal}(L/\text{Fix}(H))$ and $\text{Fix}(H) = \text{Fix}(\overline{H})$.*
- (4) *If $k \subseteq E \subseteq L$ is an intermediate field, then E/k is Galois if and only if $\text{Gal}(L/E) \trianglelefteq \text{Gal}(L/k)$, and $\text{Gal}(E/k) \cong \text{Gal}(L/k)/\text{Gal}(L/E)$ as topological groups.*

Proof.

- (1) Clearly these maps are inclusion reversing. If $H \leq \text{Gal}(L/k)$ is closed, then $\text{Gal}(L/\text{Fix}(H)) = \overline{H} = H$. For an intermediate field $k \subseteq E \subseteq L$, L/E is Galois, and so $E = \text{Fix}(\text{Gal}(L/E))$.

- (2) We prove the top two items, as the bottom two follow by applying the bijection. Since $\text{Gal}(L/EK)$ fixes both E and K , we clearly have

$$\text{Gal}(L/EK) \leq \text{Gal}(L/E) \cap \text{Gal}(L/K).$$

Taking fixed fields,

$$EK = \text{Fix}(\text{Gal}(L/EK)) \supseteq \text{Fix}(\text{Gal}(L/E) \cap \text{Gal}(L/K)) \supseteq E, K.$$

Since EK is the smallest field containing both E , and K , we must have equality. Similarly, we clearly have

$$\text{Gal}(L/E \cap K) \geq \langle \text{Gal}(L/E), \text{Gal}(L/K) \rangle,$$

and so

$$\text{Gal}(L/E \cap K) \geq \overline{\langle \text{Gal}(L/E), \text{Gal}(L/K) \rangle}.$$

Taking fixed fields,

$$E \cap K = \text{Fix}(\text{Gal}(L/E \cap K)) \subseteq \text{Fix}(\overline{\langle \text{Gal}(L/E), \text{Gal}(L/K) \rangle}) \subseteq E, K,$$

so we have equality (note that we need both subgroups to be closed in order to conclude equality).

- (3) We have already shown $\overline{H} = \text{Gal}(L/\text{Fix}(H))$. Taking fixed fields again gives $\text{Fix}(\overline{H}) = \text{Fix}(H)$.
- (4) The algebraic argument here is no different from in the finite case, so we only give an overview: $\text{Gal}(L/E) \trianglelefteq \text{Gal}(L/k)$ if and only if E is fixed (as a set) by every automorphism of L , if and only if E/k is normal, if and only if E/k is Galois. If E/k is Galois, restriction gives a map $\text{Gal}(L/k) \rightarrow \text{Gal}(E/k)$ with kernel $\text{Gal}(L/E)$. Now, this map is continuous with respect to the Krull topologies, and so the induced map $\text{Gal}(L/k)/\text{Gal}(L/E) \rightarrow \text{Gal}(E/k)$ is continuous. We show that this is an open map. Indeed, if F/k is finite, then the image of $\sigma \text{Gal}(L/F)$ is all those elements of E/k that agree with σ on $E \cap F$, which is to say the image of $\sigma \text{Gal}(L/F)$ is $\sigma \text{Gal}(E/E \cap F)$, which is open.

■

We also have an analog of Proposition 5.2.11.

Proposition 5.5.9. *If L/k is Galois and E/k is any extension (where L and E are contained in some larger field), then LE/E is Galois and $\text{Gal}(LE/E) \cong \text{Gal}(L/L \cap E)$ as topological groups.*

Proof. Since L is the splitting field of some collection of polynomials over k , LE is the splitting field of that same collection of polynomials over E , and so LE/E is Galois. Now, the map $\text{Gal}(LE/E) \rightarrow \text{Gal}(L/k)$ given by restriction to L is injective (any element fixing L and E must fix all of LE) and has image $\text{Gal}(L/L \cap E)$. We claim this map is also a homeomorphism. If $F/L \cap E$ is finite, the preimage of $\sigma \text{Gal}(L/F)$ is $\sigma \text{Gal}(LE/FE)$ and FE/E is finite. If F/E is finite, the image of $\sigma \text{Gal}(LE/F)$ is $\sigma \text{Gal}(L/L \cap F)$, and $L \cap F/L \cap E$ is finite. Thus, the map is indeed a homeomorphism.

■

The infinite Galois correspondence allows us to classify how subgroups look in the Krull topology.

Theorem 5.5.10. *Let L/k be Galois.*

- (1) *The closed subgroups of $\text{Gal}(L/k)$ are precisely those of the form $\text{Gal}(L/E)$.*
- (2) *The open subgroups of $\text{Gal}(L/k)$ are precisely those of the form $\text{Gal}(L/F)$, where F/k is finite.*
- (3) *The normal closed subgroups of $\text{Gal}(L/k)$ are precisely those of the form $\text{Gal}(L/E)$, where E/k is Galois.*
- (4) *The normal open subgroups of $\text{Gal}(L/k)$ are precisely those of the form $\text{Gal}(L/F)$, where F/k is finite and Galois.*

Proof.

- (1) We know that all subgroups of the form $\text{Gal}(L/E)$ are closed. If $H \leq \text{Gal}(L/k)$ is closed, then $H = \text{Gal}(L/\text{Fix}(H))$ can be written in the desired form.

- (2) We know that subgroups of the form $\text{Gal}(L/F)$ for F/k finite are open by the definition of the Krull topology. If H is any open subgroup, then H contains some base open set around the identity $\text{Gal}(L/F)$ for F/k finite. Since H is a subgroup, this means that $\sigma \text{Gal}(L/F) \subseteq H$ for all $\sigma \in H$, and so $\tau \text{Gal}(L/F) \cap H = \emptyset$ for all $\tau \notin H$. Thus, H is closed (since its complement is open), say $H = \text{Gal}(L/E) \supseteq \text{Gal}(L/F)$. We then see that $E \subseteq F$, so E/k is finite, as desired.
- (3) A closed normal subgroup must be of the form $\text{Gal}(L/E)$, and E/k must be Galois by the Galois correspondence. All sets of this form are closed by (1) and normal again by the Galois correspondence.
- (4) An open normal subgroup must be of the form $\text{Gal}(L/F)$ for F/k finite, and F/k must be Galois by the Galois correspondence. All sets of this form are open by (2) and normal again by the Galois correspondence.

Corollary 5.5.11. *A subgroup of $\text{Gal}(L/k)$ is open if and only if it is closed with finite index.*

Proof. This follows immediately by recalling that $[\text{Gal}(L/k) : \text{Gal}(L/F)] = [F : k]$ when F/k is finite. ■

We may hope that a subgroup with finite index is automatically closed, but this turns out to be false in general. We will construct an example now. Let $H \leq G = \text{Gal}(\mathbb{Q}_{\sqrt{p}}/\mathbb{Q})$ consist of all automorphisms that act as the identity on $\sqrt{p}$ for all but finitely many p , and note that H is dense in G (indeed, we've seen that $\text{Fix}(H) = \mathbb{Q}$). We may view G/H as an $\mathbb{F}_2$ -vector space and find some subspace G'/H such that $(G/H)/(G'/H)$ has finite dimension n (by taking the span of all but finitely many elements of a basis, for example). Then G' is not open, as it would otherwise be closed, contradicting the density of H . But $[G : G'] = 2^n$ is finite.

Now, if N is a closed normal subgroup, then we know $\text{Fix}(N)$ is Galois over k . We may hope this is true even when N is not necessarily closed (that is, we may hope the closure of a normal subgroup is still normal). This is true, and immediate from the next result.

Corollary 5.5.12. *For any subset $S \subseteq \text{Gal}(L/k)$,*

$$\overline{S} = \bigcap_{\substack{H \leq \text{Gal}(L/k) \\ \text{open}}} SH = \bigcap_{\substack{N \trianglelefteq \text{Gal}(L/k) \\ \text{open}}} SN.$$

Proof. We first show that the intersection $\bigcap_H SH$ is closed. For each $s \in S$, sH is closed (as H is closed, and multiplication by an element is a homeomorphism). Since H has finitely many left cosets (since it is open), $SH = \bigcup_{s \in S} sH$ is closed. Since this is true for all H , $\bigcap_H SH$ is closed.

Suppose $C \supseteq S$ is closed, and take $g \notin C$. Then there is some open H such that gH is disjoint from C . In particular, gH contains no elements of S , and so we must have $g \notin SH$ (since SH is a union of cosets of H). This means that $\bigcap_H SH \subseteq C$, so $\bigcap_H SH = \overline{S}$.

Now, clearly $\bigcap_H SH \subseteq \bigcap_N SN$. But since every open subgroup contains an open normal subgroup, we have $\bigcap_H SH \supseteq \bigcap_N SN$, and so there is equality. ■

5.5.4 More on the Krull Topology

We explore some more properties of the Krull topology.

Proposition 5.5.13. *Let L/k be Galois. The Krull topology on $\text{Gal}(L/k)$ is totally disconnected.*

Proof. Let C be a connected subset of $\text{Gal}(L/k)$, and let $\sigma \in C$. For every open subgroup H , σH is clopen, so we must have $C \subseteq \sigma H$ (as it would otherwise disconnect C). Thus, $C \subseteq \bigcap_H \sigma H = \{\sigma\}$, since $\{\sigma\}$ is closed. ■

Theorem 5.5.14. *Let L/k be Galois. The Krull topology on $\text{Gal}(L/k)$ is compact.*

Proof. Define a map

$$\Phi : \text{Gal}(L/k) \rightarrow \prod_{F/k \text{ fin}} \text{Gal}(F/k)$$

given by $\sigma \mapsto (\sigma|_F)_F$, where the product runs over finite Galois extensions. This group on the right is given the product topology, where each $\text{Gal}(F/k)$ is given the discrete topology (which is the same as the Krull topology). The map Φ is clearly a homomorphism. Further, Φ is injective: every element of L is contained in a finite Galois extension of k , and so the only way for $\Phi(\sigma) = \Phi(\tau)$ is if σ and τ agree everywhere. Additionally, since each of the component maps $\text{Gal}(L/k) \rightarrow \text{Gal}(F/k)$ is continuous, the product map Φ is also continuous.

We now show that Φ is homeomorphic into its image (which is given the subspace topology). Since Φ is continuous and injective, we must show it is open. Take a base open set $\sigma \text{Gal}(L/F_0)$, and shrink so that F_0/k is Galois. Then $\Phi(\sigma \text{Gal}(L/F_0))$ consists of precisely the elements of $\Phi(\text{Gal}(L/k))$ that agree with σ on F_0 . That is to say,

$$\Phi(\sigma \text{Gal}(L/F_0)) = \Phi(\text{Gal}(L/k)) \cap \left(\{\sigma|_{F_0}\}_{F_0} \times \prod_{F \neq F_0} \text{Gal}(F/k) \right),$$

which is open in the subspace topology.

Finally, we show that $\Phi(\text{Gal}(L/k))$ is closed. Note that $\Phi(\text{Gal}(L/k))$ consists precisely of the sequences $(\sigma_F)_F$ such that $\sigma_{F'} = (\sigma_{F''})|_{F'}$ whenever $F' \subseteq F''$ (since, given any such sequence, we can “glue” together an element of $\text{Gal}(L/k)$). If we have a sequence $(\sigma_F)_F \notin \Phi(\text{Gal}(L/k))$, then we can find fields $F' \subseteq F''$ such that $\sigma_{F'} \neq (\sigma_{F''})|_{F'}$. Thus, if we consider the set of sequences $(\tau_F)_F$ that agree with σ on F' and F'' , this is an open subset of $\prod_F \text{Gal}(F/k)$ (as we are only imposing constraints on finitely many coordinates) that is disjoint from $\Phi(\text{Gal}(L/k))$. Thus, $\Phi(\text{Gal}(L/k))^c$ is open, and so $\Phi(\text{Gal}(L/k))$ is closed.

Since $\prod_F \text{Gal}(F/k)$ is compact by Tychonoff’s theorem, and $\Phi(\text{Gal}(L/k))$ is a closed subset, it is also compact. Since Φ is a homeomorphism of $\text{Gal}(L/k)$ onto $\Phi(\text{Gal}(L/k))$, we see that $\text{Gal}(L/k)$ is compact, as desired. ■

Our proof in particular tells us that the Krull topology on $\text{Gal}(L/k)$ is the same as the topology it gets under the subspace topology of $\prod_F \text{Gal}(F/k)$. We also note that, since $\Phi(\text{Gal}(L/k))$ consists of exactly those sequences $(\sigma_F)_F$ with the compatibility relation $\sigma_{F'} = (\sigma_{F''})|_{F'}$ for $F' \subseteq F''$, we may view

$$\text{Gal}(L/k) = \varprojlim \text{Gal}(F/k).$$

By contrast, $L/k = \varinjlim F/k$.

This view allows us to relatively easily compute some Galois groups. For instance, we know that $\overline{\mathbb{F}_q}/\mathbb{F}_q = \varinjlim \mathbb{F}_{q^n}/\mathbb{F}_q$, and this gives us that

$$\text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) = \varprojlim \text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \varprojlim \mathbb{Z}/n\mathbb{Z},$$

where the relating maps $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ are present when $m \mid n$ and are given by reducing modulo m .